The Infinite, the Infinitesimal and the Imaginary

Introduction to Differential and Integral Calculus

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Outline of the Workshop

What is the interest of differential and integral calculus

beyond mathematics?

- It provides a formal framework for addressing Zeno's paradoxes concerning the notions of ***change*** and ***continuity*** that have haunted mathematics, physics, and philosophy since at least the time of the Greeks—paradoxes that, as Bergson put it, motivated the emergence of metaphysics.

- By doing so, it offers a mathematical formalization and conceptualization of the notion of ***change***.

- The mathematical taming of change required opening the doors of mathematics to the notion of ***actual infinity***.

- In the same vein, it is the mathematical theory that underpins the **predictive power** of the natural sciences and provides the ***formal grammar of physical theories*** (e.g., classical and quantum mechanics, electromagnetic theory, general relativity, etc.).

- From the standpoint we shall adopt in this workshop, it also bears on what we might call the philosophy of the category of ***quantity***.

The (Western) Fourfold Genealogy of Calculus

♣ **The Greek moment** (5th–3rd centuries BCE):

.Zeno of Elea: the paradoxes of continuous motion;

.Aristotle: the notion of continuity and the rejection of actual infinities;

.Eudoxus of Cnidus and Archimedes of Syracuse: the method of exhaustion as a proto-limit process.

♥ **The Golden Age** (17th–18th centuries):

.Newton;

.Leibniz: calculus based on the notion of *infinitesimal.*

♠ **The Arithmetization Enlightenment** (19th century) or the War on Infinitesimals: Cauchy, Weierstrass, Riemann, Cantor.

♦ **Contemporary Offsprings** (20th century):

.Cartan's *theory of differential forms;*

.Robinson's *nonstandard analysis;*

.Grothendieck's *theory of schemes* in algebraic geometry,

.*Synthetic differential geometry* (Lawvere, Kock).

The fundamental act of rationalization

that grounds the so-called « mathematization of nature »

is the ***quantification*** of natural phenomena,

that is, the assignment of numbers of a certain kind

through the measurement of different ***quantities*** (e.g. weight, size, etc.).

The central concept we shall consider here is that of

quantity,

together with its various generalizations.

First, we shall consider

constant quantities

also known as

numbers.

We shall start with the natural numbers

$$\mathbb{N} = \{0,1,2,3...\}$$

and consider its different

finite and transfinite extensions.

We shall then consider ***variable quantities***

described by

functions.

Numbers of a certain kind B

that vary along an ***extension*** or arena of variability A

will be described by functions of the form:

$$f : A \to B$$

.

The ***evaluation*** of a function f on point a of A

yields a number denoted $f(a)$.

Finally, we shall consider quantities

that encode the rates of variability of functions

known as

differential forms.

Differential forms « derive » from functions

— and from other differential forms —

in the sense that they encode the rates of variability of the latter.

The definition of differential forms requires

introducing the notion of

derivative.

Whereas functions can be evaluated on points

to yield numbers,

differential forms can be « evaluated »

on higher dimensional objects

— like curves, surfaces, etc. —

and the corresponding generalized form of « evaluation » is called

integration.



Finally, we shall generalize the notion of differential forms

to differential forms that do not encode the variability of another form

i.e. that do not « derive » from another form.

We shall see that these forms can be used to characterize

an important topological property of spaces,

namely the presence of ***holes***.

The whole trajectory of the workshop

will take us from the ***infinite*** ∞

— the first extension of the notion of quantity that we shall consider —

to the ***hole*** ◯.

Numbers

The fundamental act of rationalization

that grounds the so-called « mathematization of nature »

is the ***quantification*** of natural phenomena,

that is, the assignment of ***quantities*** of different kinds

— tipically, ***numbers*** —

depending on the phenomenon at stake

by ***counting***, ***measuring***, etc.

« Meti […] números nas coisas e era ciência. »

P. Leminski, *Catatau*.

Number as an Operator of Enlightened Disenchantment

« The mythologizing equation of Forms with numbers in Plato's last writings expresses the longing of all demythologizing: **number became enlightenment's canon**. […] Bourgeois society is ruled by equivalence. It makes dissimilar things comparable by reducing them to abstract quantities. **For the Enlightenment, anything which cannot be resolved into numbers** […] **is illusion**. […] All gods and qualities must be destroyed. »

M. Horkheimer and T.W. Adorno, *Dialectics of Enlightenement. Philosophical Fragments*, tr. E. Jephcott, pp.4-5.

« Nunca quis meter um número na minha cachola, queria viver num mundo de qualidade, agora tardio. »

P. Leminski, *Catatau*.

« Nous vivons le temps du despotisme du nombre […] et cependant […] nous ne disposons d'aucune idée récente, active, de ce que c'est qu'un nombre. […] L'impact factuel du nombre n'escorte qu'un silence du concept. […] les nombres […] servent, au sens strict, à tout […] Mais que sont-ils nous l'ignorons […] »

A. Badiou, *Le Nombre et les nombres*, Paris: Éditions du Seuil, 1990, §0.1, p.11.

Rather than trying to save the world

from the “icy waters”

of disenchanted numerical quantification,

we shall try to show that the evolution

of our understanding of the notion of *number*

is an open living adventure of human understanding

— that is, a dehiscence of life itself —

that traverses human histories and geographies.

The thesis according quantification

is concomitante with a disenchantment

presupposes that mathematics

is not itself a living inflorescence of life.

« Ô mathématiques sévères, je ne vous ai pas oubliées, depuis que vos savantes leçons, plus douces que le miel, filtrèrent dans mon cœur comme une onde rafraîchissante. J'aspirais instinctivement, dès le berceau, à boire à votre source, plus ancienne que le soleil, et je continue encore de fouler le parvis sacré de votre temple solennel, moi, le plus fidèle de vos initiés. Il y avait du vague dans mon esprit, un je ne sais quoi épais comme de la fumée ; mais je sus franchir religieusement les degrés qui mènent à votre autel, et vous avez chassé ce voile obscur, comme le vent chasse le damier. Vous avez mis, à la place, une froideur excessive, une prudence consommée et une logique implacable. À votre lait fortifiant, mon intelligence s'est rapidement développée et a pris des proportions immenses, au milieu de cette clarté ravissante dont vous faites présent, avec prodigalité, à ceux qui vous aiment d'un sincère amour. »

Lautréamont, *Chants de Maldoror*.

« Continuo com capacidade de raciocínio – já estudei matemática que é a loucura do raciocínio – mas agora quero o plasma – quero me alimentar diretamente da placenta. »

Clarice Lispector, *Água viva*

« […] la “création” dans mon travail de mathématicien, c’était avant tout là qu’elle se plaçait : dans cette attention intense pour appréhender, dans les replis obscurs, informes et moites d’une chaude et inépuisable matrice nourricière, les premières traces de forme et de contours de ce qui n’était pas né encore et qui semblait m’appeler, pour prendre forme et s’incarner et naître… Dans le travail de découverte, cette attention intense, cette sollicitude ardente sont une force essentielle, tout comme la chaleur du soleil pour l’obscure gestation des semences enfouies dans la terre nourricière, et pour leur humble et miraculeuse éclosion à la lumière du jour. »

A. Grothendieck, *Récoltes et semailles*.

Metaphysical Oppositions

Sublated by the History of the Mathematical Notion of *Quantity*.

- The ***constant*** and the ***variable***.
- The ***finite*** and the ***infinite***.
- The ***discrete*** and the ***continuous***.
- The ***relative*** and the ***universal***.
- The ***local*** and the ***global***.

The central concept we shall consider in this workshop

is the first incarnation of the notion of ***quantity***,

namely the ***constant quantities*** known as ***numbers***.

We shall start with the ***natural numbers***

$$\mathbb{N} = \{0,1,2,3...\}$$

and consider some of its different

finite and transfinite extensions.

Natural Numbers

$$\mathbb{N} = \{0,1,2,3...\}$$

The inclusion of ***zero*** as a number

was a non-trivial conceptual and formal step,

starting in India

(introduced by Brahmagupta in the 7th century BC)

and — after being rejected as a number by the Greeks —

transmitted through the Arabic world

(cf. al-Khwarizmi, 10th century)

into Europe in the 11th century by Fibonacci.

Refs.: R. Kaplan & E. Kaplan, *The Nothing That Is: A Natural History of Zero*, Oxford University Press, 2000; Seife, *Zero: The Biography of a Dangerous Idea*, Penguin Books, 2000.

« God made the integers,

all else is the work of man »

Leopold Kronecker (1823-1891)

Numbers in Greek Mathematics (I)

In Greek mathematics,

numbers are understood exclusively

as ***discrete numbers***

composed of units.

Continuity (i.e. the non-discrete) does not belong to the realm of arithmetics,

but to the realm of ***geometry***.

The opposition between the ***discrete*** and the ***continuous***

grounds the separation between ***arithmetics*** and ***geometry***.

Numbers in Greek Mathematics (II)

Number as a plurality of units in Euclid

« 1. An unit is that by virtue of which each of the things that exist is called one.

2. A number is a multitude composed of units. »

Euclid, *The Elements*, Book VII, Definitions, tr. by T.L. Heath, p.277.

The Discreteness of Being

According to Euclid's definition,

numbers have an ontological scope

given by the fact that anything that exist

is *one thing*.

Numbers in Greek Mathematics (III)

Numbers according to Aristotle

« […] el *ser* de lo uno es *ser* un principio del numero […] lo que es indivisible según la cantidad […] se llama 'unidad' »

Aristoteles, *Metafísica*, 1016b, 15-25, tr. E. Sinnott, Ediciones Colihue, p.349.

« […] todo numero significa una cantidad y […] la unidad […] [es] lo indivisible según la cantidad. »

Aristoteles, *Metafísica*,1089b, 35, *op.cit.*, p.1069.

The Concept of *Natural Number* in the XIX-XX Centuries

- **Logicist definition** (Frege, Russell).

- **Axiomatic or formalist definition** (Peano, Hilbert).

- **Set-theoretic definition** (Dedekind, Cantor, Zermelo, von Neumann Gödel).

The Logicist Definition of *Natural Numbers* (I)

« […] the apparatus of general logical notions […] suffices, without new indefinables or new postulates, to establish the whole theory of cardinal integers as a special branch of Logic. »

B. Russell, *Principles of Mathematics*, 1903, § 107, p.111.

The Logicist Definition of *Natural Numbers* (II)

The logicist definition of natural numbers

exhibit numbers as the ***abstract property***

shared by sets

that can be put in a one-to-one correspondence.

In other terms, natural numbers are defined

as instances of an abstract notion

— called ***cardinal number*** —-

given by the equinumerosity relation.

The Logicist Definition of *Natural Numbers* (III)

Definition: A number associated to a concept *F*

is the set of concepts

whose extension is equinumerous to the extension of *F*.

(Cardinal) numbers are not beings,

but features of (the extensions of) concepts.

They have a logical (rather than ontological) nature.

Ref.: G. Frege, *The Foundations of Arithmetic* [*Die Grundlagen der Arithmetik*], tr. J.L. Austin, Evanston: Northwestern University Press, 1976.

The Logicist Definition of *Natural Numbers* (IV)

Frege's Definition of Numbers

• Given a predicate $P(x)$, we can define a set E_P, the *extension* of $P(x)$, whose elements are the objects a such that $P(a)$ is true.

Note: this presupposes that the set E_P exists for every $P(x)$ (called ***axiom schema of unrestricted comprehension***).

• Two predicates $P(x)$ and $Q(x)$ have the same number if it is possible to establish a bijection between E_P and E_Q (which are then said to be ***equinumerous***).

• The ***number of a predicate*** $P(x)$ is the set of all concepts $Q(x)$ such that their extensions are equinumerous to the extension E_P of $P(x)$.

The Logicist Definition of *Natural Numbers* (V)
The Definition of Zero

For instance, the number 0 is the number

associated to the concept

« being non identical to itself »,

whose extension is empty,

where we are accepting that « there is no entity without identity » (Quine).

« I could have used for the definition of nought any other concept under which no object falls. But I have made a point of one which can be proved to be such on purely logical grounds […] » G. Frege, *The Foundations of Arithmetic*, op.cit., § 74.

Hence, this definition is purely logical

since it depends on a purely logical principle, the identity principle $a = a$.

The Logicist Definition of *Natural Numbers* (V)
The Definition of One

• Frege defines 1 as « the number which belongs to the concept 'identical with 0' », G. Frege, op.cit., § 77, p.90.

• This means that 1 is the set of concepts whose extension is equinumerous to the extension of the concept 'identical with 0'.

• Now, what is the extension of the concept 'identical with 0'?

• Since 0 is the cardinal of the empty set ∅, there is no set other than ∅ that is identical to ∅. Hence, the extension of the concept « Identical to 0 » is given by a set that only contains ∅ (i.e. a « unique » element). There is only one set without elements, namely the empty set ∅.

• All in all, 1 is the collection of all concepts whose extension can be put in a one-to-one correspondence with a set with a « unique » element.

Cf. G. Frege, *The Foundations of Arithmetic*, op.cit., § 77, p.90.

Russell's Paradox
(communicated to Frege in 1903)

Russell exhibits a concept

$P =$ « Being a set that is not an element of itself »

whose extension does not exist (i.e. it is not even empty).

Let's call A the extension of P, i.e. a set of sets. Does A belong to A?

If we assume that $A \in A$, then A contains a set which is an element of itself (which is contradictory with the definition of P).

If we assume that $A \notin A$, then there is a set that is not an element of itself that does not belong to A. Hence, A cannot be the extension of P.

Indeed, it is contradictory to think

that the objects (sets) that fall under this concept form a set.

Such a concept cannot be numbered.

Peano's Axiomatic Definition (I)
(1889)

Peano does not define explicitly

what a natural number is,

but rather specifies a set of axioms

they have to fulfill

(« implicit definition », cf. Schlick, *General Theory of Knowledge*, 1918).

« Toute procedure purement axiomatique introduit des signes *non définis*, dont il n'y aura d'autre presentation dans la pensée que la codification de leur usage par les *axiomes*. »

A. Badiou, *Le Nombre et les nombres*, op.cit., §5.7, p.67.

Peano's Axiomatic Definition (II)
(1889)

- 0 is a natural number.

- Every natural number n has a *successor* $S(n)$:

$1 \doteq S(0)$
$2 \doteq S(1) = S(S(0))$
$3 \doteq S(2) = S(S(1)) = S(S(S(0)))$

- $n \neq n'$ cannot have the same successor (i.e. S is an injection).

- There is no n such that $S(n) = 0$, i.e. 0 is the unique natural number that is not a successor.

- Axiom of Induction (see below).

« I regard the whole of arithmetic as a necessary, or at least natural, consequence of the simplest arithmetic act, that of counting, and counting itself as nothing else than the successive creation of the infinite series of positive integers in which each individual is defined by the one immediately preceding; the simplest act is the passing from an already-formed individual to the consecutive new one to be formed. »

R. Dedekind, *Continuity and Irrational Numbers*, 1872, p.4.

Cantor's *First Principle of Generation*

« The sequence (I) of positive integers 1,2,3…, n, … has its ground of origin [Entstehungsgrund] in the repeated positing [Setzung] and uniting [Vereinigung] of underlying unities [Einheiten], which are regarded as alike; the number n is the expression for a definite finite number [bestimmte endliche Anzahl] of such positings following one another in a sequence; it is also the expression for the unification [Vereinigung] of the posited unities [gesetzten Einheiten] into a whole [zu einem Ganzen]. The formation of the finite real integers [i.e. natural numbers] thus rests upon the principle of adding a unity to an already formed and existing number; I call this principle (which, as we shall soon see, plays an essential role in the generation of the higher integers) the *first principle of generation*. The number [Anzahl] of the numbers n […] formed in this way is infinite and there is no greatest among them. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

Addition of Natural Numbers

- $a + 0 = a$ (i.e. 0 is the neutral element for the operation +).
- $a + S(b) = S(a + b)$

Examples:

$$2 + 1 = 2 + S(0) = S(2 + 0) = S(2)$$

(adding 1 amounts to apply once the successor operator)

$$2 + 2 = 2 + S(1) = S(2 + 1) = S(2 + S(0)) = S(S(2 + 0)) = SS(2) = 4$$

(adding 2 amounts to apply twice the successor operator)

All in all, adding *n* to *a* amounts to apply the successor operators to *a* *n* times.

« Addition is the combination of any arbitrary repetitions

of the above-mentioned simplest act into a single act »

R. Dedekind, *Continuity and Irrational Numbers,* p.4.

Additive Algebraic Structure of $\mathbb{N}$

It can be shown that the operation $+$ is

- **closed**:

$$a + b \in \mathbb{N} \text{ if } a, b \in \mathbb{N}.$$

- **associative** :

$$(a + b) + c = a + (b + c).$$

- **commutative** :

$$a + b = b + a$$

It follows that $(\mathbb{N}, +, 0)$ is a *commutative monoid,*

where 0 is the neutral element for the addition: $n + 0 = n$.

Note: the operation $+$ is not invertible.

Multiplication of Natural Numbers

- $a \times 0 = 0.$
- $a \times S(b) = a \times b + a$

Examples:

$$2 \times 1 = 2 \times S(0) = 2 \times 0 + 2 = 0 + 2 = 2$$

(the number 1 is the neutral element for the multiplication)

$$2 \times 2 = 2 \times S(1) = 2 \times 1 + 2 = 2 \times S(0) + 2 = 2 \times 0 + 2 + 2 = 4$$

Hence, $n \times n'$ means adding $n + n + \ldots + n$ with n' summands

or $n' + n' + \ldots + n'$ with n summands since $\times$ is commutative).

Multiplicative Algebraic Structure of $\mathbb{N}$

It can be shown that the operation $\times$ is

- **closed**:

$$a \times b \in \mathbb{N} \text{ if } a, b \in \mathbb{N}.$$

- **associative** :

$$(a \times b) \times c = a \times (b \times c).$$

- **commutative** :

$$a \times b = b \times a$$

It follows that $(\mathbb{N}, \times, 1)$ is a *commutative monoid,*

where 1 is the neutral element for the addition: $n \times 1 = n$.

Note: the operation $\times$ is not invertible.

The addition $+$ and the multiplication $\times$ are related by the ***distributivity law***

$$a \times (b + c) = (a \times b) + (a \times c)$$

stating that multiplication $\times$ distributes over addition $+$.

$(\mathbb{N}, +, \times, 0, 1)$ is a ***commutative semi-ring***.

The Axiom of Induction

A predicate $P(n)$ can be demonstrated to be true for every n by

1) showing that $P(0)$ is true.

2) showing that $P(n) \Rightarrow P(n + 1)$ is true.

Example of Mathematical Induction

Prove that

$$\sum_{i=0}^{n} i = 0 + 1 + 2 + \ldots + n = \frac{n(n+1)}{2}$$

Peano and the Infinite

« […] la récurrence permet […] de légiférer dans une totalité infinie *sans parler de son infinité*. Il est clair en effet qu'il y a une infinité de nombres entiers. Parler de 'tous' ces nombres veut donc dire parler d'une infinité actuelle. Mais dans le dispositif axiomatique de Peano, cette infinité n'est pas introduite comme telle. L'axiome de récurrence permet, à partir d'une *vérification* (1 possède la propriété) et d'une *démonstration* implicative (si n possède la propriété, *alors* $n + 1$ la possède aussi) , de conclure que 'tous les nombres possèdent la propriété', sans avoir à s'interroger sur l'extension de ce 'tous'. Le quantificateur universel masque ici la pensée d'un infini actuel : l'infini demeure une forme latente, inscrite au quantificateur sans être délivrée dans la pensée. Peano introduit ainsi le concept de nombre sans transgresser le vieil interdit qui frappe l'infini actuel, interdit qui surplombe encore notre pensée […]. L'axiomatique de Peano évite l'infini, ou la mention explicite de l'infini. »

A. Badiou, *Le Nombre et les nombres*, Paris: Éditions du Seuil, 1990, §5.4, p.64.

Poincaré on Mathematical Induction

« Pourquoi donc [le jugement sur lequel repose le raisonnement par récurrence peut] s'impose-t-il à nous avec une irrésistible évidence ? C'est qu'il n'est que l'affirmation de la puissance de l'esprit qui se sait capable de concevoir la répétition indéfinie d'un même acte dès que cet acte est une fois possible. […] L'induction mathématique, c'est-à-dire la démonstration par récurrence, s'impose […] nécessairement, parce qu'elle n'est que l'affirmation d'une propriété de l'esprit lui-même. »

H. Poincaré, *La science et l'hypothèse*, Ch. I *Sur la nature du raisonnement mathématique.*

« Le sens dont je parle et qui mène l'esprit à se prévoir lui-même, à imaginer l'ensemble de ce qui allait s'imaginer dans le détail, et l'effet de la succession, ainsi résumée, est la condition de toute généralité. »

Paul Valéry, *Introduction à la méthode de Léonard de Vinci*.

General properties of $\mathbb{N}$

• Natural numbers provide the general scheme of ***discreteness***.

« […] 'le nombre" [entier] est apte à saisir la structure des agrégats 'discontinus', ou 'discrets' : les systèmes, souvent finis, formés d"'éléments' ou 'objets' pour ainsi dire isoles les uns par rapport aux autres, sans quelque principe de 'passage continu' de l'un à l'autre », A. Grothendieck, *Récoltes et semailles*.

• The set $\mathbb{N}$ is endowed with a (reflexive, antisymmetric, transitive, and total) relation $\leq$ called a *total order*. Hence they can be used to « measure » a degree of *greatness*.

• According to this order, $\mathbb{N}$ has a smallest element (namely, $0 \in \mathbb{N}$), but not a biggest one. In other terms, every $n \in \mathbb{N}$ has a successor, but not every $n \in \mathbb{N}$ is a successor of another number (being the exception $0 \in \mathbb{N}$, which is not a successor).

• Natural numbers introduce an example of a ***potential infinite***, i.e. of and endless progression.

• They can be used to *count* (cardinals) and *order* (ordinals).

Rather than asking the ontological question

What are numbers?,

we shall ask the pragmatic question

What can we do with (natural) numbers?

Cardinals vs. Ordinales

Cardinal numbers

specify the « size » of a finite set X, its number of elements

(they provide an answer to the question ***How many elements a set X have?***)

Ordinal numbers

specify the position (zeroth, first, second, third…) of an element in a *well-ordered set*

and the *length/order type* of the later:

(they provide an answer to the question ***What is the length of a well-ordered set*** $(X, \leq)$***?***)

[Definition: A *well-ordered* is a set X endowed with total order $\leq$

such that each subset has a minimal element]

« Dans la vision ordinale, le nombre est pensé comme le maillon d'une chaîne, il est élément d'un ordre total. Dans la vision cardinale, il est plutôt la marque d'une 'quantité pure' obtenue par abstraction de domaines d'objets ayant 'même quantité'. Le nombre ordinal est sous le schème d'une *suite*, le nombre cardinal sous celui d'une *mesure*. »

A. Badiou, *Le Nombre et les nombres*, *op.cit.*, p.45.

For finite sets, these notions of cardinal and ordinal number coincide,

that is, the ***length*** or ***order type*** of $(X, <)$

— i.e. the least ordinal that is not a label for an element of the set —

for any order $<$

coincides with the cardinal of the set.

For instance, in the ordered set $X = \{a < b < c\}$,

a is the zeroth element, b the first and c the second

(which means that 3 is the least ordinal number that is not a label for an element of the set).

The ordered set X has length or order type 3 (ordinal) and has 3 elements (cardinal).

The Potential Infinite

The Potential Infinite of $\mathbb{N}$

There is an infinite sequence 0, 1, 2, 3… of natural numbers,

but every natural number *n* is finite.

The term ***potential infinite***

— also called *improper infinite* [*Uneigentlich-Unendliche*] by Cantor —

does not designate a precise number,

but rather the fact that the successor operation *S*

(Cantor's *First Principle of Generation*)

that consists in adding a unity

can be continued forever.

However, this endless process only produces (bigger and bigger) finite numbers.

« L'infini math[ématique] est une manière d'énoncer le mot *Pouvoir*. 'Je puis ajouter 1 quel que soit *n*.' C'est détacher l'acte de sa matière. »

P. Valery, *Cahiers II*, Bibliothèque de la Pléiade, p.819.

« Pourquoi pas un tour de spire de plus, dit le coquillage? Ainsi la pensée. Ainsi l'infini »

P. Valery (XIX, 147).

Potential Infinity According to Aristotle

"For generally the infinite is as follows:

there is always another and another to be taken.

And the thing taken will always be finite, but always different."

Aristotle, *Physics*, 206a27-29.

The « bad infinity » of the infinite progression according to Hegel (I)

« Something becomes an other, but the other is itself a something, hence it likewise becomes an other, and so on and so forth ad infinitum. […] This progression to the infinite is, however, not the true infinite […] [but] the bad infinity of the infinite progression. […] Engaging in the contemplation of this infinite progression is tedious because the same thing is incessantly repeated here. […] there is nothing here but a superficial alternation that remains stuck in the finite. […] If it is then further said that the infinite cannot be reached, then this is quite right, but only because the determination of being something abstractly negative is read into it. Philosophy does not waste its time with such empty and merely transcendent [*Jenseitigen*] things. What philosophy deals with is always something concrete and absolutely present. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §94.

The « bad infinity » of the infinite progression according to Hegel (II)

« I amass colossal numbers,
Millions of mountains,
I pile time upon time
And world upon worlds galore,
And when from this terrifying height
With vertigo I look to you again:
If all the mighty numbers
Were increased thousand-fold,
They would not even be a part of you.

So here we have, first, that constant propelling of quantity and, more specifically, of number beyond itself that Kant describes as horrifying, although the actual horror is perhaps only the boredom of a limit constantly being set and then sublated so that, as a result, one does not make any headway. But, furthermore, the poet mentioned aptly adds to this description of the bad infinity the conclusion:

I remove them, and you lie before me entire

– thereby making it explicit that the true infinite is not to be regarded as something merely beyond the finite and that, to attain consciousness of it, we must renounce that *progressus in infinitum*. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §104.

Aquinas on Infinity

Actual infinite does not exist

because all what exists is (according to the Bible)

ordered « in measure and number »,

but there are no infinite numbers.

« It is written, 'Thou hast ordered all things in measure, and number, and weight' (Wisdom 11:21). [...] every kind of multitude must belong to a species of multitude. Now the species of multitude are to be reckoned by the species of numbers. But no species of number is infinite […] Hence it is impossible for there to be an actually infinite multitude […] But a potentially infinite multitude is possible; because the increase of multitude follows upon the division of magnitude; since the more a thing is divided, the greater number of things result. Hence, as the infinite is to be found potentially in the division of the continuous […] by the same rule, the infinite can be also found potentially in the addition of multitude. »

St. Thomas Aquinas, *Summa Theologica*, Part I, Question 7, Article 4.

Note: Cantor will later make an explicit reference to this paragraph.

Now, what is the obstruction

to the conception

of an actual infinite number?

Equinumerocity via Bijections

Two sets A and B have the same number of elements

(i.e. the same cardinal)

if it is possible to establish

a one-to-one or bijective correspondence

$$f : A \to B$$

between their elements,

i.e. such that for every $y \in B$, there is a unique $x \in A$

such that $f(x) = y$.

Cf.; Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638; Cantor; denied by Bolzano *Paradoxes of the infinite*, 1851, §19-2.

Countable Sets

The cardinality of $\mathbb{N}$

— denoted $|\mathbb{N}|$ —

is called (after Cantor) ***aleph*-0**

and denoted $\aleph_0$.

A set X is *countable*

— i.e. its cardinality is $\aleph_0$, $|X| = \aleph_0$ —

if there is a bijection

$$f : X \to \mathbb{N}$$

i.e. if the elements of X can be put in a one-to-one correspondence with the set $\mathbb{N} = \{0,1,2,3,...\}$.

Galileo's Paradox

Now, it is possible to establish a bijection

$$f : \{0,1,4,9,16,25,...\} \to \mathbb{N}$$

$$n^2 \mapsto n$$

Hence, there are as many natural numbers

0,1,2,3,4,...

as square numbers

0,1,4,9,16,...

Therefore, a strict part (the set of square numbers)

is as big as the whole (ℕ),

which contradicts Euclid's Common Notion 5 « The whole is greater than the part ».

Galileo on Infinity (I)

• **Simplicio**: Here a difficulty presents itself which appears to me insoluble. Since it is clear that we may have one line greater than another, each containing an infinite number of points, we are forced to admit that, within one and the same class, we may have something greater than infinity, because the infinity of points in the long line is greater than the infinity of points in the short line. This assigning to an infinite quantity a value greater than infinity is quite beyond my comprehension.

• **Salviati**: This is one of the difficulties which arise when we attempt, with our finite minds, to discuss the infinite, assigning to it those properties which we give to the finite and limited; but this I think is wrong, for we cannot speak of infinite quantities as being the one greater or less than or equal to another. To prove this I have in mind an argument which, for the sake of clearness, I shall put in the form of questions to Simplicio who raised this difficulty. I take it for granted that you know which of the numbers are squares and which are not.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

Galileo on Infinity (II)

• **Simplicio**: I am quite aware that a squared number is one which results from the multiplication of another number by itself; thus 4, 9, etc., are squared numbers which come from multiplying 2, 3, etc., by themselves.

• **Salviati**: Very well; and you also know that just as the products are called squares so the factors are called sides or roots; while on the other hand those numbers which do not consist of two equal factors are not squares. Therefore if I assert that all numbers, including both squares and non-squares, are more than the squares alone, I shall speak the truth, shall I not?

• **Simplicio**: Most certainly.

• **Salviati**: If I should ask further how many squares there are one might reply truly that there are as many as the corresponding number of roots, since every square has its own root and every root its own square, while no square has more than one root and no root more than one square.

• **Simplicio**: Precisely so.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

Galileo on Infinity (III)

• **Salviati**: But if I inquire how many roots there are, it cannot be denied that there are as many as the numbers because every number is the root of some square. This being granted, we must say that there are as many squares as there are numbers because they are just as numerous as their roots, and all the numbers are roots. Yet at the outset we said that there are many more numbers than squares, since the larger portion of them are not squares. Not only so, but the proportionate number of squares diminishes as we pass to larger numbers, Thus up to 100 we have 10 squares, that is, the squares constitute 1/10 part of all the numbers; up to 10000, we find only 1/100 part to be squares; and up to a million only 1/1000 part; on the other hand in an infinite number, if one could conceive of such a thing, he would be forced to admit that there are as many squares as there are numbers taken all together.

• **Sagredo**: What then must one conclude under these circumstances?

• **Salviati**: So far as I see we can only infer that the totality of all numbers is infinite, that the number of squares is infinite, and that the number of their roots is infinite; neither is the number of squares less than the totality of all the numbers, nor the latter greater than the former; and finally *the attributes "equal," "greater," and "less," are not applicable to infinite, but only to finite, quantities*. When therefore Simplicio introduces several lines of different lengths and asks me how it is possible that the longer ones do not contain more points than the shorter, I answer him that one line does not contain more or less or just as many points as another, but that each line contains an infinite number.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

The Infinite Dilemma

Either

♣ we deny the concept of an infinite number and the possibility of an actual infinity (as Aristotle, Aquinas, Galileo and Leibniz did);

♥ or we limit the validity of Euclid's Common Notion 5 « The whole is greater than the part » to finite numbers (as Cantor did).

The Cartesian *Indefinite* (I)

Descartes warns us

that we cannot infer the existence of an actual infinite

(a notion that can only be applied to God)

from the mere absence of limits.

« […] the Cardinal of Cusa and other theologians have supposed the world to be infinite […] my opinion is easier to accept than theirs, because I say not that the world is infinite but only that it is *indefinite*. There's a quite notable difference between the two: because if we say that something is infinite we should have a reason that tells us that it is so; and we can't have such a reason except in the case of God; but to say that a thing is indefinite, all we need is *not* to have a reason showing that it has bounds. »

R. Descartes, *Letter to Chanut*, June 6th, 1647.

The Cartesian *Indefinite* (I)

« [...] we must never discuss the infinite, but must simply consider those things in which we notice no limits as indefinite; as, for instance, the extension of the world, the divisibility of parts of matter, the number of stars, etc. Thus we shall never be wearied by any debates concerning the infinite. [...] inasmuch as we are finite, it would be absurd for us to attempt to determine anything concerning the infinite [...] surely only those who judge their own mind to be infinite ought to think about such things. Moreover, we shall not affirm that all those things in which we have been able to find no limit after some consideration are infinite; but shall view them as indefinite. [...] And we shall say that these things [the extension of the world, the divisibility of parts of matter, the number of stars] are indefinite rather than infinite: both in order to reserve the term ‘infinite' for God alone; because in Him alone, in every respect, we not only recognize no limits [to His perfection], but also in a positive sense understand that there are none […] »

R. Descartes, *Principles of Philosophy*, § 26 and 27, 1644.

Leibniz on the Infinite

I. The Infinite is not a Number

« I concede the infinite plurality of terms, but this plurality itself does not constitute a number or a single whole. It means […] that there are more terms than can be designated by a number. Just so there is a plurality or a complex of all numbers, but this plurality is not a number or a single whole. »

G.W. Leibniz, Letter to Bernoulli, February 21, 1699.

Leibniz on the Infinite

II. The Infinite *qua* Number is a Useful Fiction

« [...] the infinite is not a whole, but only a fiction, since otherwise the part would be equal to the whole. » (Leibniz, quoted in Arthur & Rabouin, 2025, p.172).

« I no more support infinitely small magnitudes than infinitely large ones, or no more infinitesimals than infinituples. For I consider both to be fictions of the mind, due to abbreviated ways of speaking, which are suitable for calculation, in the way that imaginary roots in algebra are. »

G.W. Leibniz, Letter to Des Bosses, March 11, 1706.

« I also deny that properly speaking there exists an infinite or an infinitely small number, or an infinite or infinitely small line—even though Euclid often speaks of an infinite line, but in a sound sense. The *infinite*, whether *continuous* or *discrete*, is properly neither one, nor a whole, nor a quantity [*quantum*]; and if we take it as such by a certain analogy, it is only, so to speak, a way of speaking; when namely, there are more things than can be comprised by any number, yet by analogy we attribute to these things a number, which we call infinite. »

G.W. Leibniz, An Observation That Ratios or Proportions Do Not Hold for Quantities Less Than Zero, and on the True Meaning of the Infinitesimal Method, quoted in R.T.W. Arthur & D. Rabouin, *Leibniz on the Foundation of the Differential Calculus*, p.245.

Leibniz on the Infinite

III. The Existence of the Actual Infinite

« I am so in favor of the actual infinite that instead of admitting that Nature abhors it, as is commonly said, I hold that Nature makes frequent use of it everywhere, in order to show more effectively the perfections of its Author. Thus I believe that there is no part of matter which is not, I do not say divisible, but actually divided; and consequently the least particle ought to be considered as a world full of an infinity of different creatures. »

G.W. Leibniz, Letter to Foucher, January 1692.

Leibniz on the Actual Infinite of Minute Perceptions

« […] at every moment there is in us an infinity of perceptions, unaccompanied by awareness or reflection; that is, of alterations in the soul itself, of which we are unaware because these impressions are either too minute and too numerous, or else too unvarying, so that they are not sufficiently distinctive on their own. But when they are combined with others they do nevertheless have their effect and make themselves felt, at least confusedly, within the whole. […] To give a clearer idea of these minute perceptions which we are unable to pick out from the crowd, I like to use the example of the roaring noise of the sea which impresses itself on us when we are standing on the shore. To hear this noise as we do, we must hear the parts which make up this whole, that is the noise of each wave, although each of these little noises makes itself known only when combined confusedly with all the others, and would not be noticed if the wave which made it were by itself. We must be affected slightly by the motion of this wave, and have some perception of each of these noises, however faint they may be; otherwise there would be no perception of a hundred thousand waves, since a hundred thousand nothings cannot make something. […] These minute perceptions […] are […] vivid in the aggregate but confused as to the parts; those impressions which are made on us by the bodies around us and which involve the infinite; that connection that each being has with all the rest of the universe. […] noticeable perceptions arise by degrees from ones which are too minute to be noticed. To think otherwise is to be ignorant of the immeasurable fineness of things, which always and everywhere involves an actual infinity. »

G.W. Leibniz, *New Essays on Human Understanding*, Preface, 53-7.

Deleuze on Leibniz's Theory of Minute Perceptions

« Et ce sont ces petites perceptions obscures, confuses, qui composent nos macroperceptions, nos aperceptions conscientes, claires et distinctes : jamais une perception consciente n'arriverait si elle n'intégrait un ensemble infini de petites perceptions […] Les petites perceptions […] sont […] des 'différentiels de la conscience'. […] Pour Maïmon comme pour Leibniz, la détermination réciproque des différentielles ne renvoie pas à un entendement divin, mais aux petites perceptions comme représentants du monde dans le moi fini […] L'infini actuel dans le moi fini […] Les petites perceptions constituent l'obscure poussière du monde inclus dans chaque monade, le sombre fond. Ce sont les rapports différentiels entre ces actuels infiniment petits qui *tirent au clair*, c'est-à-dire qui constituent une perception claire […] Toujours les rapports différentiels sélectionnent les petites perceptions qui entrent dans chaque cas, et produisent ou tirent la perception consciente qui en sort. Aussi le calcul différentiel est-il le mécanisme psychique de la perception […] *Je produis un accord* chaque fois que je peux établir dans un ensemble d'infiniment petits des rapports différentiels qui rendront possible une intégration de l'ensemble, c'est-à-dire une perception claire et distinguée. C'est un filtre, une sélection. »

G. Deleuze, *Le pli. Leibniz et le baroque*, pp.115, 118-9 & 178.

Gauss against actual infinity

« I protest against the use of an infinite quantity as an actual entity; this is never allowed in mathematics. The infinite is only a manner of speaking, in which one properly speaks of limits to which certain ratios can come as near as desired, while others are permitted to increase without bound. »

C.F. Gauss, Letter to Heinrich Schumacher, July 12, 1831.

The *horror infiniti* or *apeirophobia*

« If, however, out of a justified aversion against such an *illegitimate* actual infinite, a certain *Horror Infiniti,* which found its classic expression and support in the mentioned letter of Gauss, has been formed in broad layers of science, under the influence of the modern Epicurean-materialistic tendency of our time, so the therewith-connected uncritical rejection of the *legitimate* actual infinite seems to me to be no trifling offense against the nature of things, which one has to take as they are, and this behavior can be understood as a kind of *shortsightedness,* which deprives one of the possibility to see the actual infinite, although it in its Supreme, Absolute Bearer has created us and preserves us, and in its secondary, transfinite forms surrounds us everywhere [allüberall] and even dwells in our mind. »

G. Cantor, *On the Various Standpoints With Regard to the Actual Infinite*, letter to Mr. G. Eneström.

The Infinite and the Sublime in Kant (I)

In Kant (*Critique of Judgement*, 1790), the sublime is the « absolutely great »

- beyond comparison;
- beyond all what can be grasped by the senses (that is, it is the suprasensible);
- beyond numerical quantification;
- beyond the progress *ad infinitum* (that is, it is an « absolute totality »).

The Infinite and the Sublime in Kant (II)

« Sublime is the name given to what is absolutely great. […] to assert […] that something is great, is quite a different thing from saying that it is absolutely great (*absolute, non comparative magnum*). The latter is what is beyond all comparison great. […] If […] we call anything not merely great, but […] absolutely […] great, that is to say, sublime, we soon perceive that for this it is not permissible to seek an appropriate standard outside itself, but merely in itself. It is a greatness comparable to itself alone. Hence it comes that the sublime is not to be looked for in the things of nature, but only in our own ideas. […] Nothing […] which can be an object of the senses is to be termed sublime […]. But precisely because there is a striving in our imagination towards progress *ad infinitum*, while reason demands absolute totality […], is the awakening of a feeling of a supersensible faculty within us […] *The sublime is that, the mere capacity of thinking which evidences a faculty of mind transcending every standard of the senses.* »

I. Kant, *Critique of judgment*, 1790, §26.

The Infinite and the Sublime in Kant (III)

« But the infinite is absolutely (not merely comparatively) great. In comparison with this all else (in the way of magnitudes of the same order) is small. But the point of capital importance is that the mere ability even to think it as a whole indicates a faculty of mind transcending every standard of the senses. For the latter would entail a comprehension yielding as unit a standard bearing to the infinite a definite ratio expressible in numbers, which is impossible. Still the mere ability even to think the given infinite without contradiction, is something that requires the presence in the human mind of a faculty that is itself supersensible. »

I. Kant, *Critique of judgment*, 1790, §26.

Hilbert on Infinite

« Just as in the limit processes of the infinitesimal calculus, the infinite in the sense of the infinitely large and the infinitely small proved to be merely a figure of speech, so too we must realize that the infinite in the sense of an infinite totality […] is an illusion. […] the infinite is nowhere to be found in reality, no matter what experiences, observations, and knowledge are appealed to. […] It neither exists in nature nor provides a legitimate basis for rational thought — a remarkable harmony between being and thought. […] The role that remains for the infinite to play is solely that of an idea — if one means by an idea, in Kant's terminology, a concept of reason which transcends all experience and which completes the concrete as a totality. »

D. Hilbert, *Über das Unendliche*, 1925.

Towards the Actual Infinite

The Leap Into the Beyond

« The leap into the beyond occurs when the sequence of numbers that is never complete but remains open toward the infinite is made into a closed aggregate of objects existing in themselves. Giving the numbers the status of objects becomes dangerous only when this is done. The belief in the absolute is deeply implanted in our breast; no wonder, then, that mathematics was bold and naive enough to perform the leap. [...] the system of numbers has become a realm of absolute existence which is 'not of this world' and from which only gleams here and there are caught and reflected in our consciousness. »

H. Weyl, *Philosophy of Mathematics and Natural Science.*

Cantor on Aquinas

« The thoughtful arguments which one finds here and with the church fathers (Augustine should be excepted) *against* the standpoint represented by me were thoroughly justified, and it was therefore right and correct to oppose actually infinite numbers, so long as a principle of *individuation*, *specification* and *ordination* of that actual infinite which I call the *transfinite* had not been found. However, all these arguments crumble as soon as one set up such a principle and it had been demonstrated as *true*. This however one finds in my works. »

G. Cantor, Nachlass Vl , p. 110.

Cantor's determinate and complete transfinite numbers

« I have been logically compelled [...] to the point of view which considers the infinitely great not merely in the form of something growing without limits [unbegrentz Wachseden] [...] but also to fix it mathematically by numbers in the determinate form of the completed infinite [Vollendet-unendlichen] following the finite there is a *Transfinitum* [...] that is, there is an unbounded step-ladder of determinate modes which in their nature are not finite but infinite, but which just like the finite, can be determined by well-defined and distinguishable *numbers*. I am convinced that the domain of definable quantities is *not* exhausted by the finite quantities, and the bounds of our knowledge may accordingly be extended without violence to our nature. [...] All things, finite of infinite are *definite* and, God excepted, can be determined by the intellect. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §4.

How can we label with ordinal numbers

the set $\mathbb{N}$ when we endow it

with the following order:

$0,2,4,\ldots,1,3,5,\ldots$?

Cantor's answer

We shall introduce a new number ω

which is the ordinal number

corresponding to the position of the term 1

in the well-ordered set

0,2,4,...,1,3,5,...

« L’idée surgit alors (ou le désir) de précipiter le cours de cette suite, d’en porter les termes à leur limite, à celle de leurs expressions imaginables, après laquelle tout sera changé. »

Paul Valéry, *Introduction à la méthode de Léonard de Vinci*.

Cantor called ω a ***limit ordinal***

in the sense that it is the limit number

towards which the series 0,1,2,3,... tends.

The successor operator S allows us

to infinitely tend towards the infinite number ω

without ever reaching it,

which means that ω cannot be generated

by applying Cantor's first principle of generation.

ω is the length or order type of the well-ordered set

$0 < 1 < 2 < 3 < \ldots$

(i.e. the first ordinal that does not label an element of the set).

Like 0,

the ordinal ω is not a successor,

i.e. it cannot be reached from a previous number

by applying the successor operation S.

However, ω has a successor, namely $\omega + 1$, even if it is transfinite.

Transfinite ordinals might not have a predecessor (e.g. ω),

but they always have a successor.

Cantor's First Transfinite Ordinal

« The number [Anzahl] of the numbers n [1,2,3,...] of class (I) formed in this way is infinite and there is no greatest among them. However contradictory it might be to speak of a greatest number of class (I), there is nevertheless nothing offensive in thinking of a new number which we shall call ω, and which will be the expression for the idea [fact] that the entire assemblage [aggregate, Inbegriff] (I) is given in its natural, orderly succession [natural succession according to a law] (Similarly n is an expression for the fact that a certain finite number [Anzahl] of unities is united to form a whole [zu einem Ganzen vereinigt wird].) We may even imagine the newly created number ω as a limit which the numbers n approach, provided that we understand nothing more thereby than that ω is the first integer which follows all numbers n. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

Limit Ordinals According to Mallarmé

A limit ordinal is a new position (*un endroit*)

beyond (*au-délà*) all finite positions.

« Aussi loin qu'un endroit fusionne avec un au-delà »

S. Mallarmé.

Cantor's *Second Principle of Generation* of natural numbers

« The logical function which gave us the two numbers ω and 2ω is obviously different from the *first* principle of generation. I call it the *second principle of generation* of real whole numbers and define them more precisely: if any definite succession of defined whole real numbers exists, for which there is no largest, then a new number is created by means of this second principle of generation which is thought of as the *limit* of those numbers, i.e., it is defined as the next number larger than all of them. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

The notion of ***order type*** of an infinite well-ordered set

(i.e. the first ordinal number that does not label an element in the set)

is order-dependent.

For instance, the order type of

0,1,2,3,...

is ω

whereas the the order type of

0,2,4,...,1,3,5,...

is $\omega \cdot 2$.

In the realm of transfinite numbers,

the notions of *cardinal number*

and *ordinal number*

do not coincide.

For instance, 0,1,2,3,... and 0,2,4,...,1,3,5,...

have the same cardinal $\aleph_0$

(where a bijection is given by $n \mapsto n$)

but different ordinals (ω and $\omega \cdot 2$ respectively).

In the realm of transfinite numbers,

ordering a set *X*

— i.e. labeling the elements *X* with the ordinals *zeroth*, *first*, *second*,…

and assigning the corresponding length/order type to *X* (*Anzahl* in Cantor's terms) —

is not the same as ***counting*** the number of elements of *X* (« power » [*Mächtigkeit*] in Cantor's terms).

The "number of elements" of such a set

is given by the cardinality

— or « power » [*Mächtigkeit*] in Cantor's terms —

of the resulting ordinal.

« It is well known that in the Middle Ages, the '*infinitum actu non datur*' (the actual-infinite does not exist) was always represented by all scholastics as an irrefutable sentence taken from Aristotle. If, however, one considers the reasons which Aristotle gives against the real existence of the infinite (see for example his *Metaphysics*, Book XI, Chapter 10), then they essentially lead back to one presupposition, involving one *petitio principii* (begging the question), based on the assumption that there are only finite numbers, which he concluded from the fact that he was only aware of things in finite quantities. I believe, however, to have proved above, and this will be shown even more clearly in the remainder of this work, that definite countings can be made on infinite sets as well as on finite sets, provided that the sets follow a certain rule which is that they can be well-ordered sets. Without such a rule regarding the ordering of the elements of a set no counting can be done on the set — this is in the nature of the concept of counting. Even in the case of finite quantities, a counting can only be carried out for a certain sequence of the counted elements, but here it shows as a special nature of finite quantities that the result of the counting — the *ordinal-number* [*Anzahl*]— is independent of the respective arrangement; while in the case of infinite quantities, as we have seen, such independence in general does not apply, but the number of an infinite set is an infinite integer which is *also determined* by the rule of counting. This is where the essential difference between the finite and the infinite lies, which is based upon nature itself and can therefore never be eradicated, but the existence of the infinite will never henceforth be denied on account of this difference […] »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §4.9.

A part can have the same cardinal as the whole

« Let M be the totality (n) of all finite numbers n, M' the totality ($2n$) of all even numbers $2n$. Here it is definitely correct to say that according to its entities M is richer than M'; indeed, M contains in addition to the even numbers, which make up M, also all the uneven numbers M''. On the other hand, it is also definitely correct that both sets M and M' […] have the same cardinal number. Both (propositions) are certain and they do not conflict with each other if one carefully observes the distinction between reality and number. One should therefore say: the set M has more reality than M', because it contains as parts M' and M'' in addition; the cardinal numbers corresponding to them are however equal. When will these easy and enlightening truths be finally acknowledged by all thinkers?

G. Cantor, *Mitteilungen zur Lehre vom Transfiniten* (quoted in P. Mancosu, *Abstraction and Infinity*, p.132).

Hilbert's Hotel (I)

In a hotel with a countably infinite number of rooms,

the fact that

all rooms are occupied

does not imply that

there is no more space for new guests.

Hilbert's Hotel (II)

« Let us take the simplest example: the set of integers. Here the statement "The part is smaller than the whole" no longer holds. This important fact can easily be illustrated by means of the example of the occupied hotel.

We now assume that the hotel has infinitely many rooms numbered 0,1,2,3,4,5,… and that each of the rooms is occupied by a single guest. All that the manager has to do in order to accommodate a new guest is to make sure that each of the old guests moves to a new room with the number one unit larger. In this way room 0 becomes available for the new guest. [I.e. the application $n \mapsto n+1$ between the sets {0,1,2,3,...} and {1,2,3,...} is a bijection, which means that a part (the set {1,2,3,...}) has as many elements as the whole {0,1,2,3,...}] One can of course make room for any finite number of new guests in the same manner; and thus, in a world with an infinite number of houses and occupants there will be no homeless.

[…] It is even possible to get space for an infinite number of new guests […]. One could, for example, ask the old guest who originally occupied room number n to move to room number $2n$. In this way infinitely many rooms with odd numbers would be left free for new guests. [I.e. the application $n \mapsto 2n$ between the sets {0,1,2,3,...} and {2,4,6,...} is a bijection, which means that a part (the set of even numbers {2,4,6,...}) has as many elements as the whole set of natural numbers {0,1,2,3,...}]»

D. Hilbert, *David Hilbert's Lectures on the Foundations of Arithmetics and Logic 1917-1933*, ed. by W. Ewald and W. Sieg, 2013, p.730.

Hilbert's Hotel (III)

Hence, all the guests in the « bigger » set of rooms {0,1,2,3,...}

can be accommodated in the « smaller » set of rooms {1,2,3,...}

through the bijective map $n \mapsto n + 1$,

thereby liberating the room with the label 0.

(Re)learning to Count (I)

$$0,1,2,3,...$$

$$0,1,2,3,...,\omega$$

$$0,1,2,3,...,\omega,\omega+1,\omega+2,...,\omega+\omega=\omega\cdot 2$$

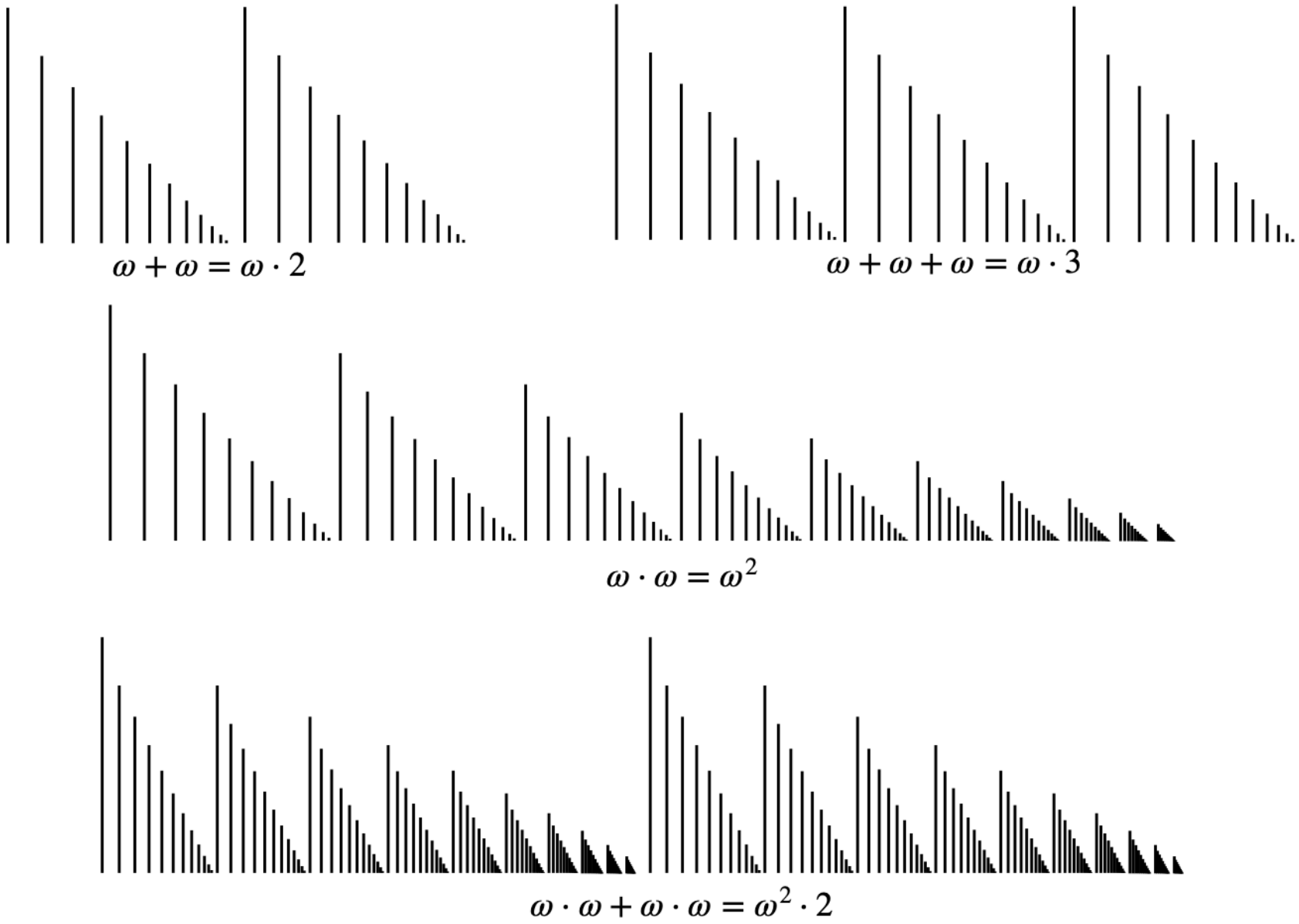
$$\omega\cdot 2+1,\omega\cdot 2+2,...,\omega\cdot 2+\omega=\omega\cdot 3$$

$$\omega,\ldots,\omega\cdot 2,\omega\cdot 3,...,\omega\cdot\omega=\omega^2$$

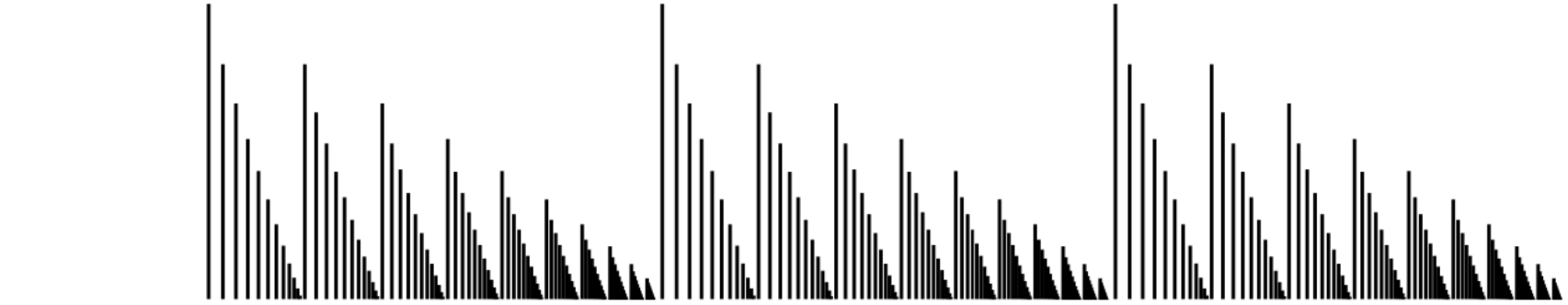
For instance, ω^2 is the order type of an encyclopedia

with countably infinite volumes,

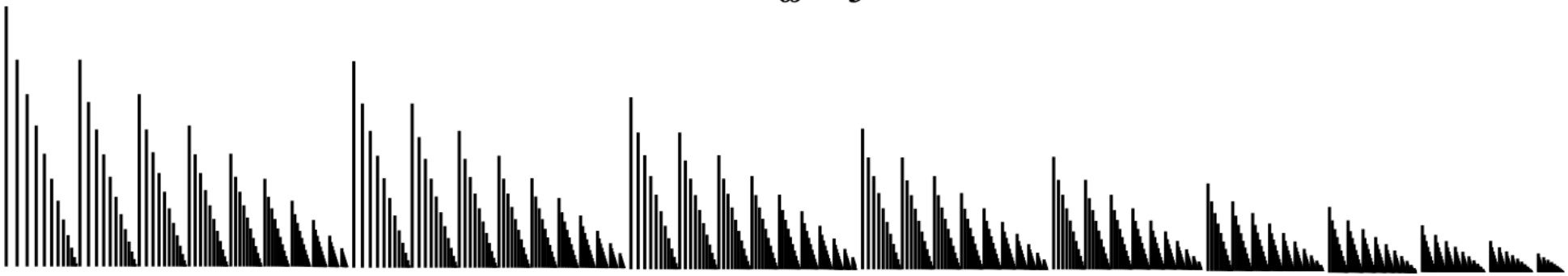
each of which has countably infinite pages.



Ref. K. Easwaran; A. Hájek; P. Mancosu; G. Oppy, Infinity, *The Stanford Encyclopedia of Philosophy (Spring 2022 Edition)*, Edward N. Zalta (ed.).



$\omega^2 \cdot 3$



$\omega^2 \cdot \omega = \omega^3$

Ref. K. Easwaran; A. Hájek; P. Mancosu; G. Oppy, Infinity, *The Stanford Encyclopedia of Philosophy (Spring 2022 Edition)*, Edward N. Zalta (ed.).

ω^3 is the order type

of a countably infinite number of encyclopedias,

each with countably infinite volumes,

each with countably infinite pages.

(Re)learning to Count (II)

Starting to count from ω^2, we obtain the following ordinals:

$$\omega^2 + 1,...,\omega^2 + \omega, \omega^2 + \omega + 1,...,\omega^2 + \omega \cdot 2,...,\omega^2 \cdot 2,...,\omega^3, \ldots, \omega^4, \ldots \omega^\omega$$

$$\omega^\omega + 1,...,\omega^\omega + \omega^\omega = 2\omega^\omega, \ldots, \omega\omega^\omega = \omega^{\omega+1}, \ldots, \omega^{2\omega}, \ldots, \omega^{\omega^2}, \ldots \omega^{\omega^\omega}$$

$$\epsilon_0 \equiv \omega^{\omega^{\omega^{\cdots}}}$$

(Re)learning to Count (III)

Starting to count from

$$\epsilon_0 \equiv \omega^{\omega^{\omega^{\cdots}}},$$

we obtain the ordinals:

$$\epsilon_1 \equiv (\epsilon_0)^{(\epsilon_0)^{(\epsilon_0)^{\cdots}}},$$

$$\epsilon_2 \equiv (\epsilon_1)^{(\epsilon_1)^{(\epsilon_1)^{\cdots}}}, \ldots, \epsilon_\omega, \epsilon_{\omega+1}, \ldots, \epsilon_{\epsilon_0}, \ldots, \epsilon_{\epsilon_1}, \ldots, \epsilon_{\epsilon_\omega}, \ldots, \epsilon_{\epsilon_{\epsilon_0}}, \ldots, \epsilon_{\epsilon_{\epsilon_1}}, \ldots, \epsilon_{\epsilon_{\epsilon_\omega}}, \ldots \epsilon_{\epsilon_{\epsilon_{\epsilon_0}}}, \ldots,$$

$$\eta_0 = \epsilon_{\epsilon_{\epsilon_{\epsilon_{\cdots}}}}$$

Cantor on the Finite, the Infinite and the Absolute (I)

« As different as the theories of these writers [Locke, Descartes, Spinoza, Leibniz] are, in their judgment of the finite and the infinite they essentially agree that the concept of a number includes the finiteness of it, and that on the other hand the true infinite or absolute, which resides in God, is not permitted any determination. As far as the latter point is concerned, I fully agree with it, as it cannot be otherwise, because the sentence: *"omnis determinatio est negatio"* (every determination is negative) is, for me, completely beyond question. On the other hand […] I see in the former a *petitio principii* (begging the question), which renders explainable many contradictions which can be found in all these authors and especially in Spinoza and Leibniz. The assumption is that, apart from the finite, and the absolute infinite that cannot be reached by any determination, there can be no new entities which are infinite but at the same time are determinate numbers, the ones that I refer to as the actual-infinite. I do not find that this assumption is justified by anything and, in my opinion, it even stands in contradiction to certain assertions put forward by the two aforementioned philosophers. What I maintain and believe to have proved […] is that after the finite there is a *transfinite* (which might also be called a *suprafinite*)—that is, there is an unlimited scale of definitive steps, which in their nature are not finite but rather infinite, but which, like finite entities, can be determined by definite, well-defined and mutually distinguishable numbers. My conviction is therefore that the finite magnitudes do not constrict the realm of definable magnitudes and that the limits of our knowledge can be extended accordingly without our nature having to apply any limitation. Instead of the Aristotelian school sentence discussed above in § 4, I therefore put another:

'*Omnia seu finita seu infinita definita sunt et excepto Deo ab intellectu determinari possunt.*'

(All things finite or infinite are *definite* and, except for God, can be determined by the intellect). »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §5.3.

Cantor on the Finite, the Infinite and the Absolute (II)

« [The Absolute is the veritable infinity whose magnitude is such that it] cannot in any way be added to or diminished, and it is therefore to be looked upon quantitatively as an absolute maximum. In a certain sense it transcends the human power of comprehension, and in particular is beyond mathematical determination. »

G. Cantor, quoted in M. Hallett, *Cantorian set theory and limitation of size*, p.13.

Cantor on the Finite, the Infinite and the Absolute (III)

« The finiteness of human understanding is very often cited as the reason why only finite numbers can be the subject of thought, but I see in this assertion again the aforementioned circular argument. […] The words 'finiteness of understanding', which one hears so often, do not, I believe, in any way apply; as limited as human nature is, in truth it is much attached to the infinite and I believe that even if it were not itself infinite in many aspects, we all have a firm confidence and certainty regarding the being of the absolute, and in which we all agree is known to be unexplainable. And in particular I take the view that the human mind has an unlimited capacity for the stepwise formation of sets of integers which are related to the infinite steps and whose *cardinal-numbers* are of increasing magnitude. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §5.4.

Cantor on the Finite, the Infinite and the Absolute (IV)

« There is no doubt for me that we will continue on this path, never reaching an insurmountable boundary, but also not reaching any sort of comprehension of the absolute. The absolute can only be recognized, but never known, not even approximately known. […] As Albrecht von Haller says of eternity: 'I attain to the enormous number, but you, O eternity, lies always ahead of me.' »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, Endnotes 2.

Cantor on the Essence of Mathematics as Freedom

« [mathematics] *only* takes into account the *fundamental* reality of its concepts in the formation of its ideas and is therefore *not* duty-bound to also examine them according to their *ephemeral* reality. Because of this exceptional standpoint, which distinguishes it from all other sciences […], it particularly deserves the name of *free mathematics* […].

Mathematics is completely free in its development and is only bound to the self-evident consideration that its concepts are both free of contradictions and that they are in fixed relationships to proven concepts that have already been previously established. In particular, […] [o]nce a number satisfies all these conditions, it can and must be considered as existing and real in mathematics. Regarding this one can now see, […] the reason why one has to consider that the rational, irrational and complex numbers exist, in exactly the same way as we consider the finite positive integers to exist.

[…] the *essence of mathematics* lies precisely in its *freedom*. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §8.3-5.

Summary of Cantor's Transfinite Revolution

Pre-Cantorian Conception of the Infinite

- There is no actual infinite, only potential infinite, i.e. the infinite of an endless progression (Aristotle).

- The infinite is « beyond measure » or determination, i.e. it cannot be measured or determined by a number (Aquinas, Galileo, Leibniz).

- It is unique, *the* infinite.

« […] what is infinite is not comparable as a greater or smaller; there cannot therefore be a relation of infinite to infinite according to orders or dignities of the infinite […] »

G.W.F. Hegel, *The Science of Logic*, p. 216.

Cantorian Conception of the Infinite

- There exists many actual infinities associated with different well-determined transfinite numbers.

- In the realm of infinite numbers, certain properties of finite numbers do not hold:

  .Coincidence between cardinals and ordinals.

  .« The whole is greater than the part » (Euclid's Common Notion 5).

- The classical dichotomy between *finite understanding* and the *divine infinite* becomes complexified into a triptych: the *Finite*, the *Transfinite*, and the *Absolute,* where the first one and the second one admit numerical determination and are fully accessible to human understanding.

Finite extensions of $\mathbb{N}$

Up to know, we have enriched $\mathbb{N}$

by adding transfinite numbers bigger than any n.

We shall now explore extensions of $\mathbb{N}$

obtained by adding

finite numbers.

Why should we

introduce new numbers?

The « Speculative » or « Dialectical » Dynamics of Reason

Given a number system A,

there are « problems » formulated

by using the numbers in A

that do not have « solutions » in A.

These « problems » are like symptoms

that inscribe within A its potential extensions.

What are the solutions of the $\mathbb{N}$-equation

$$x + 1 = 0?$$

An equation formulated using numbers in $\mathbb{N}$

does not have a solution in $\mathbb{N}$

or, equivalently,

the result of an operation

between numbers in $\mathbb{N}$ is not in $\mathbb{N}$.

« What we cannot speak about

we must pass over in silence. »

L. Wittgenstein, *Tractatus Logico-Philosophicus*, §7.

Historical and Dialectical Mathematics

What we cannot speak about

we must expand our « transcendental »

(linguistic, numerical) resources…

… in order to avoid falling prey

under the « precritical syndrom »,

which consists in mistaking a limitation of our « transcendental resources »

for an insight into necessity

(D.C. Dennett, *Consciousness Explained*).

Lacan against Wittgenstein

« Lacan's formula, according to which whatever is excluded from the symbolic reappears in the real, can be interpreted here as follows: in certain conditions, the excluded proper to an *already produced* mathematical structure reappears as the instigating mark of a real (historical) process of *production* of a different structure. »

A. Badiou, *Infinitesimal Subversion*.

The Maximalist Ethos of the Mathematician

We shall always introduce

all the « ideal elements » we need

in order

- to guarantee the closure of a given operational domain;
- to find solutions to a given equation (history of number systems);
- to avoid exceptions (points at infinity in projective geometry).

The new « ideal elements » were successively described as

fictitious *(« fictions of the mind »)*,

ideal (i.e. non real),

irrational,

imaginary,

ghostly elements

introduced to fulfill a desire for completeness, elegance or symmetry.

Invariable, these pure « inventions » of the mind

firstly introduced as mere « symbols » obeying certain rules

gained « reality » through systematic use.

« Casi inmediatamente, la realidad cedió en más de un punto. Lo cierto es que anhelaba ceder. »

J.L. Borges, *Tlön, Uqbar, Orbis Tertius*.

« The expression ‘ideal elements’ is only justified from the point of view of the original system. In the new system, we no longer distinguish between real and ideal elements. »

D. Hilbert, *Die Rolle von idealen Gebilden*.

« […] the sharp distinction between the intuitive [Anschaulichen] and the non-intuitive [Nicht-Anschaulichen] […] can apparently not be drawn so strictly […]’. »

P. Bernays, *Abhandlungen zur Philosophie der Mathematik,* 1976.

Cavaillès on the Mathematical Expansion of Intuition

« The conditions imposed by the strict observance finitists on mathematical existence show a misunderstanding of what makes the very reality of the object. There is no absolute at the outset: what can be said of the point, the continuum, the integer, except that they are elements and origins of sequences in which they take on their meaning and which exceed them? An irreducible intuition is only a thoughtless halt. The Kantian theory of mathematics still obsesses logicians and mathematicians, but because of its superficiality and its illegitimate interpretation in psychological terms. Taken in its authenticity, it remains inadequate in so far as it confuses the dialectical moment of the position of the concept and the transcendental moment of its schematisation. [...] At each stage of mathematical progress, new concepts are denied their right to exist, because they cannot provide a representation in the pre-existing intuitive system. This was the case for the negative numbers, for the imaginaries, for infinitesimals, and for the infinite at the time of Kant. The solution each time is not, as is superficially believed, the discovery of a translation in an immutable spatial intuition (representation of negative numbers on a straight line, of imaginary numbers in the plane) [...]. The solution consists in a transformation of the intuitive zone, i.e. of the rules which lay down the use of […] the schematic system. [...] In other words, intuition in its quiddity progresses in parallel with the dialectical sequence of concepts. [...] [T]here is no absolute transcendental […]. »

J. Cavaillès, 'Transfini et continu', in *Œuvres Completès de Philosophie des Sciences,* Paris: Hermann, 1994.

From « blind or symbolic » knowledge to « intuitive » knowledge

A syntax can be enriched

by adding new symbols (e.g. ω, i)

whose « intuitive » meaning

— far from relying on an *a priori* intuition —

are progressively constituted *a posteriori*

by a « blind » or purely « symbolic » manipulations

according to the rules they obey.

« […] les mathématiciens […] écrivent avec plus de généralité qu'ils ne voient. Et essayent ensuite de voir. »

P. Valery, *Cahiers II*, Bibliothèque de la Pléiade, p.794.

« Le continu, les infinis sont des abus de langage et d'écriture, des développements fiduciaires sans garanties. »

P. Valery, *Cahiers II*, Bibliothèque de la Pléiade, p.829.

***Blind* or *symbolic knowledge* vs *intuitive knowledge* in Leibniz**

« […] we don't usually grasp the entire nature of a thing all at once […] but in place of the things themselves we make use of signs […] I usually call such thinking, which is found both in algebra and in arithmetic and, indeed, almost everywhere, *blind* or *symbolic*. And indeed, when a notion is very complex, we cannot consider all of its component notions at the same time. When we can […] I call knowledge *intuitive*. »

G.W. Leibniz, *Meditations on Knowledge, Truths, and Ideas*, in *Philosophical Essays*, pp.24-5.

By adding solutions to the $\mathbb{N}$-equations

$$x + n = 0$$

for all $n \in \mathbb{N}$

(which amounts to introduce an additive inverse

for each $n \in \mathbb{N}$, denoted $-n$)

we define the entire numbers

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}.$$

Algebraic Structure of $\mathbb{Z}$

$(\mathbb{Z}, +, \times, 0, 1)$ is a ***ring***.

$(\mathbb{Z}, +, 0)$ is a group (rather than a monoid) since the addition $+$ has now an inverse.

$(\mathbb{Z}, \times, 1)$ is a monoid, since the multiplication $\times$ does not have an inverse.

Whereas the idea of introducing a number

associated with nothing (the *zero*)

was already a non-trivial conceptual leap,

the idea of introducing ***negative numbers***

representing quantities that are « ***less than nothing*** »

met with strong resistance

well into the modern period.

Conceptual History of Negative Numbers (I)

- Diophantus (200 - c.284 AC, *Arithmetica*) rejected negative solutions.

- Fibonnacci (XII century): interpretation of negative quantities in terms of debts.

- Cardano, *Ars Magna,* 1545 (XVI century). Even if he made a systematic use of negative numbers, he considered them as fictions (*numeri ficti*) and he wrote:

« Subtraction is made only of the smaller from the bigger. In fact, it is entirely impossible to subtract a bigger number from a smaller one. »

Conceptual History of Negative Numbers (II)

- In his *Géometrie* (1637), Descartes called negative solutions of equations "false".

- « […] too much truth is paralyzing (I know some who cannot understand that to take four from nothing leaves nothing). »

B. Pascal, *Pensées*, Section II, fr. 72, 1670.

- Coordinates on a line in one of the two possible directions with respect to an arbitrary zero, like dates before Christ (John Wallis, 1616 - 1703):

« But it is also Impossible, that any Quantity […] can be Negative. Since that is not possible that any Magnitude can be Less than Nothing, or any Number Fewer than None. Yet is not that Supposition (of Negative Quantities) either Unuseful or Absurd; when rightly understood. […] when it comes to a Physical Application, it denotes as Real a Quantity as if the Sign were +; but to be interpreted in a contrary sense. »

J. Wallis, *Algebra*, 1673, in D.E. Smith, A Source Book in Mathematics, Dover, 1959.

Cf. D. Rabouin, *Negatives as fictions in 16th and 17th century mathematics*, Hist. Math. 69 (2024) 41-61.

Enlightening the Negative Numbers

« NÉGATIF, adj. (*Algeb.*) quantités *négatives, en Algèbre*, sont celles qui sont affectées du signe -, & qui sont regardées par plusieurs mathématiciens, comme plus petites que zéro. Cette dernière idée n'est cependant pas juste, comme on le verra dans un moment. [...] Il faut avouer qu'il n'est pas facile de fixer l'idée des quantités négatives, & que quelques habiles gens ont même contribué à l'embrouiller par les notions peu exactes qu'ils en ont données. Dire que la quantité *négative* est au-dessous du rien, c'est avancer une chose qui ne se peut pas concevoir. [...] Quand on considère l'exactitude & la simplicité des opérations algébriques sur les quantités *négatives*, on est bien tenté de croire que l'idée précise que l'on doit attacher aux quantités négatives doit être une idée simple, & n'être point déduite d'une métaphysique alambiquée. Pour tâcher d'en découvrir la vraie notion, on doit d'abord remarquer que les quantités qu'on appelle *négatives*, & qu'on regarde faussement comme au-dessous du zéro, sont très souvent représentées par des quantités réelles, comme dans la Géométrie, où les lignes *négatives* ne different des positives que par leur situation à l'égard de quelque ligne au point commun. »

D'Alembert, *Encyclopedie*, 1er ed. (1765), article *Négatif*, Tome 11, pp. 72-4.

Conceptual History of Negative Numbers (III)

• I. Kant, *Attempt to Introduce the Concept of Negative Magnitudes into Philosophy* (1763):

Interpretation of negative numbers

in terms of « real oppositions »

— e.g. forces pointing in opposite directions —

that produce real effects

— e.g. a state of uniform motion —

in contrast with « logical oppositions »

(where the affirmation of both A and $\neg A$ is just a contradiction).

Kant's remark points to a new use of numbers beyond

counting

ordering

measuring (e.g. lengths)

labeling or **coordinating** (e.g. coordinates on the real line),

namely,

changing the orientation (e.g. of a force vector)

by multiplying by negative number.

This last use of numbers leads to the use of numbers

as **scalars in a vector space**.

The cardinality of $\mathbb{Z}$ is $\aleph_0$

($\mathbb{Z}$ is countable)

Indeed we can establish a bijection between $\mathbb{N}$

and $\mathbb{Z}$ ordered as

$0,1,-1,2,-2,3,-3,...$

Hence, the intuition that there are twice

as many whole numbers $0,1,2,3,...$

as entire numbers $...-2,-1,0,1,2,...$

is misleading

(once again, a strict part has the same « size » as the whole).

Rational Numbers

The $\mathbb{Z}$-equation

$$2x - 1 = 0$$

does not have a solution in $\mathbb{Z}$

since the multiplication $\times$ does not have an inverse.

By adding solutions to any $\mathbb{Z}$-equation of the form

$$nx - 1 = 0$$

— which amounts to introduce multiplicative inverses

for each $n \in \mathbb{Z}$, denoted n^{-1}—

we define the ***rational numbers***.

Representation of Rational Numbers as Fractions

Formally, a ***rational number*** $q \in \mathbb{Q}$

is an equivalence class of pairs

(n, m) with $n, m \in \mathbb{Z}$

— usually denoted $\frac{n}{m}$ —

where the equivalence relation $\sim$ is given by

$(n, m) \sim (n', m')$ if $nm' = n'm$.

Example: $\frac{1}{2}$ and $\frac{3}{6}$ are *fractions* that represent the same rational number

since $1 \times 6 = 3 \times 2$.

Positional Representation of Rational Numbers With Respect to a Given Base

For instance, the rational number represented by the fraction $\frac{2536}{100}$

admits a ***decimal positional representation*** by writing it as

$$2 \times \mathbf{10^1} + 5 \times \mathbf{10^0} + 3 \times \mathbf{10^{-1}} + 6 \times \mathbf{10^{-2}}$$

We could say that the following sequence of numbers between 0 and 9

$$(\ldots,0,2,5.3,6,0,\ldots)$$

— usually written in decimal notation 25,36 —

are « coordinates » of $\frac{2536}{100}$ in the « base »

$$(\ldots,10^2,10^1,10^0.10^{-1},10^{-2},10^{-3},\ldots).$$

The decimal representation

of a rational number

is not intrinsic

since it depends on the chosen base-10.

If we change the base to (for instance) a base-2

$$(\ldots,2^3,2^2,2^1,2^0.2^{-1},2^{-2},2^{-3},\ldots)$$

or

$$(\ldots 8,4,2,1.\frac{1}{2},\frac{1}{4},\frac{1}{8},\ldots)$$

the coordinates of the corresponding rational number $q \in \mathbb{Q}$

will change accordingly to a sequence of 0's and 1's.

For instance, the rational number $\frac{925}{100}$ can be rewritten as

$$1 \times \mathbf{2^3} + 0 \times \mathbf{2^2} + 0 \times \mathbf{2^1} + 1 \times \mathbf{2^0} + 0 \times \mathbf{2^{-1}} + 1 \times \mathbf{2^{-2}}$$

or

$$(\ldots,0,0,1,0,0,1.0,1,0,0,...)$$

or

$$1001.01$$

Algebraic Structure of $\mathbb{Q}$

$(\mathbb{Q}, +, \times, 0, 1)$ is a ***field***

since both $+$ and $\times$

have now inverses

with respect to the neutral elements 0 and 1 respectively.

How many rational numbers are?

Or, what is the cardinality of $\mathbb{Q}$?

Cantor's Snake

In spite of its density,

the cardinality of $\mathbb{Q}$ is still $\aleph_0$

i.e. $\mathbb{Q}$ is countable:

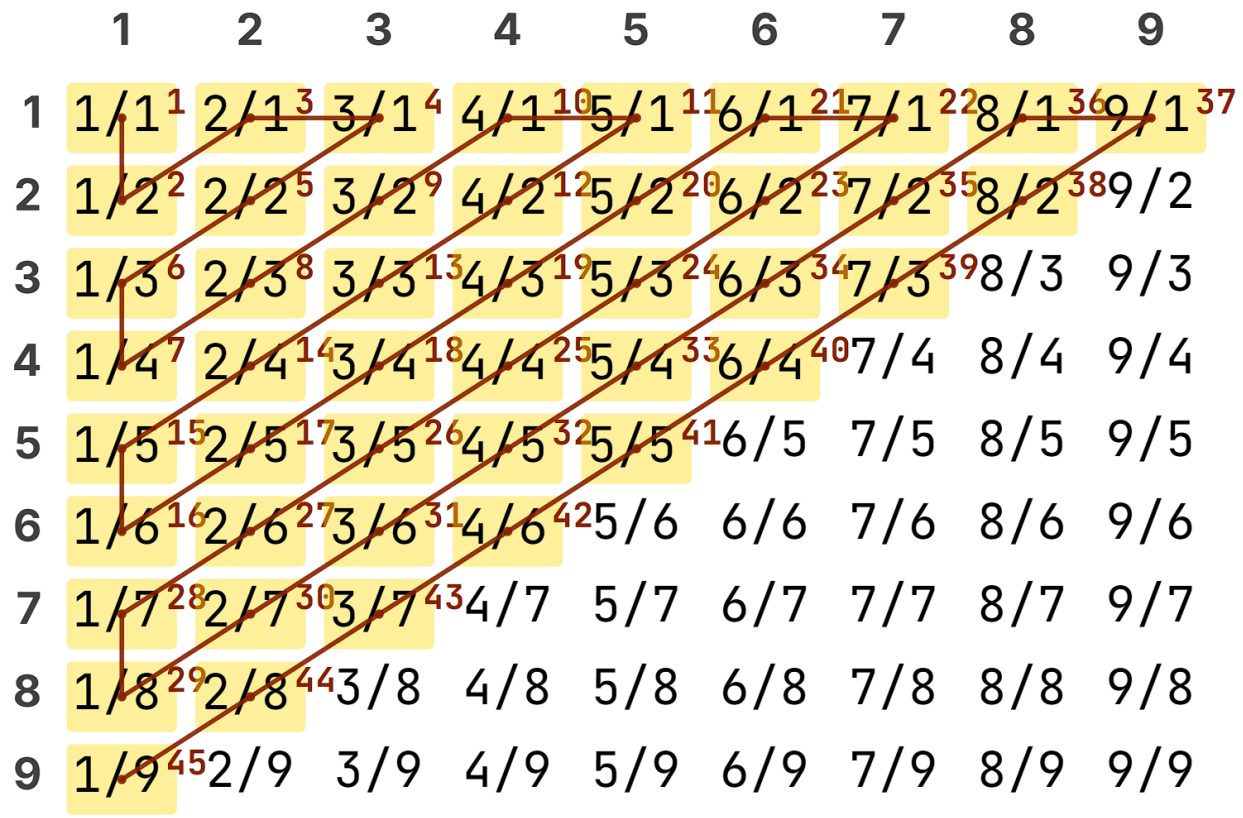
there are as many fractions

$$\pm\frac{n}{m}$$

as natural numbers n.

	1	2	3	4	5	6	7	8	9
1	1/1	2/1	3/1	4/1	5/1	6/1	7/1	8/1	9/1
2	1/2	2/2	3/2	4/2	5/2	6/2	7/2	8/2	9/2
3	1/3	2/3	3/3	4/3	5/3	6/3	7/3	8/3	9/3
4	1/4	2/4	3/4	4/4	5/4	6/4	7/4	8/4	9/4
5	1/5	2/5	3/5	4/5	5/5	6/5	7/5	8/5	9/5
6	1/6	2/6	3/6	4/6	5/6	6/6	7/6	8/6	9/6
7	1/7	2/7	3/7	4/7	5/7	6/7	7/7	8/7	9/7
8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8	9/8
9	1/9	2/9	3/9	4/9	5/9	6/9	7/9	8/9	9/9


1 2 3 4 5 6 7 8 9
1 1/1 1 2/1 3 3/1 4 4/1 10 5/1 6/1 7/1 8/1 9/1
2 1/2 2 2/2 5 3/2 9 4/2 5/2 6/2 7/2 8/2 9/2
3 1/3 6 2/3 8 3/3 4/3 5/3 6/3 7/3 8/3 9/3
4 1/4 7 2/4 3/4 4/4 5/4 6/4 7/4 8/4 9/4
5 1/5 2/5 3/5 4/5 5/5 6/5 7/5 8/5 9/5
6 1/6 2/6 3/6 4/6 5/6 6/6 7/6 8/6 9/6
7 1/7 2/7 3/7 4/7 5/7 6/7 7/7 8/7 9/7
8 1/8 2/8 3/8 4/8 5/8 6/8 7/8 8/8 9/8
9 1/9 2/9 3/9 4/9 5/9 6/9 7/9 8/9 9/9


1 2 3 4 5 6 7 8 9
1 1/1 1 2/1 3 3/1 4 4/1 10 5/1 11 6/1 21 7/1 22 8/1 36 9/1 37
2 1/2 2 2/2 5 3/2 9 4/2 12 5/2 20 6/2 23 7/2 35 8/2 38 9/2
3 1/3 6 2/3 8 3/3 13 4/3 19 5/3 24 6/3 34 7/3 39 8/3 9/3
4 1/4 7 2/4 14 3/4 18 4/4 25 5/4 33 6/4 40 7/4 8/4 9/4
5 1/5 15 2/5 17 3/5 26 4/5 32 5/5 41 6/5 7/5 8/5 9/5
6 1/6 16 2/6 27 3/6 31 4/6 42 5/6 6/6 7/6 8/6 9/6
7 1/7 28 2/7 30 3/7 43 4/7 5/7 6/7 7/7 8/7 9/7
8 1/8 29 2/8 44 3/8 4/8 5/8 6/8 7/8 8/8 9/8
9 1/9 45 2/9 3/9 4/9 5/9 6/9 7/9 8/9 9/9

Since n can be as large as we wish, its inverse

$$\frac{1}{n}$$

can be as small as we wish.

Thus, we have introduced numbers

that can be *potentially infinitely small*

just as n can be *potentially infinitely large*.

However, because there is no largest n,

there can be no smallest $\frac{1}{n}$.

$\mathbb{Q}$ is dense

Between any two rational numbers,

there is an infinite number

of other rational numbers.

Demonstration.

Given two rational numbers $q < p \in \mathbb{Q}$,

it is always possible to choose n big enough

$$\text{so that } \frac{1}{n} < p - q.$$

$$\text{Hence, } q < q + \frac{1}{n} < p.$$

The ***continuous***

is that which is not ***discrete***,

like a line traced without ever lifting the pen from the paper,

a set in which no gap separates any two elements,

a gapless extension.

« […] ese laberinto, que consta de una sola línea recta y que es indivisible, incesante. »

J.L. Borges, *La muerte y la brújula*.

By adding numbers between any two numbers

and iterating this operation,

it seems that we can fill the line.

Can mere *density*

endow a set of points with

the nature of a ***continuum***

that is, of a ***gapless extension***?

« When points are situated in such a way that there are no two between which there is not an intermediate point, then, by that fact, there is continuous extension. »

G.W. Leibniz, Letter to Des Bosses, May 29, 1716.

« If we try to form a clear idea of what we call a '*continuous extension*' or '*continuum*', we are forced to declare that a continuum is present when, and only when, we have an aggregate of simple entities (instants or points or substances) so arranged that each individual member of the aggregate has, at each individual and sufficiently small distance from itself, at least one other member of the aggregate for a neighbour. »

B. Bolzano, *Paradoxes of the Infinite*, 1851, p.129.

However… Rational Density is Not Enough to Guarantee Continuity

« Of the greatest importance, however, is the fact that in the straight line L there are infinitely many points which correspond to no rational number. […] the ancient Greeks already knew and had demonstrated that there are lengths incommensurable with a given unit of length, e. g., the diagonal of the square whose side is the unit of length. If we layoff such a length from the point o upon the line we obtain an end-point which corresponds to no rational number. […] The above comparison of the domain $\mathbb{Q}$ of rational numbers with a straight line has led to the recognition of the existence of gaps, of a certain incompleteness or discontinuity of the former, while we ascribe to the straight line completeness, absence of gaps, or continuity. In what then does this continuity consist? »

R. Dedekind, *Continuity and Irrational Numbers*, 1872, pp.8-10.

Irrational numbers

We can rephrase the Greeks' discovery of irrational numbers

by saying that there are $\mathbb{Q}$-polynomial equations

— e.g. $x^2 - 2 = 0$ —

whose solutions are not in $\mathbb{Q}$,

i.e. cannot be written as $\frac{n}{m}$

for integers $n, m \in \mathbb{Z}$.

There are *algebraic numbers*

— numbers that are solutions of $\mathbb{Q}$-equations —

that are not rational.

Continuous Fractions

Even if « irrational » numbers cannot be expressed as fractions, they can be nonetheless be expressed in a very « rational » form as « continuous fractions », like for instance:

$$\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \dots}}}}$$

$$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \dots}}}}} \quad \text{(golden ratio)}$$

Hence, ***density***

— the fact that between two rational numbers

there are infinite rational numbers —

does not guarantee ***completeness*** (or *gaplessness*).

$\mathbb{Q}$ is full of (algebraic) gaps.

Transcendental Gaps in $\mathbb{Q}$

Besides $\mathbb{Q}$-polynomial equations with no $\mathbb{Q}$-solutions (algebraic gaps)

there other kinds of « symptoms »

of the limitations of $\mathbb{Q}$,

like sequences $a_0, a_1, a_2, \ldots$ of rational numbers

that verify ***Cauchy's condition***

— that for any $\epsilon > 0$, there is a positive integer N

such that for all $m, n > N$ we have

$$|a_m - a_n| < \epsilon,$$

which means that the difference

between terms tends to 0 as n increases —

but that do not converge to any algebraic number.

Gaps in $\mathbb{Q}$ that are not algebraic are called *transcendental*.

Zeno's First Paradox: The Dichotomy

« The first [paradox] asserts the non-existence of motion on the ground that that which is in locomotion must arrive at the half-way stage before it arrives at the goal. »

Aristotle, *Physics*, 239b11-13.

« [...] if there is motion, it is necessary that a moving thing will entirely traverse an infinite number of things in a finite time; but this is impossible; consequently, there is no motion. »

Simplicius, *Commentary on Aristotle's* Physics, 1013,3-14.

Zeno's Arrow

The $\mathbb{Q}$-sequence

$$a_n = \sum_{k=1}^{n} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \cdots$$

where $a_1 = 0.5, a_2 = 0.75, a_3 = 0.875, a_3 = 0.9375, \ldots$

converges to the rational number 1.

Generalized Zeno's Arrows with Ghostly Targets

There are convergent sequences a_n of rational numbers

— which we might call ***generalized Zeno's arrows*** —

that do not converge to any algebraic number.

These generalized Zeno's arrows strike $\mathbb{Q}$-nonexistent targets.

Examples of Generalized Zeno's Arrows striking Ghostly Targets

The $\mathbb{Q}$-sequence

$$a_n = \sum_{k=0}^{n} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots$$

tends to a « transcendental » (i.e. non-algebraic) number denoted e.

The $\mathbb{Q}$-sequence

$$a_n = 4 \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} = 4(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots)$$

tends to a « transcendental » (i.e. non-algebraic) number denoted π.

Construction of the Real Numbers $\mathbb{R}$ from Cauchy Sequences

Every sequence a_n of rational numbers that verifies ***Cauchy's condition***

— that for any $\epsilon > 0$, there is a positive integer N

such that for all $m, n > N$ we have

$$| a_m - a_n | < \epsilon$$ —

defines a ***real number***.

Every generalized Zeno's arrow becomes the index of a « real » target.

The decimal expansion of any real number x

defines a Cauchy sequence (x_n)

whose n-th term x_n is the truncation to n

of the decimal expansion of x.

Example: π is given by the Cauchy sequence 3,3.1,3.14,3.141,3.1415,...

Dedekind's argument against the completeness of $\mathbb{Q}$.

There are partitions of the set $\mathbb{Q}$ of rational numbers

into two nonempty classes

L (the *left set* of the cut)

— without a largest element —

and R (the *right set* of the cut)

— without a smallest element —

such that every member L is smaller than every member R

$l < r$ for every $l \in L$ and $r \in R$

and yet no rational number lies between the two classes.

Such a « Dedekind cut » is a manifestation of a gap in $\mathbb{Q}$, i.e. of the fact that $\mathbb{Q}$ is not a gapless continuum.

Such a « Dedekind cut » associated to a gap in $\mathbb{Q}$ defines an irrational number.

$\sqrt{2}$ as a Dedekind cut

For instance, $\mathbb{Q}$ can be « cut » in the two sets

$$L = \{q / q < 0 \text{ or } q^2 < 2\}$$

and

$$R = \{q / q > 0 \text{ and } q^2 > 2\},$$

but there is no rational number that is

larger than every member of L

and smaller than every member of R.

Such a « cut » defines a gap in $\mathbb{Q}$.

Construction of the Real Numbers $\mathbb{R}$ from Dedekind's Cuts

The set of real numbers $\mathbb{R}$

can be defined as the set of ***Dedekind cuts*** of $\mathbb{Q}$,

where a Dedekind cut of $\mathbb{Q}$

is a non-empty proper subset L of $\mathbb{Q}$ such that

- **Downwards closeness**: for all $q \in L$, if $p \in \mathbb{Q}$ and $p < q$, then $p \in L$.
- **No maximal element**: if $q \in L$, there exists $q' \in L$ such that $q < q'$.

Every rational number q is defined by the Dedekind cut

$$L_q = \{p \in \mathbb{Q} / p < q\}$$

Real Numbers

are either

algebraic

—numbers that are solutions to polynomial equations —

like $\sqrt{2}$ and the golden ratio φ, etc.;

or ***transcendental***

— real numbers that are not solutions to polynomial equations —

like π or e.

In turn, there are real numbers

that are *computable* (e.g. $\sqrt{2}, \pi, e$)

— i.e. numbers that can be approximated

to any desired precision

by a finite, effective algorithm —

and others that are *non-computable*.

Having departed from the shores

of the « natural » and the « rational »

to embrace the « fictitious » and the « irrational »,

we now find ourselves in the realm of the « real »

that is, the gapless,

that which unfolds seamlessly,

without ever needing to leap over a discontinuity.

The Continuum Hypothesis

How many points are there

on the real line $\mathbb{R}$?

Moreover, Cantor proved in 1874

that the algebraic numbers

— including irrationals like $\sqrt{2}$ —

are also countable.

Idea of the demonstration:

- There are $\mathbb{Z}^{n+1}$polynomials with coefficients in $\mathbb{Z}$ of degree $n \geqslant 1$, each of which has at most n solutions.

- This yields a countable number $n\mathbb{Z}^{n+1}$ of algebraic numbers arising as solutions of polynomial equations of degree n (since a Cartesian product of countable sets is countable).

- The union of all these countable numbers for all n is countable.

Thus, even when we leave the shores of $\mathbb{Q}$

to embrace all the algebraic numbers,

we are still within the realms of the countable.

But there are still the transcendental numbers

to might spoil the countability of $\mathbb{R}$…

G. Cantor proved (1874, 1891) that

The *continuum*

is strictly greater

than the *countable/discrete*.

In other terms, it is not possible to establish

a bijection between $\mathbb{R}$ and $\mathbb{N}$.

First Proof of the Uncountability of $\mathbb{R}$

(G. Cantor, *Über eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen*, 1874)

• **Cantor's Excess Theorem**: the cardinal $|\mathscr{P}(X)|$ of the set of parts (subsets)

$$\mathscr{P}(X) = \{0,1\}^X$$

of a set X

(where $\{0,1\}^X$ is the set of maps $X \to \{0,1\}$)

is bigger than the cardinal $|X|$ of X:

$$|\mathscr{P}(X)| = 2^{|X|} > |X|$$

that is, there are always more parts (subsets) than elements.

• **« The continuum […] is the set of parts of the discrete »** (A. Badiou, *Being and Event*):

$$\mathbb{R} = \mathscr{P}(\mathbb{N}) = \{0,1\}^{\mathbb{N}}$$

• Hence, the continuum is greater than the countable:

$$|\mathbb{R}| = 2^{\aleph_0} > |\mathbb{N}| = \aleph_0$$

Excess of Parts over Elements for Finite Sets

For finite sets

$$|\mathscr{P}(X)| = 2^{|X|}$$

i.e., the number of parts $|\mathscr{P}(X)|$ grows exponentially

with the number of elements $|X|$.

A part A of X can be defined by specifying

if each $x \in X$ belongs or not to A,

which defines a « characteristic function » $\chi_A : X \to \{0,1\}$

such that $\chi_A(x) = 1$ if $x \in A$ and $\chi_A(x) = 0$ if $x \notin A$.

Now, there are $2 \times \ldots \times 2$ (times the number of elements of X) $= 2^{|X|}$ functions $X \to \{0,1\}$.

Since $2^{|X|} > |X|$, the number of parts $|\mathscr{P}(X)|$ is always greater

than the number of elements $|X|$ for X finite.

Cantor's Excess Theorem

or Why there are always more parts than elements

- Let's assume a one-to-one correspondence

$$S : X \to \mathscr{P}(X)$$

between elements and parts.

- Let's define the following part of X:

$$Q = \{x \in X / x \notin S(x)\}$$

- Since S is assumed to be one-to-one, there should be a $q \in X$ such that $S(q) = Q$.

- Does q belong to Q?

  .q cannot be in $Q = S(q)$ since Q contains only the elements x that are not in $S(x)$.

  .But if q is not in Q, then it should be in Q since Q contains all the elements x that are not in $S(x)$.

- Hence, a one-to-one correspondence between X and $\mathscr{P}(X)$ cannot exist.

- $X \subset \mathscr{P}(X)$ where the inclusion if given by $x \mapsto \{x\}$.

Consequence of Cantor's Excess Theorem

Since the set $\mathscr{P}(X)$ of parts/subsets of X

has more elements than X

— even when X is infinite —,

there is no largest infinity.

The Continuum is the Set of Parts of the Discrete (I)

$$\mathbb{R} = \mathscr{P}(\mathbb{N}) = \{0,1\}^{\mathbb{N}}$$

which implies:

$$|\mathbb{R}| = 2^{|\mathbb{N}|} = 2^{\aleph_0}$$

This can be proved by showing that

$$\mathbb{R} \hookrightarrow \mathscr{P}(\mathbb{N})$$

and that

$$\mathscr{P}(\mathbb{N}) \hookrightarrow \mathbb{R},$$

which implies

— by the by the Cantor–Schröder–Bernstein theorem —

$$\mathbb{R} = \mathscr{P}(\mathbb{N}).$$

The Continuum is the Set of Parts of the Discrete (II)

Proof that $\mathbb{R} \hookrightarrow \mathscr{P}(\mathbb{N})$

Since $\mathbb{Q}$ is countable,

there is a bijection

$$b : \mathbb{Q} \to \mathbb{N}.$$

Each $x \in \mathbb{R}$ defines the following subset of $\mathbb{N}$:

$$A_x = b(\{q \in \mathbb{Q} / q < x\})$$

In other terms, we take the Dedekind cut that represents x (which is a set of rationals)

and then use the bijection

$$b : \mathbb{Q} \to \mathbb{N}$$

to convert this set of rationals into a set of natural numbers, i.e. a part of $\mathbb{N}$.

The Continuum is the Set of Parts of the Discrete (II)

Proof that $\mathscr{P}(\mathbb{N}) \hookrightarrow \mathbb{R}$

Each subset

$$A \subset \mathbb{N}$$

defines a « characteristic function

$$\chi_A : \mathbb{N} \to \{0,1\},$$

i.e. an infinite sequence of 0's and 1's indexed by $\mathbb{N}$.

Each such sequence defines a real number

in the interval [0,1] written in base-3 as

$$0.\chi_A(0)\chi_A(1)\chi_A(2)\chi_A(3)\ldots$$

Second Proof of the Uncountability of $\mathbb{R}$

Cantor's Diagonal Argument

(*Über eine elementare Frage der Mannigfaltigkeitslehre,* 1891)

- Let's assume that it is possible to establish a one-to-one correspondence between the real numbers in the interval [0,1] and $\mathbb{N}$.

- This means that it is possible to make a list indexed by $\mathbb{N}$ of all $x \in [0,1]$.

- It is always possible to construct « diagonal » real numbers in the interval [0,1] that are not in the list.

List all real numbers between 0 and 1

1) 0.81728056931661013616090881330367580918331390128519747920…
2) 0.58787619690593538894245451824849205989889286583490896030…
3) 0.33327924801963223542552639721823659830585587541248796990…
4) 0.65513924153272811245900919954349430689648453470508978210…
5) 0.66560141186785141558028741544170954400044602585952229340…
6) 0.07772339164214430688263308157460042035244021202327992180…
7) 0.03821153525967325839348345297560221305429837867916572150…
8) 0.21951900414452078193905288558439117290841235286267143230…
9) 0.90602421167967824425409891543774832733240440346204144550…
10) 0.71788275612658991468952573508616920659669778308438007900…
11) 0.13184577057263264368716835976498388364636780620729660200…
12) 0.88105502948384574948184695942626437579151752259136938120…
13) 0.81019501082193672630738755035503061529558794514296463800…
14) 0.12876529962860790788912142446412832571767103179130564000…
15) 0.78070719203767892711304362606160498892216148913396369520…
16) 0.84994003422777524274714141283138509286627529374312003950…
17) 0.26200197034813093277282836237523147304415860495136885220…
18) 0.71527255403422136941506293794252051103341692070235909400…
19) 0.04867539094699200058213078457796579497911324824851009810…
20) 0.40493497450073615836984915388854910799132956274007248840…
… …

Select the digits on the diagonal

1) 0.**8**1728056931661013616090881330367580918331390128519747 92…
2) 0.5**8**7876196905935388942454518248492059898892865834908960 3…
3) 0.33**3**279248019632235425526397218236598305855875412487969 9…
4) 0.655**1**392415327281124590091995434943068964845347050897821…
5) 0.6656**0**14118678514155802874154417095440004460258595222934…
6) 0.07772**3**391642144306882633081574600420352440212023279921 8…
7) 0.038211**5**3525967325839348345297560221305429837867916572 15…
8) 0.219519**0**04144520781939052885584391172908412352862671432 3…
9) 0.9060242**1**16796782442540989154377483273324044034620414455…
10) 0.71788275**1**26589914689525735086169206596697783084380079 0…
11) 0.131845770**5**726326436871683597649838836463678062072966020…
12) 0.8810550294**8**38457494818469594262643757915175225913693812…
13) 0.810195010821**9**3672630738755035503061529558794514296463 80…
14) 0.12876529962860**7**907889121424464128325717671031791305640 0…
15) 0.780707192037678**9**27113043626061604988922161489133963695 2…
16) 0.84994003422777**5**2427471414128313850928662752937431200395…
17) 0.26200197034813093**2**77282836237523147304415860495136885 22…
18) 0.715272554034221369**4**150629379425205110334169207023590940…
19) 0.048675390946992000**5**821307845779657949791132482485100981…
20) 0.40493497450073615836**9**8491538885491079913295627400724884…
… …

Add 1 to the digits : 0.99421461228401934067 —> Not in the list

1) 0.917280569316610136160908813303675809183313901285197479 2…
2) 0.597876196905935388942454518248492059898892865834908960 3…
3) 0.334279248019632354255263972182365983058558754124879699…
4) 0.655239241532728112459009199543494306896484534705089782 1…
5) 0.665611411867851415580287415441709544000446025859522293 4…
6) 0.077724391642144306882633081574600420352440212023279921 8…
7) 0.038211635259673258393483452975602213054298378679165721 5…
8) 0.219519014144520781939052885584391172908412352862671432 3…
9) 0.906024212679678244254098915437748327332404403462041445 5…
10) 0.717882756226589914689525735086169206596697783084380079 0…
11) 0.131845770582632643687168359764983883646367806207296602 0…
12) 0.881055029484845749481846959426264375791517522591369381 2…
13) 0.810195010821036726307387550355030615295587945142964638 0…
14) 0.128765299628617907889121424464128325717671031791305640 0…
15) 0.780707192037679927113043626061604988922161489133963695 2…
16) 0.849940034227775342747141412831385092866275293743120039 5…
17) 0.262001970348130942772828362375231473044158604951368852 2…
18) 0.715272554034221360415062937942520511033416920702359094 0…
19) 0.048675390946992000682130784577965794979113248248510098 1…
20) 0.404934974500736158379849153888549107991329562740072488 4…
… …

The real number thus constructed

(by adding 1 to the elements in the diagonal)

is not in the original list since

- it differs from the first digit of the first number,
- it differs from the second digit of the second number,
- it differs from the n digit of the nth number.

The continuum $\mathbb{R}$ is bigger that the countable/discrete,

i.e. there are more real numbers than natural numbers.

Now, what is the cardinality $|\mathbb{R}|$ of the *continuum* $\mathbb{R}$?

How bigger is it?

$\aleph_1$

• Let's consider the first ordinal — denoted ω_1 — that is not countable (i.e. isomorphic to $\mathbb{N}$).

• Since every ordinal can be understood as the collection of all the preceding ones, ω_1 is the collection of all countable ordinals.

• $\aleph_1$ is the cardinal of ω_1.

The **Continuum Hypothesis** (Cantor, 1878) states that

$$|\mathbb{R}| = \aleph_1$$

i.e. that the cardinality of the continuum $|\mathbb{R}|$

is the smallest cardinal after $\aleph_0$.

The validity of this conjecture

is independent of ZFC-set theory.

(K. Gödel, 1940; P. Cohen, 1963).

We do not know how many real numbers are.

Philosophical Interpretations of the Continuum

Main antagonism in the interpretation of the continuum

- Either the continuum as such is essentially inintelligible and has to be constructed out of discrete entities (Cantor) or postulated as a kind of pre-mathematical a priory (Brouwer). Only the discrete is intelligible.

- Or the continuum is the very condition of intelligibility (Leibniz), making for instance possible the predictive capacities (via differential and integral calculus) of modern science. Only the continuous is intelligible.

I. **The continuum is made of points**

The continuum is made of

discrete extensionless point-like « elements »,

i.e. a continuous extension can be generated by an accumulation of points

(Bolzano, Cauchy, Dedekind, Cantor).

The arithmetization of love according to Proust

“Car ce que nous croyons notre amour, notre jalousie, n’est pas une même passion continue, indivisible. Ils se composent d’une infinité d’amours successifs, de jalousies différentes et qui sont éphémères, mais par leur multitude ininterrompue donnent l’impression de la continuité, l’illusion de l’unité. La vie de l’amour de Swann, la fidélité de sa jalousie, étaient faites de la mort, de l’infidélité, d’innombrables désirs, d’innombrables doutes, qui avaient tous Odette pour objet. S’il était resté longtemps sans la voir, ceux qui mouraient n’auraient pas été remplacés par d’autres. Mais la présence d’Odette continuait d’ensemencer le coeur de Swann de tendresses et de soupçons alternés.”

M. Proust, *Du côté de chez Swann, A la recherche du temps perdu.*

« Je trouve vicieux et fantastique de considérer les grandeurs comme *formées* de *points*. On peut bien déterminer des *points* sur cette grandeur ou lui appartenant, mais non la construire par des points. On le dit, et même on le *fait* — mais en y ajoutant du *non-point*, — la liaison. On ne peut considérer plus d'un point. Si on veut en considérer 2, on trace une droite de l'oeil. »

P. Valery, *Cahiers II*, Bibliothèque de la Pléiade, p.820.

« On ne *peut* construire une figure par accumulation d'actes ponctuelles (ou *marques*). Mais *on le dit*. C'est *détacher du possible réel un possible verbal* (lequel, d'ailleurs, est en accord avec le fait réel du tracement 'continu' d'un trait — qui procède sans égard à *tous* les *points*). C'est une unité d'action motrice. »

P. Valery, *Cahiers II*, Bibliothèque de la Pléiade, p.828.

II. The continuum is made of infinitesimals

The continuum is made of « elements »

which are not extensionless points,

by rather ***infinitesimals***,

i.e. hypothetical new kind of numbers

that are bigger than zero

but smaller than any real number

(Peirce, Robinson).

III. The continuum is an indivisible unity dynamically generated by motion

« […] a continuum cannot be composed of indivisibles, e.g. a line of points. […] nothing continuous was divisible into partless parts. […] Further, every continuum is divisible into divisibles *ad infinitum*. »

Aristotle, *Physics*, Book VI, 231.

« Diogenes the Cynic, having heard these puzzles once, said nothing in reply to them, but stood up and walked and by means of self-evidence itself resolved the paradoxes in the arguments. »

Simplicius, *Commentary on Aristotle's* Physics, 1012,22-1013,2

Newton on Continuous Magnitudes

Continuity is dynamically generated by « fluent quantities »

« I don’t here consider Mathematical Quantities as composed of Parts extremely small, but as generated by a continual motion. Lines are described, and by describing are generated, not by any apposition of Parts, but by a continual motion of Points. […] Time by a continual flux, and so in the rest. These Geneses are founded upon Nature, and are every Day seen in the motion of Bodies. »

I. Newton, *Quadrature of Curves*.

IV. The continuum is beyond symbolic reason

The continuum is a primitive notion

that cannot be fully apprehended by

— a necessarily discrete —

symbolic formalism

(Brouwer, Bergson, Thom).

Brouwer's « Intuitionism » (1881-1966)

The continuum is an *Ur*-intuition

associated with the (the flow of) time

— to the exclusion of space

(as a consequence of the emergence of non-Euclidean geometries) —

as the *a priori* form of experience that generates both

the sequence of natural numbers,

— associated with the « succession » of one sensation after another

(that Brouwer calls « two-oneness ») —

and the *continuum* understood as the in-between of two successive instants.

« However weak the position of intuitionism seemed to be after this period of mathematical development, it has recovered by abandoning Kant's apriority of space but adhering the more resolutely to the apriority of time. This neo-intuitionism considers the falling apart of moments of life into qualitatively different parts, to be reunited only while remaining separated by time, as the fundamental phenomenon of the human intellect, passing by abstracting from its emotional content into the fundamental phenomenon of mathematical thinking, the intuition of the bare two-oneness. This intuition of two-oneness, the basal intuition of mathematics, creates not only the numbers one and two, but also all finite ordinal numbers, inasmuch as one of the elements of the two-oneness may be thought of as a new two-oneness, which process may be repeated indefinitely; this gives rise still further to the smallest infinite ordinal number ω. Finally this basal intuition of mathematics, in which the connected and the separate, the continuous and the discrete are united, gives rise immediately to the intuition of the linear continuum, i.e., of the 'between,' which is not exhaustible by the interposition of new units and which therefore can never be thought of as a mere collection of units. »

L.E.J. Brouwer, *Intuitionism and formalism*, 1912.

Thom on the Continuum (I)

« I would accept that, in any kind of description, we have a discrete element, because a true continuum has no points. We are unable to specify anything in the continuum. The continuum is something which cannot be described. It is a sort of unsayable. It is a world in which one lives outside of symbolic description. But nevertheless, it exists, despite the fact that we cannot describe it in any sense. »

R. Thom, Round-Table discussion, in Casacuberta and Castellet (eds), *Mathematical Research Today and Tomorrow Viewpoints of Seven Fields Medalists*, p.102.

Thom on the Continuum (II)

« People who say that the discrete is more important than the continuous always forget that discrete entities are carved out from within a continuous medium. It is in this sense that there is an ontological primacy of the continuous. [...] People who believe in pure discreteness are obsessed with writing, with the discrete event symbolized by a letter… »

R. Thom, *Théorie des catastrophes, Mathématiques et sciences humaines*, tome 59, 1977, 3-81, p.38.

Thom on the Continuum (III)

« Now, among the primary intuitions, there is one that is essential: that of time. And time, in the flow of its passing, appears to us as fundamentally continuous: between two instants *a* and *b*, one can always imagine an intermediate instant *c*. Consequently, we are led to consider as 'ontologically better founded' a theory such as Differential Calculus, which aims at representing continuous geometric entities—such as curves and surfaces—than a purely discrete theory, like Set-theoretic Logic, for example. This priority of the continuum that we postulate goes against the traditional approach, which treats the real line as a constructed object—by a procedure that is in fact not very constructive—out of the integers and then the rationals (via Dedekind cuts or the completion of Cauchy sequences).

[...] Of the continuum in its pure state, nothing can be said; it is an object as unsayable as divinity may be for a mystic. In order to make the continuum an object of representation, in order to be able to say something about it, one must necessarily make 'marks' in it, trace 'figures' within it, which stand out against an undifferentiated background. Every communicable representation thus involves a discrete figuration. The permanent temptation, then, is to forget the background and keep only the figure: an excessive simplification strongly suggested to us by the discrete character of the phonological combinatorics of human language, and into which logicians—and, following them, quite a few formalist mathematicians—have rushed headlong. »

R. Thom, *Les réels et le calcul différentiel ou la mathématique essentiel, Séminaire de Philosophie et Mathématiques*, 1982, fascicule 3, Les réels et le calcul différentiel, p. 1-20.

Le différentiel en deçà de la representation ou la mystification du calcul

« […] les interprétations finitistes modernes ne trahissent pas moins la nature du différentiel que les anciennes interprétations infinitistes, parce qu'elles laissent échapper toutes deux la source extra-propositionnelle ou sub-représentative, c'est-à-dire le 'problème', d'où le calcul tire son pouvoir. »

G. Deleuze, *Différence et répétition*, p. 339.

Cantor on the (Arithmetization of the) Continuum

« Leucippus, Democritus and Aristotle regard the continuum as a composite, which consists of endlessly divisible particles (*ex partibus sine fine divisibilibus*), whereas Epicurus and Lucretius consider that it is a composite of atoms that are finite things. From this a great dispute later arose among philosophers, some of whom followed Aristotle, while others followed Epicurus, while others, in an attempt to steer clear of the dispute, sided with Thomas Aquinas in the claim that the continuum consists neither of an infinite number nor of a finite number of parts, but of no parts at all. […] Here we observe the *medieval-scholastic* origin of a view that we still find to be held today, according to which the continuum is an indivisible concept or, as others put it, a notion that is purely a priori, and hardly amenable as a basis for any definition. Every attempt at an arithmetical *determination* of this *mystery* is regarded as an inadmissible interference and rejected with all due intensity, and those of a diffident disposition get the impression that the 'continuum' is not a *mathematical-logical notion* but rather a *religious dogma*. […] in my opinion, the use of the *concept of time* or the *intuition of time* is not applicable when discussing the much more fundamental and general concept of the continuum. In my opinion, *time* is a notion which requires for any clear explanation of it, a presupposition of the concept of continuity, a concept that is independent of it. […] It is also my conviction that one cannot do anything with the so-called *form of intuition* of space in order to gain information about the continuum […] Hence I have no choice but to attempt, with the help of the real number concepts defined in § 9, to form a purely arithmetical concept of a point continuum that is as general as possible. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical*

Continuity as a Rational Principle

Natura non facit saltus

« Nothing takes place suddenly, and it is one of my great and best confirmed maxims that nature never makes leaps. I called this the Law of Continuity […] It implies that any change from small to large, or vice versa, passes through something which is […] in between […] just as one could never traverse a certain line or distance without first traversing a shorter one. […] To think otherwise is to be ignorant of the immeasurable fineness of things, which always and everywhere involves an actual infinity. »

Gottfried Leibniz, *New Essays on Human Understanding*, Preface, 56-57.

Badiou on Leibniz's Continuity Principle

« […] presentation is continuous, and it does not tolerate interruption […]: 'The present is always pregnant with the future and no given state is naturally explainable save by means of that which immediately preceded it. If one denies this, the world would have hiatuses, which would overturn the great principle of sufficient reason, and which would oblige us to have recourse to miracles or to pure chance in the explanation of phenomena.' The event is thus excluded on the following basis: the complete language is the integral calculus […], whilst a local approximation already authorizes its differential calculus. »

A. Badiou, *Being and Event*, Meditation 30: Leibniz, p.319.

« Le secret […] ne peut être que dans les relations qu'ils trouvèrent […] *entre des choses dont nous échappe la loi de continuité*. »

P. Valery, *Introduction à la méthode de Léonard de Vinci*, Éditions Gallimard, 1957.

Tropical Discontinuities

« You wait for explanations, and, like Descartes, you receive only ever-new pieces of information, without connection, and from sentence to sentence, from word to word, you never know what to expect, ready to throw in the towel, to withdraw from this incessant abortion of the continuous, from this permanent frustration of your expectation […] »

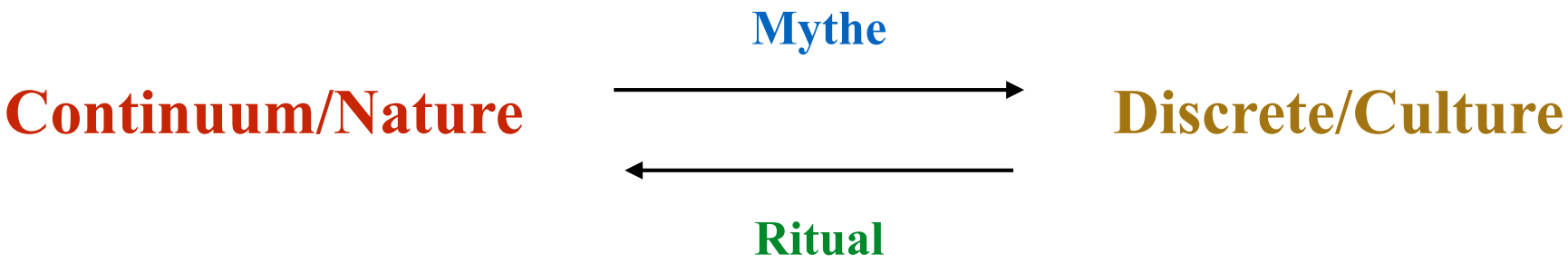
J.-.C. Goddard, *Scabby Black Brazilian,* Urbanomic, 2024.

Declensions of the *Discrete-Continuum* dialectic

- Pitch vs. Rythm (Stokhausen).
- Nature vs. Culture (Lévi-Strauss).
- Singing vs. spoken language.
- Gradualism vs. saltationism in evolutionary biology.
- Bergson vs. Bachelard (*durée* vs. cinematographic model in H. Bergson.
- Smooth Schrödinger evolution vs. quantum leaps (Quantum Mechanics).
- Waves vs. particles (Quantum Mechanics).
- Fields vs. Particles (Quantum Field Theory).

The *continuum/discrete* opposition

in Lévi-Strauss' structural anthropology



Refs.: G. Schrempp, *Magical Arrows: The Maori, the Greeks, and the Folklore of the Universe*, 1992, University of Wisconsin Press; P.A. Lolli, *O contínuo e o discreto em Lévi-Strauss: transformações ameríndias*, Tellus, ano 12, n. 22, p. 81-105, jan./jun. 2012 Campo Grande, MS.

On the *continuum/nature* - *discrete*/*culture* opposition

« I shall establish the existence of an isomorphism between two oppositions; that of nature and culture and that of continuous and discrete quantities. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.28.

« […] the transition from nature to culture corresponds, in native thought, to the transition from the continuous to the discontinuous. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.279.

« It is as if South American thought […] invested chromaticism [« in the very general sense of the use of small intervals »] with a primordial maleficence of such a kind that long intervals, which are necessary in culture for it to exist and in nature for it to be 'thinkable' to men, can only be the result of the self-destruction of a primeval continuity, whose power is still to be felt at those rare points where it has survived its own self-destructiveness […] »

C. Lévi-Strauss, *The Raw and the Cooked*, p.280.

The Mythological Transition from Continuous to Discrete Quantity (I)

« It would therefore seem that the two myths, taken together, refer to three domains, each of which was originally continuous, but into which discontinuity had to be introduced in order that each might be conceptualized. In each case, discontinuity is achieved by the radical elimination of certain fractions of the continuum. Once the latter has been reduced, a smaller number of elements are free to spread out in the same space, while the distance between them is now sufficient to prevent them overlapping or merging into one another. Men had to become less numerous for the most closely related physical types to be clearly distinguishable. For if we admit the existence of clans or tribes bearing *insignificant* gifts—that is whose distinctive originality is as weak as one cares to imagine—we should run the risk of finding that between any two given clans or communities there was an unlimited number of other clans or other tribes, each one of which would differ so slightly from its immediate neighbors that they all would ultimately merge into each other; whereas in any field a system of significances can be constructed only on the basis of discrete quantities. […] In all these instances, therefore, a discrete system is produced by the destruction of certain elements or their removal from the original whole. […] it is clear that the myths I am comparing all offer an original solution to the problem of the change-over from continuous quantity to discrete quantity. »

C. Lévi-Strauss, *The Raw and the Cooked*, pp.52-53.

The Mythological Transition from Continuous to Discrete Quantity (II)

« […] mythic thought proceeds in the same way as language, which chooses phonemes from among the natural sounds of which a practically unlimited range is to be found in childish babbling. For, as in the case of language, the empirical material is too abundant to be all accepted indiscriminately or to be all used on the same level. […] it must be whittled down. Only a few of its elements are retained—those suitable for the expression of contrasts or forming pairs of opposites. But, as with language, the discarded elements are not thereby eliminated. They are always present in latent fashion, behind those that have been singled out […] the virtually unlimited totality of the elements always remains available. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.341.

The Ritualistic Transition from Discrete to Continuous Quantity (I)

« Pour acquérir un rituel […] il a fallu qu'un humain abjurât les distinctions claires et nettes en vigueur dans la culture et la société ; que […] il revînt à l'état de nature marqué par la promiscuité des sexes et la confusion entre les degrés de parenté […] Ainsi donc, tandis que le mythe tourne résolument le dos au continu pour découper et désarticuler le monde au moyen de distinctions, de contrastes et d'oppositions, le rite suit un mouvement en sens inverse : parti des unités discrètes qui lui sont imposées par cette conceptualisation préalable du réel, il court après le continu et cherche à le rejoindre, bien que la rupture initiale opérée par la pensée mythique rende la tâche impossible à jamais. […] le rite se réfère bien à ces catégories, lois ou axiomes. Mais le rite ne les crée pas et s'emploie plutôt, sinon à les désavouer, au moins à oblitérer temporairement les distinctions et oppositions qu'ils édictent, en faisant apparaître entre celles-ci toutes sortes d'ambiguïtés, de compromis et de passages. »

C. Lévi-Strauss, *Mythologiques 4 : L'homme nu.*

The Ritualistic Transition from Discrete to Continuous Quantity (II)

« […] pour ce qui est du rituel, l'anxiété accompagnatrice […] tient […] à la crainte que les découpages opérés sur le réel par la pensée discrète en vue de le conceptualiser ne permettent plus de rejoindre […] la continuité du vécu. Donc une anxiété qui […] résulte de ce que le pensé, du seul fait qu'il est du pensé, creuse un écart croissant entre l'intellect et la vie. Le rituel n'est pas une réaction à la vie, il est une réaction à ce que la pensée a fait d'elle. »

C. Lévi-Strauss, *Mythologiques 4 : L'homme nu.*

Complex Numbers

History of Complex Numbers

- Gerolamo Cardano, *Ars Magna* (1545).

- Raphael Bombelli, *L'Algebra* (1572).

- Descartes, *La géométrie* (1637): introduction of the term « imaginary ».

- Euler's formula (1748).

- Complex plane: Caspar Wessel (1797), Jean Robert Argand (1806), John Warren (1828), Carl Friedrich Gauss (1831).

Towards Complex Numbers

The solution of the $\mathbb{R}$-equation

$$x^2 + 1 = 0$$

is not in $\mathbb{R}$

(since $x^2 > 0$ for any $x \in \mathbb{R}$).

We shall declare that the symbol

$$i$$

— called the ***imaginary unit*** —

is a solution of the equation

$$x^2 + 1 = 0,$$

which means that i is such that

$$i^2 = -1$$

or

$$i = \sqrt{-1}$$

Given the property $\sqrt{ab} = \sqrt{a}\sqrt{b}$,

the introduction of $i = \sqrt{-1}$ allows us to find

the square roots of any negative quantity.

For instance, $\sqrt{-4} = \sqrt{(-1)4} = \sqrt{-1}\sqrt{4} = i2$

At first, *i* is a « blind » or symbolic extension

deprived of any intuition.

How can we endow

the symbol *i*

with some form of intuitive understanding?

Besides using numbers to *count*, *order* and *measure*,

we can also use numbers to specify

how much of each « ingredient » is present

in a particular composition

(like the different ingredients in a cooking recipe),

In particular, we can use numbers

as ***scalars*** in a ***vector space***.

A digression on Vector Spaces

A ***k-vector space*** $(V, +, k = \mathbb{R} \text{ or } \mathbb{C})$

is an additive commutative group $(V, +)$

— a set V of vectors equipped with an invertible addition $+$

(and the corresponding neutral element, the null vector $\overrightarrow{0}$) —

equipped with a scalar multiplication

$$k \times V \to V$$

by elements of a field k ($\mathbb{R}$ or $\mathbb{C}$)

called *scalars*,

which refers to the fact that vectors can be « scaled » (e.g. stretched).

A k-vector space can be visualized

as a set V of « arrows »

based at a common origin

where vectors can be « scaled » by numbers in k and added,

such that the addition $\overrightarrow{v} + \overrightarrow{w}$ of two vectors $\overrightarrow{v}$ and $\overrightarrow{w}$

is the vector corresponding

to the diagonal of the parallelogram spanned by $\overrightarrow{v}$ and $\overrightarrow{w}$.

Linear combinations

Given two vectors $\vec{v}$ and $\vec{w}$,

we can define the ***linear combination***

$$a\vec{v} + b\vec{w},$$

where $a, b \in k$.

The vectors $\vec{v}$ and $\vec{w}$ are scaled

by the numbers a and b respectively

and the resulting scaled vectors $a\vec{v}$ and $b\vec{w}$ are added,

which yields the vector $a\vec{v} + b\vec{w}$.

Basis of a Vector Space

Given a vector space $(V, +, k)$,

it is possible to define a set of (non-collinear) vectors $(\vec{e}_1, \ldots, \vec{e}_n)$

— called a ***basis*** —

such that any vector v can be written

as a linear combination of the elements of the basis

$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2 + \ldots + v_n \vec{e}_n$$

The set of numbers $(v_1, v_2, \ldots, v_n)$ are the ***coordinates*** of $\vec{v}$ in the basis $(\vec{e}_1, \ldots, \vec{e}_n)$.

The number n of elements in the basis is the ***dimension*** of the vector space.

Real Vector Spaces

In a $\mathbb{R}$-vector space, multiplying a vector $\overrightarrow{v}$ with a scalar k

stretches ($|k| > 1$) or shrinks ($|k| < 1$) the vector

while changing ($k < 0$) or not ($k > 0$) its orientation.

In particular, multiplying by the scalar -1

keeps the length of the vector $\vec{v}$,

but changes its orientation.

In other terms, multiplying by -1

rotates the vector in $180°$ or π radians.

Scaling by $i = \sqrt{-1}$ or How to Factorize the Inversion of a Vector

Since $-1 = i \times i$, we have

$$(-1)\overrightarrow{v} = (ii)\overrightarrow{v} = i(i\overrightarrow{v})$$

(where we have used that in a vector space $(kk')v = k(k'v)$).

Since multiplying a vector $\overrightarrow{v}$ by -1 inverts the orientation of $\overrightarrow{v}$,

multiplying by i can be understood as a rotation in $90°$ or $\frac{\pi}{2}$ rad.

The imaginary unit i is a new number,

such that multiplying a vector by i

rotates the vector in 90°

(without stretching or shrinking its length).

Complexification as an Enhancement of Continuity

It is not possible to rescale a unit vector $\overrightarrow{1}$ on $\mathbb{R}$

by means of a continuously varying scalar $k(t)$

so that the transformed vector is $-\overrightarrow{1}$.

On the contrary,

$\overrightarrow{1}$ can be transformed into $-\overrightarrow{1}$

by means of a continuous rotation

obtained by multiplying $\overrightarrow{1}$

with a continuously varying complex scalar.

« [...] that curiously elegant form of continuity that is provided by *complex numbers* [...] Perhaps, in complex numbers will ultimately be found the resolution between the discrete and continuous. »

R. Penrose, Foreword to Schrödinger, *Nature and the Greeks*, p.x.

The field $\mathbb{C}$ of complex numbers

is an extension of $\mathbb{R}$ such that each « scalar » $z \in \mathbb{C}$

not only specifies a ***dilatation*** or ***stretching*** of vectors

by an amount r

but also a ***rotation*** in an angle θ.

Whereas all the previous number systems

encode the concept of ***magnitude***

(how much?, how many?, how big?),

the complex numbers enrich the concept of

magnitude with an ***orientation***.

Complex numbers can therefore be understood as

oriented magnitudes.

Complex numbers become pertinent

in situations in which we combine certain « ingredients »

in such a way that it is not only important

to specify *how much* of each ingredient is present,

but also the *orientation* of each ingredient

over a full circle $0 \leq \theta < 2\pi$.

Example: Fourier Synthesis

In Fourier synthesis,

circular motions are combined

to yield a periodic signal,

where the corresponding linear combination

requires to specify the amplitude of each motion

as well as their initial orientation.

Hence, the « scalars » will be complex numbers $z = (r, \theta)$,

where r will fix the amplitude of the corresponding component

and θ the initial orientation.

We can describe a complex number as a pair

$$z = (r, \theta),$$

where $0 \leq r$

is the *radius* (or *length* or *modulus*) of z

and $0 \leq \theta < 2\pi$

is the *angle* (or *argument*).

z « scales » vectors

by multiplying the length of $\overrightarrow{v}$ by r

and by rotating $\overrightarrow{v}$ counterclockwise in θ.

Since complex numbers depend

on two independent real variables

encoding a ***scaling*** of the length of vectors and a ***rotation***,

they are not geometrically schematized as points in a line (like $\mathbb{R}$),

but rather points in a plane,

the so-called ***complex plane***.

In order to identify each complex number z

with a point in a plane,

we can understand the plane

as a $\mathbb{C}$-vector space of dimension 1

in which the horizontal axis is identified with the real line $\mathbb{R}$.

This means that $\mathbb{C}$ is an extension of $\mathbb{R}$

as a plane is an extension of a line.

The real number 1 can then be identified with the vector $\overrightarrow{1}$

pointing from the origin to the number 1 on the real axis.

A complex number $z = (r, \theta)$ can then be identified

with the vector in the plane

obtained by « scaling » the vector $\overrightarrow{1}$ with $z = (r, \theta)$:

$$z \leftrightsquigarrow z\overrightarrow{1}.$$

Now, « scaling » the vector $\overrightarrow{1}$ by $z = (r, \theta)$

amounts to stretch $\overrightarrow{1}$ by a factor r and to rotate it in θ anti-clockwise.

Hence, the complex number $z = (r, \theta)$ can be identified

with a vector of length r rotated in θ from the real axis anti-clockwise.

In particular, the imaginary unit i

can be associated with the point in the plane

obtained by scaling the vector $\overrightarrow{1}$ with

the complex number z given by $r = 1$ and $\theta = \frac{\pi}{2}$,

which yields a vector pointing from the origin

to the number 1 on the vertical axis.

« If one formerly contemplated this subject from a false point of view and therefore found a mysterious darkness, this is in large part attributable to clumsy terminology. Had one not called $+1$, -1, $\sqrt{-1}$ positive, negative, or imaginary (or even impossible) units, but instead, say, direct, inverse, or lateral units, then there could scarcely have been talk of such darkness. »

C. F. Gauss, Theoria residuorum biquadraticorum. Commentatio secunda, *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores* 7 (1831), 89148.

The Complex Plane is in fact a « Line »

Since any vector in the complex plane

can be obtained by properly scaling $\overrightarrow{1}$

with a complex number z ,

the complex plane is a $\mathbb{C}$-vector space of dimension 1,

that is, as a line.

Complex Numbers in Cartesian Coordinates

We can now consider the complex plane as a $\mathbb{R}$-vector space of dimension 2.

By choosing the vectors $1 \leftrightsquigarrow (1,0)$ and $i \leftrightsquigarrow (0,1)$

as a basis we can express each z as a linear combination

$$z = a(1,0) + b(0,1)$$

where $a, b \in \mathbb{R}$ are the real coordinates

or (in short)

$$z = a + bi.$$

From Polar to Cartesian Coordinates

Since $a = rcos(\theta)$ and $b = rsin(\theta)$

the complex number $z = a + bi$ can be reexpressed

in terms of the polar coordinates (r, θ) as

$$z = r(cos(\theta) + isin(\theta))$$

or

$$z = re^{i\theta}$$

where we have used Euler's formula

$$e^{i\theta} = cos(\theta) + isin(\theta).$$

From Cartesian to Polar Coordinate

The polar coordinates of a complex number

$$z = a + bi$$

are given by the expressions

$$r = \sqrt{a^2 + b^2}$$

and

$$\theta = arctang(\frac{b}{a}).$$

Coordinates in the Complex Plane

A complex number z can be schematized

as a point in a Cartesian plane determined

either by ***Cartesian coordinates*** (a, b)

— in which case, $z = a + bi$ —

or in ***polar coordinates*** (r, θ)

— in which case, $z = re^{\theta}$ —

Complex Conjugate

The complex conjugate of a complex number

$$z = a + bi$$

is $\bar{z} = a - bi$,

which is the specular reflection of z

with respect to the real axis.

Algebraic properties of $\mathbb{C}$

- Complex number can be added:

$$(a + bi) + (c + di) = (a + b) + (b + d)i$$

where the additive inverse of $a + bi$ is $-a - bi$.

- Complex number can be multiplied:

$$(a + bi) \times (c + di) = ac + adi + bic + bdi^2 = (ac - bd) + (ad + bc)i$$

where the multiplicative inverse of z is

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2}.$$

$\mathbb{C}$ is a *field*.

The Number e

The Number e
I. Simple Interest

By depositing a capital C

with an annual rate of interest r written as a decimal number

(e.g. $r = 0.2$ for a 20 % of interest)

the bank gives back

an increased capital C' given by

$$C' = C + Cr = C(1 + r)$$

after a year.

The Number e
II. Compound Interest

If the annual rate is r and the year is divided in n parts,

the bank will give back

$$C' = C + C\frac{r}{n} = C(1 + \frac{r}{n}) \text{ after the first } n\text{-part of the year}$$

$$C'' = C'(1 + \frac{r}{n}) = C(1 + \frac{r}{n})^2 \text{ after the second } n\text{-part of the year}$$

$$\text{and } \widetilde{C} = C(1 + \frac{r}{n})^n \text{ after the last } n \text{ part of the year.}$$

The Number e
III. Compound Interest

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = 2.71828...$$

is the factor by which the initial capital is multiplied

after a year with an annual interest of 100%.

Instead of getting twice the initial capital,

one gets 2.71828... the initial capital.

Euler's Formula

$$e^{i\theta} = cos(\theta) + isin(\theta).$$

$$e^{i\pi} + 1 = 0$$

When exponential growth

becomes purely imaginary,

be sure you've entered a loop.

Properties of $\mathbb{C}$
I. The Fundamental Theorem of Algebra

Every polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.

Properties of $\mathbb{C}$
II. $\mathbb{C}$ is not endowed with a linear order.

Complex numbers do not entirely fall

within the scope of the definition of magnitude as

"that which can be *increased* or *decreased*"

(G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §99).

Indeed, a complex number can vary

only with respect to its orientation, i.e. by keeping its radius constant.

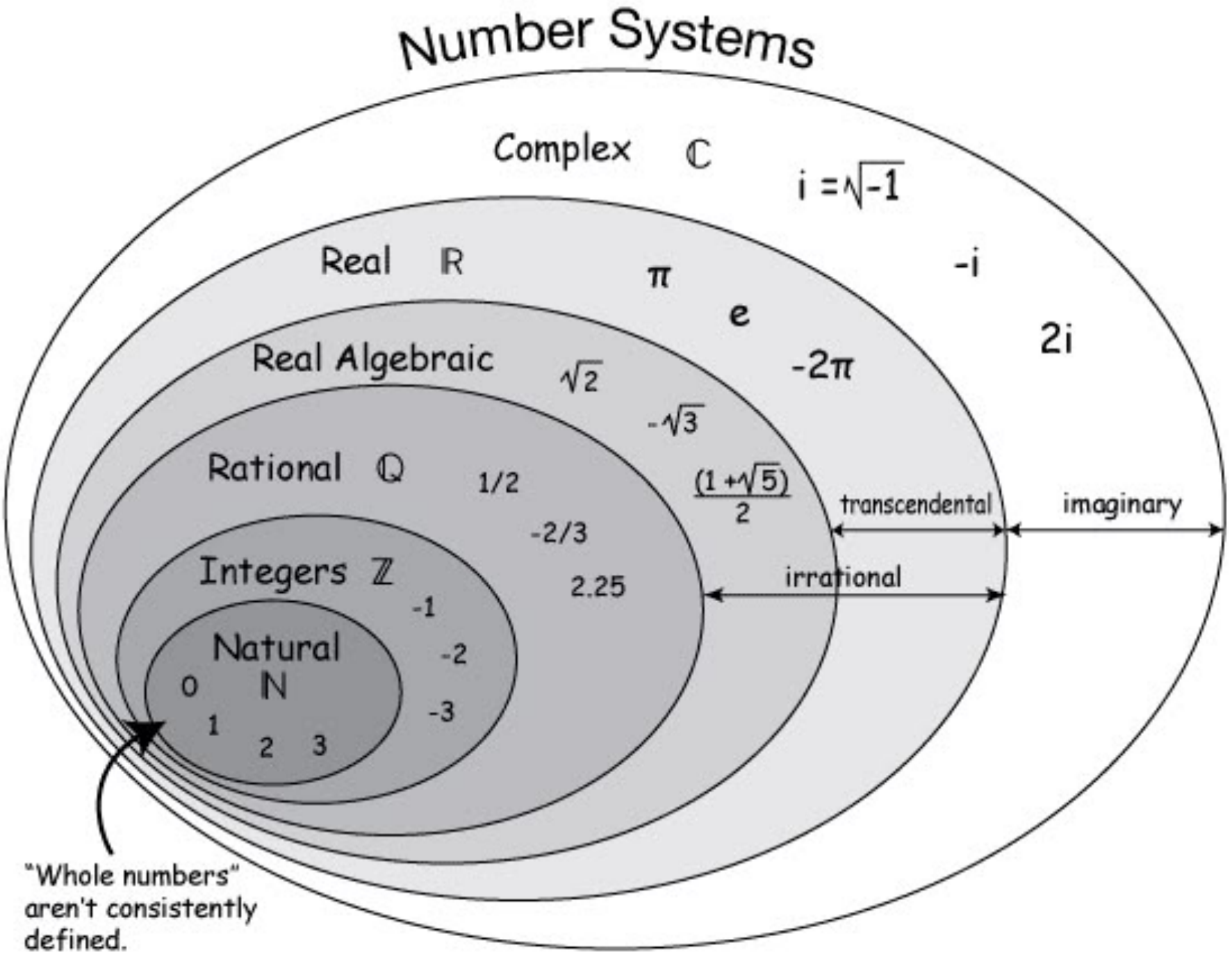
Are complex numbers really numbers?

« Mais les nombres complexes […] sont-ils des nombres ? On peut, je crois, raisonnablement soutenir qu'à partir du moment où nous quittons toute 'linéarité', quand nous abandonnons la dimension 1, nous avons affaire à des constructions *à partir* des Nombres plutôt qu'à des Nombres. Au fond, l'essence intime des complexes est géométrique, c'est le 'plan complexe' qui délivre la vérité de ces 'nombres'. […] Ce sont en fait des nombres *qui ne nombrent pas* […] Il me paraît en outre déraisonnable de parler de 'nombres' quand il n'est pas même possible, en liaison avec le champ opératoire considéré de dire que tel 'nombre' est plus grand ou plus petit qu'un autre. · Bref : un corps *de nombres* doit à mon avis être un corps *ordonné*, ce que ne sont [pas] les complexes […] Finalement, je restreindrai le concept de Nombre […] à ce qui se laisse déployer selon l'intuition d'une ligne.»

A. Badiou, *Le Nombre et les nombres*, *op.cit.*, p.278.

Summary of Number Systems

- Natural numbers $\mathbb{N}$
- Entire numbers $\mathbb{Z}$
- Rational numbers $\mathbb{Q}$
- Irrational numbers:
    - .algebraic
    - .transcendental :
        - .computable,
        - .non-computable.
- Real numbers $\mathbb{R}$.
- Complex numbers $\mathbb{C}$.


Number Systems
Complex ℂ
i = √-1
-i
2i
Real ℝ
π
e
-2π
Real Algebraic
√2
-√3
Rational ℚ
1/2
(1+√5)/2
-2/3
2.25
transcendental
imaginary
irrational
Integers ℤ
-1
-2
-3
Natural
ℕ
0
1
2
3
"Whole numbers" aren't consistently defined.

Generalizations in the Evolution of Number Systems

- From the ***finite*** to the ***infinite***.
- From the ***discrete*** to the ***continuum***.
- From ***magnitudes*** to ***oriented magnitudes***.

From the Relative to the Absolute

The Algebra-Geometry Duality

In the XVIIth century, Fermat and Descartes

independently invented analytic geometry,

that is, the study of the correspondence

between *algebraic expressions* and *geometric figures*

by using ambient spaces endowed with coordinate systems.

Ref.: C.B. Boyer, *History of Analytic Geometry*, (2004) [1956], Dover Publications,

From the algebraic equation at stake,

one can define a equation

and the solutions of the equation

can be « schematized »

as points in a coordinate space.

Examples

- The algebraic expression $x - 1$ defines the equation $x - 1 = 0$ whose unique solution is $x = 1$, which can be schematized as a point on a line in which a coordinate system has been defined.

- The algebraic expression $y - 2x$ defines the equation $y - 2x = 0$ whose unique solution is a set of pairs (x, y), each of which defines a point in a plane encoded with coordinates. The resulting geometric figure is a straight line passing by the origin of the coordinate system.

Conics

Conics

— i.e. elipses, circles, parabolas and hyperbolas —

are geometric figures in the plane

associated to algebraic expressions of the form

$$ax^2 + bxy + cy^2 + dx + ey + f$$

What is the geometric figure that « schematizes »

the expression $x^2 + y^2 - 1$?

In other terms, what are the solutions to the equation

of a circle of radius 1

$x^2 + y^2 = 1$?

Transcendental Relativity

The geometric ‘figure’ that ‘schematizes’ a given algebraic expression,

far from being absolutely defined,

depends on the kind of numbers

in which one searches for the solutions of the corresponding equation.

The solution to a problem depends

on the « transcendental framework »

— in the Kantian sense of the term —

in which one searches for such a solution.

Solutions to the Circle Equation

- Solutions in $\mathbb{Z}^2$ (entire circle)

$$\{(0,1),(0,-1),(1,0),(-1,0)\};$$

- Solutions in $\mathbb{Q}^2$ (rational circle):

$$x = \frac{q^2-1}{q^2+1}; y = \frac{2q}{q^2+1}, \text{ with } q \in \mathbb{Q}$$

- Solutions in $\mathbb{R}^2$ (real circle)

$$x = cos(\theta); y = sin(\theta), \text{ with } \theta \in \mathbb{R}.$$

- Solutions in $\mathbb{C}^2$ (complex circle).

« [...] the unit circle is the set of all solutions in the x-y plane to the equation

$$x^2 + y^2 = 1.$$

The set of solutions of this equation depends on the type of number that one understands x and y to refer to. [...] There are many more abstract types of numbers [than the integers, the rationals, the real, and the complex numbers] and each gives rise to its own circle. In the classical set theoretic framework, there is thus an integer circle, a rational circle, a real circle, a complex circle and myriads of others […] »

M. Barr, C. McLarty, C., & C. Wells, *Variable Set Theory*, 1986.

Such a transcendental relativity

was sublated by A. Grothendieck's introduction

of the notion of ***functor of solutions***

in the framework of his refoundation

of *algebraic geometry*.

Ref. A. Grothendieck & J.A. Dieudonné, *Éléments de Géométrie Algébrique I*, 1971, Springer.

Each transcendental framework A

is an algebraic structure

— a ring or a k-algebra —

such that it makes sense to replace the x

in the equations with element in A.

A ***transformation*** between different transcendantales frames

(also called a *morphism*)

$$f : A \rightarrow B$$

is a function between the corresponding sets of numbers

that preserve the algebraic structure (e.g. the ring structure),

that is, a ***structure preserving function***.

Morphisms Between Rings

Given two rings A and B, a morphism between them is a function

$$f : A \to B$$

that preserves the ring structure, that is:

- Addition:

$$f(a + a') = f(a) + f(a').$$

- Multiplication:

$$f(a \times a') = f(a) \times f(a').$$

- Neutral element for the multiplication:

$$f(1_A) = 1_B.$$

Note: It follows from these conditions that the additive inverses and the neutral element for the addition are also preserved.

Transcendental Parallax

A morphism

$$f : A \to B$$

between different « transcendental frames »

— i.e. a structure-preserving function between the underlying sets —

sends solutions to solutions.

For instance, if $a, b \in A$ solve the circle equation

$$x^2 + y^2 = 1$$

and if

$$f : A \to B$$

is a morphism of rings,

then $f(a)$ and $f(b)$ are elements in B

that also solve the circle equation.

Indeed,

$$f(a)^2 + f(b)^2 = f(a)f(a) + f(b)f(b) = f(a^2) + f(b^2) = f(a^2 + b^2) = f(1) = 1$$

Grothendieck's key idea

is that the 'solution' of a polynomial equation

$$p(x) = 0$$

is not the set of its roots in some privileged transcendental frame,

but rather the functorial 'application' that assigns

a different set of solutions to each possible transcendental frame A

in such a way that it takes into account the « parallax effects »

associated to transformations between different transcendental frames or perspectives.

Typically, problems of transcendental relativity have been addressed either by:

- Searching for invariants (***invariantism***);
- Canonizing or hypostasizing a given transcendental perspective (***provincialism***).

Grothendieck's innovative solution

to the problem of transcendental relativity

amounts to consider all the perspectives at once

by taking into account the parallax deformations

arising from transformations

among the different transcendental standpoints.

« […] le signe d'une intelligence chamanique de première ligne est la capacité à voir simultanément selon deux perspectives incompatibles. »

E. Viveiros de Castro, *Métaphysiques cannibales,* 2009, Paris: PUF.

The solution to the circle equation

can be described as a ***variable set*** V_{S^1}

that assignes a set of solutions $V_{S^1}(A)$

to each transcendental frame A

that co-varies (or varies « covariantly »)

when the transcendental frame is changed.

The Functor of Solutions

In category-theoretical terms,

such a variable set is given by a (covariant) ***functor***

$$V_{S^1} : Ring \to Set$$

from the category of rings *Ring*

to the category of sets *Set* such that

- $V_{S^1}(A)$ is the set of solutions of the circle equation in the ring A;

- $V_{S^1}(f) : V_{S^1}(A) \to V_{S^1}(B)$, where $f : A \to B$ a morphism of rings, is a function between the solutions in A and the solutions in B induced by the morphism of rings f.

The functor of solutions

associated to an algebraic expression

encodes all the corresponding solutions

in all possible transcendental frames

as well as the « parallax effects »

associated to transformations between transcendental frames.

By considering non-standard solutions

in well-chosen transcendental frames,

Grothendieck managed in describing (for instance)

differential/infinitesimal structures

associated to geometric figures.

Infinitesimals as Nilpotent Elements

For $a < 1$ small,

a^2 is even smaller $a << 1$.

Hence Grothendieck associates infinitesimals

with nilpotent non-zero elements $x \in A$ such that

$$x^2 = 0.$$

Solutions Given by Dual Numbers

Let's consider the $\mathbb{C}$-algebra

$$A = \mathbb{C}[\epsilon]/(\epsilon^2),$$

whose elements are polynomials of the form

$$p(\epsilon) = a + b\epsilon$$

subjected to the condition $\epsilon^2 = 0$.

Now, the pair $(1,\epsilon) \in A^2$ is a solution to the circle equation:

$$1^2 + \epsilon^2 = 1.$$

Generic Solution with the Tangent Vector

Let's consider a solution of the form $(x_0 + u\epsilon, y_0 + v\epsilon) \in A^2$.

In order for this to be a solution we have

$$(x_0 + u\epsilon)^2 + (y_0 + v\epsilon)^2 = 1$$

By expanding, we find the equations

$$x_0^2 + y_0^2 = 1$$

— which defines a point of the circle —

$$x_0 u + y_0 v = 0$$

—which defines a tangent vector perpendicular to the position vector —

The solution $(x_0 + u\epsilon, y_0 + v\epsilon) \in A^2$ can be rewritten as

$$(x_0, y_0) + \epsilon(u, v)$$

i.e., as a vector (x_0, y_0) whose head is a point in the real circle

plus a vector $\epsilon(u, v)$ of infinitesimal length

—- since $\|\epsilon(u, v)\| = \epsilon\sqrt{u^2 + v^2}$) —-

perpendicular to (x_0, y_0).

The tangent vector is represented by the nilpotent part of the coordinate.

For instance, the solution $(1,\epsilon) \in A^2$ represents a point in the circle

whose coordinates are (1,0) enriched with a tangent vector.

Solutions in a Finite Field

The finite field $\mathbb{F}_3 = \{0,1,2\}$ is defined by

$a = b \bmod 3$ if $a = 3n + b$ where $n \in \mathbb{N}$,

which implies that

$$4 = 1$$

Hence (2,0) and (0,2) are solutions of the circle equation.

Indeed,

$$2^2 + 0^2 = 4 = 1$$

Generalizations in the Evolution of the Notion of Quantity

- From the ***finite*** to the ***infinite***.
- From the ***discrete*** to the ***continuum***.
- From ***magnitudes*** to ***oriented magnitudes***.

All these generalizations remain within the realm of ***constant quantities***.

We shall now make the leap to ***variable quantities***.

Functions

« As Engels remarked in.the period when set theory and the arithmetization of analysis did not yet dominate mathematical thinking [*Anti-Dühring*, in the section on Quantity and Quality], the introduction of the advance from constant quantities to variable quantities is a mathematical expression of the advance from metaphysics to dialectics […] Every notion of constancy is relative, being derived perceptually or conceptually as a limiting case of variation. »

F.W. Lawvere, *Continuously Variable Sets; Algebraic Geometry = Geometric Logic*.

Functions as Variable Numbers

V*arying numbers*

of a certain kind B (e.g. $\mathbb{R}$ or $\mathbb{C}$)

that vary along a *domain* A

can be described by functions of the form

$$f : A \to B.$$

Evaluation of a function at a point

For each $a \in A$,

$f(a) \in B$

denotes the unique *value* in B

obtained by ***evaluating***

f at a.

Defining such a function $f : A \to B$ requires specifying

- The ***domain*** A over which the varying number varies.
- The kind of varying number B (aka as ***codomain***).
- The value $f(a) \in B$ for each $a \in A$.

These values can be defined

— but not necessarily —

by means of a ***functional expression*** $f(x)$,

like x^2 or $cos(x)$.

Graphs of Functions

A function $f(x) : \mathbb{R} \to \mathbb{R}$

defines a *graph* given by a curve in a Cartesian plane $(x, y) \in \mathbb{R}^2$

whose points have coordinates $(x, f(x))$

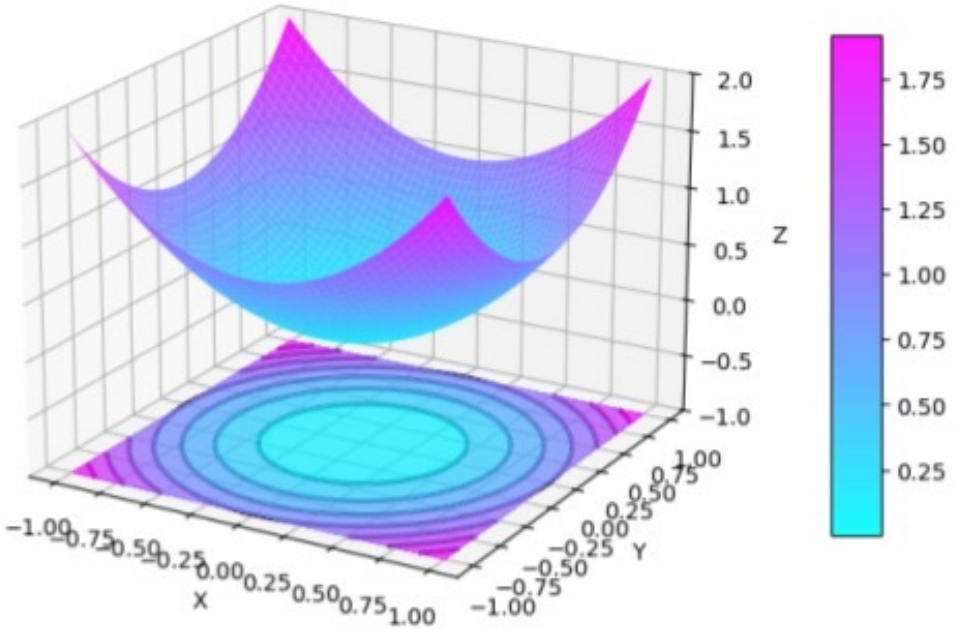
(but not any curve in $\mathbb{R}^2$ comes from a function; e.g. a circle).

A function $g(x, y) : \mathbb{R}^2 \to \mathbb{R}$

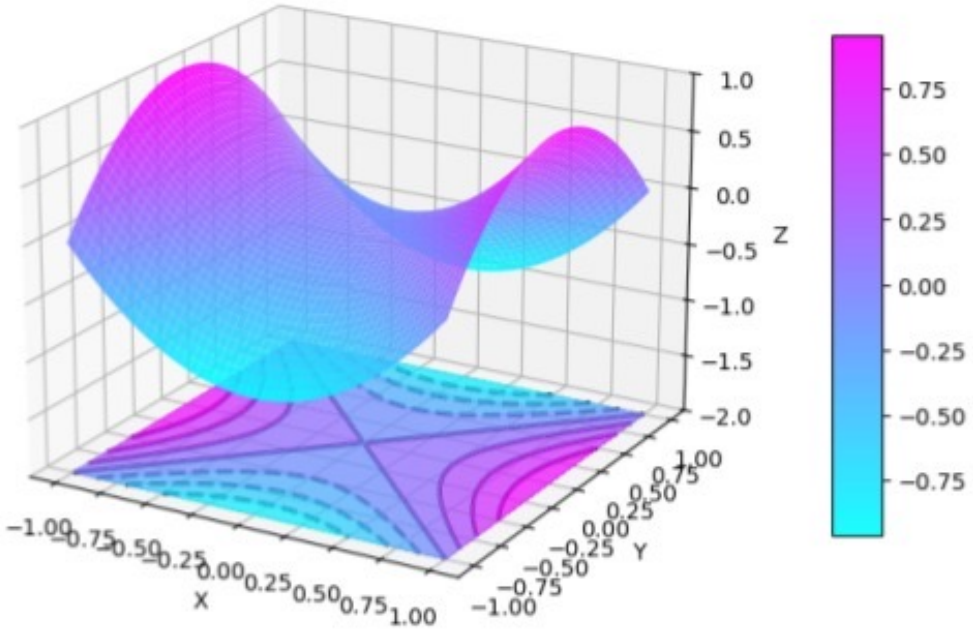
defines a graph given by *surface* in a Cartesian space $(x, y, z) \in \mathbb{R}^3$

whose points have coordinates $(x, y, g(x, y))$.

$f(x, y) = x^2 + y^2$



$f(x, y) = x^2 - y^2$



Terminology

In the expression

$$y = f(x) = a_2x^2 + a_1x + a_0$$

- x is called ***independent variable***.

- y is called ***dependent variable***.

- a_0, a_1, a_2 are called ***parameters***, where different values of the parameters define different functions.

The domain of variation A of a function f can be

- a set of numbers like $\mathbb{R}$ or $\mathbb{C}$.
- a Cartesian product of the form $\mathbb{R}^n$ or $\mathbb{C}^n$.
- a *manifold*, that is, a geometric object like a curve, a sphere or a torus.

Brief history of the notion of function

- Leibniz introduced the term *function* (letter of 1673).

- Bernoulli (s. XVI-XVII) and Euler (1748): Functions are given by *functional expressions* involving variables and constants .

- Fourier (*Théorie analytique de la chaleur*, 1822): generalized concept of function which are neither continuous nor given by functional expressions.

- Dirichlet (1837): functions are generalized to single-valued mappings from a domain to a codomain, i.e. as relations in which every first element has a unique second element.

Polynomial Functions

- Constant functions (aka as numbers):

$$f(x) = a_0$$

- Linear functions:

$$f(x) = a_0 + a_1 x$$

- Quadratic functions:

$$f(x) = a_0 + a_1 x + a_2 x^2$$

- Polynomials of degree n:

$$f(x) = a_0 + a_1 x + \ldots + a_n x^n$$

Exponential Functions with Natural and Entire Exponents

$$f_b(n) = b^n = b \times \ldots \times b \text{ with } n \text{ factors, } n \in \mathbb{N} - \{0\}$$

this definition satisfies $b^{n+m} = b^n b^m$.

This means that the exponential function f_b

maps additions to multiplications:

$$f_b(n+m) = f_b(n) f_b(m).$$

This property implies

$$b^0 = 1 \text{ and } b^{-n} = \frac{1}{b^n},$$

which allows to extend f_b to entire arguments $n \in \mathbb{Z}$.

Exponential Functions with Rational and Real Exponents

In turn, $(b^n)^m = b^{nm}$,

from which we can define

$$b^{\frac{n}{m}} = (b^{\frac{1}{m}})^n$$

which extends the notion of exponential function

to rational exponents $\frac{n}{m} \in \mathbb{Q}$.

For $x \in \mathbb{R}$,

$$b^x = lim_{n\to\infty} b^{q_n}$$

where $\{q_n\}$ is a monotone increasing sequence

of rational numbers converging to x.

In other terms, we define b^x as the value towards which b^{q_n} infinitely approaches

as the rationals q_n infinitely approach x.

Derivatives

Partial Summary

We have thus far considered two kinds of quantities:

♣ ***Constant quantities*** aka ***numbers***, with respect to which no notion of evaluation is involved.

♥ ***Varying quantities*** aka ***functions***, which can be evaluated on 0-dimensional extensions aka points.

We shall now focus on the notion of *variation* itself.

How does a function $f(x) : A \to B$ ***vary***

when its argument x ***moves across its domain*** A***?***

As we shall see, the *rate of variations* of a function $f(x)$

is encoded by another function, the ***derivative*** *of* $f(x)$.

So, it seems that the notion of *rate of variation* of a function

can still be accommodated

within the range of kinds of quantities we know

(numbers and functions).

Nonetheless, we shall see that the notion of derivative

Leads us to introduce a new kind of quantity

after *numbers* and *functions*,

namely ***differentials***.

Differently from numbers (no evaluation involved)

and functions (which can be evaluated on points),

differentials can be « evaluated »

on 1-dimensional objects,

namely intervals $[a, b]$ in $\mathbb{R}$

or, more generally, curves.

This generalized notion of evaluation is called ***integration.***

Given a function $s(t)$ that describes

the position of a moving object

at different instants of time t,

the rate of change of $s(t)$ with respect to time

defines a physical quantity known as ***velocity***,

which measures the distance traveled per unit of time.

The velocity $v(t)$

— as for instance, the velocity indicated

by the speedometer of a car —

is a function

that specifies the ***velocity*** of the object at each t.

This functional description of the velocity

presupposes that the velocity is a quantity

that can be defined at each temporal instant.

In other terms, the functional description of the velocity

seems to presuppose the notion of ***instantaneous velocity***.

The Paradoxical Notion of Instantaneous Velocity

The notion of an

instantaneous

— no lapse of time —

velocity

— change through a finite lapse of time —

seems to be an oxymoron.

Motion needs time to happen.

Zeno's (Third) Paradox

« The flying arrow is at rest. »

Aristotle, *Physics*, 239b30-32.

If time is composed of thicknessless instants,

then no motion can occur *within* an instant,

which means, in particular, that the very notion of instantaneous velocity

is self-contradictory,

and that motion itself becomes impossible.

« […] nothing is moved in a now […] »

Aristotle, *Physics*, 234-22.

If time is composed of nows, nothing moves.

Aristotle's Answer to Zeno

« The third [paradox] is that already given above, to the effect that the flying arrow is at rest, which result follows from the assumption that time is composed of moments: if this assumption is not granted, the conclusion will not follow. »

Aristotle, *Physics*, 239b30-32

« Zeno's reasoning, however, is fallacious, when he says that if everything when it occupies an equal space is at rest, and if that which is in locomotion is always in a now, the flying arrow is therefore motionless. This is false; for time is not composed of indivisible nows any more than any other magnitude is composed of indivisibles. »

Aristotle, *Physics*, 239b5-9

L'aube éléatique de la métaphysique

« La métaphysique date du jour où Zénon d'Élée signala les contradictions inhérentes au mouvement et au changement, tels que se les représente notre intelligence. À surmonter, à tourner par un travail intellectuel de plus en plus subtil ces difficultés soulevées par la représentation intellectuelle du mouvement et du changement s'employa le principal effort des philosophes anciens et modernes. C'est ainsi que la métaphysique fut conduite à chercher la réalité des choses au-dessus du temps, par-delà ce qui se meut et ce qui change, en dehors, par conséquent, de ce que nos sens et notre conscience perçoivent. »

H. Bergson, *La pensée et le mouvement*.

Infinitesimals

The immediate solution

to circumvent the paradox of instantaneous velocities

is to reintroduce a finite lapse of time Δt,

which leads to the notion of ***mean velocity***.

Mean velocities

Given an object whose motion is described by a function $s(t)$,

we can compute the *mean* or *average velocity* between any two instants t and $t + \Delta t$:

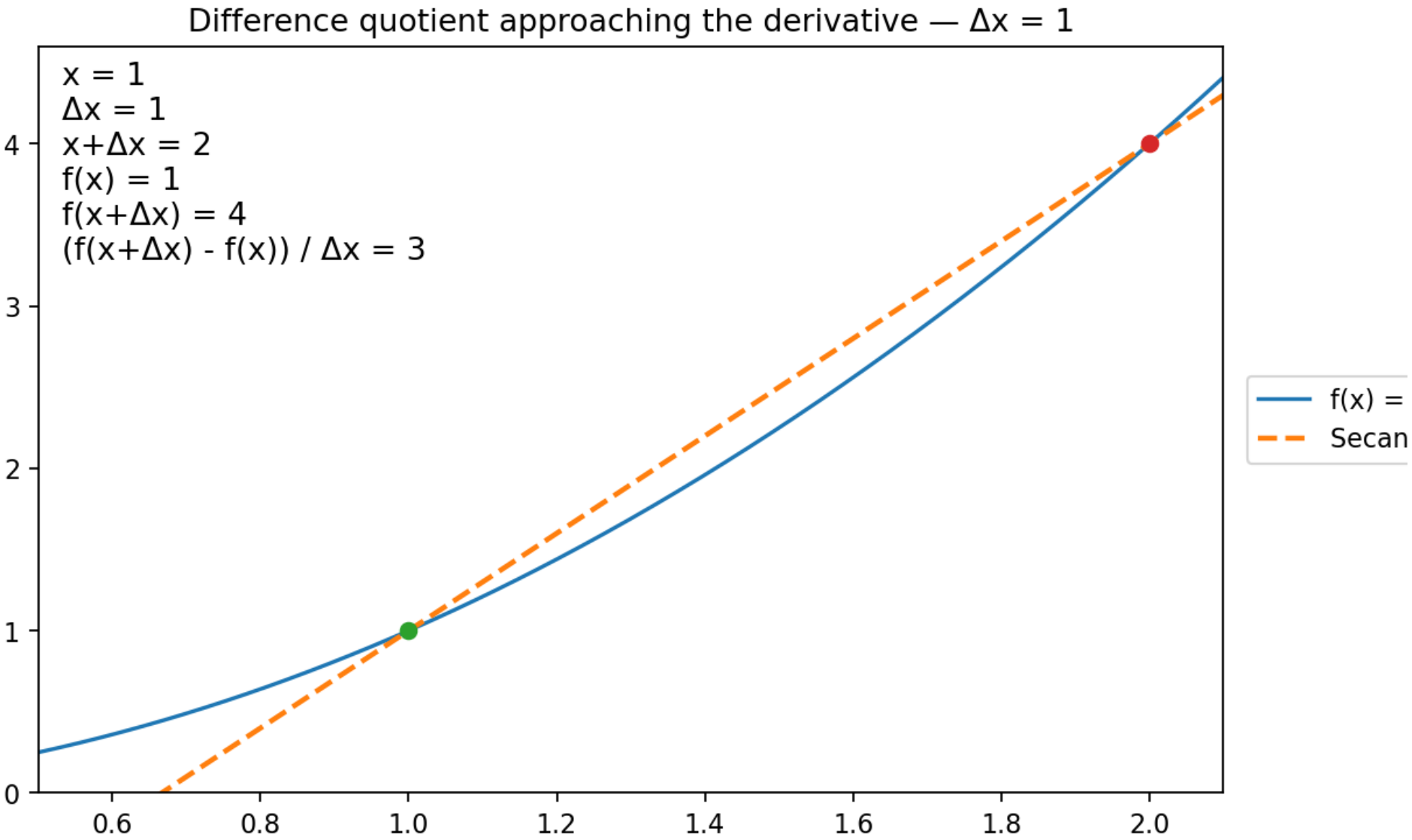
$$\langle v(t) \rangle_{\Delta t} = \frac{\Delta s}{\Delta t} = \frac{s(t + \Delta t) - s(t)}{\Delta t}$$

Graphically, the number $\frac{\Delta s}{\Delta t}$ is the *slope*

of the straight line (called *secant*)

that passes trough the events $(t, s(t))$ and $(t + \Delta t, s(t + \Delta t))$

in the graph $s(t)$ vs. t.


Difference quotient approaching the derivative — Δx = 1
x = 1
Δx = 1
x+Δx = 2
f(x) = 1
f(x+Δx) = 4
(f(x+Δx) - f(x)) / Δx = 3
0
1
2
3
4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
f(x) =
Secan

Limitation of the notion of *mean velocity*

The actual « instantaneous velocity » might change

between t and $t + \Delta t$.

The mean velocity $\langle v \rangle_{\Delta t}$ is only a mean, an average.

The Dilemma leading to the Notion of Infinitesimal

We can try to overcome this limitation

by taking a very small Δt.

But here we are confronted to a dilemma:

If Δt actually becomes zero, the expression $\dfrac{x(t+\Delta t)-x(t)}{\Delta t}$

would take the indeterminate form $\dfrac{0}{0}$

($\dfrac{0}{0} = a$ if and only if $0 = a \times 0$ with is satisfied by any a)

If Δt remains a finite difference,

we come back to the definition of the notion of mean velocity

and we cannot guarantee that the velocity does not change within Δt.

Leibniz's solution: Infinitesimals

In order to sublate this dilemma,

Leibniz introduces a new notion of number

— noted dt and called ***infinitesimal*** —

defined by the property of being bigger than zero

but smaller than any finite real number r:

$$0 < dt < r$$

Note that no real number satisfies this property,

since no matter how small a real number r is,

there always exist smaller real numbers $r' < r$.

Leibniz's Definition of Derivatives

If we hold space for this new notion of number,

then the instantaneous velocity

associated to a function $s(t)$ at t can be defined

as a quotient between infinitesimals:

$$\frac{ds}{dt}$$

where

$$ds \doteq s(t + dt) - s(t)$$

is the infinitesimal distance traversed

in the infinitesimal temporal lapse dt.

An ***infinitesimal*** is a hypothetical

new extension of the notion of number

that is so small that nothing « real » can be smaller,

an « ultimate difference » in Russell's terms

(cf. B. Russell, *Principles of Mathematics*).

As with the previous extensions of the notion of number,

the ontological status of *infinitesimals*

were the object of a heated debate.

“[…] one has never been able to form more than an imperfect idea of these elements — a kind of singular beings which at times play the role of true quantities, at times must be treated as absolutely null, and which, through their equivocal properties, seem to occupy a middle ground between magnitude and zero, between existence and nothingness.”

L. Carnot, *Réflexions sur la métaphysique du calcul infinitésimal*, 1797.

Infinitesimals as an intermediate state between *being* and *nothing* in Hegel

« These magnitudes [infinitesimals] are so determined that they are in their vanishing – not before this vanishing, for they would then be finite magnitudes; not after it, for then they would be nothing. Against this pure thought, it is objected and endlessly repeated that these magnitudes are either something or nothing; that there is no intermediary state between being and nothing ("state" is here an inappropriate, barbaric expression). Assumed here is again the absolute separation of being and nothing. But we have shown against this that being and nothing are in fact the same, or to speak in the language cited, that there is nothing which is not an intermediary state between being and nothing. Mathematics owes its most brilliant successes to precisely that determination which the understanding rejects. »

G.W.F. Hegel, *The Science of Logic*, §21.92.

« Das Differential verbindet in sich die Eigenschaften des Nichts und des Etwas […] Es ist in einem die Größe, wie sie ins Größenlose verfließt […] »

[« The differential combines in itself the qualities of nothing and something [...] It is […] the magnitude as it flows into that which has no magnitude […] »]

F. Rosenzweig, *Der Stern des Erlosung*, p.29.

What are these infinitesimals

that are smaller than anything « real »

without being nothing,

these evanescent quantities

that live on the very verge of nothingness

without ever falling into it?

Infinitesimals as « atomic elements » of the *continuum*.

Infinitesimals seem to provide the notion

of a minimal lapse of time

so that motion can occur within it

and therefore they seem to be an appropriate substitute

to the timeless notion of instant

as the atomic element of the *continuum*.

« [Infinitesimals are] the minimal differential element which would seem to be inscribed as such

in the fabric of continuity. »

A. Badiou, *Infinitesimal Subversion*.

Anaxagoras against Infinitesimals

« Among the small there is no smallest, but always something smaller. For what is cannot cease to be no matter how far it is being subdivided. »

Anaxagoras (around c. 550 BCE).

Simplicius of Cilicia, *Commentary on Aristotle's Physics*, B3.

Leibniz on Infinitesimals as « Mental Fictions »

« Speaking philosophically, I no more support infinitely small magnitudes than infinitely large ones, or no more infinitesimals than infinituples. For I consider both to be fictions of the mind, due to abbreviated ways of speaking, which are suitable for calculation, in the way that imaginary roots in algebra are. »

G.W. Leibniz, *Letter to Des Bosses*, 11 March 1706.

Berkeley on Infinitesimals

« And what are these Fluxions? The Velocities of evanescent Increments? And what are these same evanescent Increments? They are neither finite Quantities nor Quantities infinitely small, nor yet nothing. May we not call them the ghosts of departed quantities? »

G. Berkeley, *The Analyst*, 1734.

From a historical perspective,

the notion of infinitesimal was considered to be

a metaphysical notion that « infected »

the foundations of calculus

(Cantor called them an “infinitary cholera-bacilli”).

In the XIXth century, Cauchy and Weiestrass grounded calculus

on rigorous foundations in which infinitesimals play no role

and which were based on a new notion,

the notion of ***limit*** (Bolzano, Cauchy).

« The gesture of rejection is constitutive: the impurity of the origin of differential calculus was the isolated marking, the trace of the infinitely small. Thus the story of this calculus is also that of the effacement of this trace. […] the history of mathematical analysis is partly entwined with another history, incessantly counteracting it: that of the repression [*refoulement*] of infinitesimals. »

A. Badiou, *Infinitesimal subversion*.

Contemporary Reinvindactions of Infinitesimals or Differentials

- **Robinson's « non-standard analysis »**.

- **Grothendieck's *theory of schemes*** in algebraic geometry, where infinitesimals are the geometric counterpart of the nilpotent elements of a commutative ring A.

For $a < 1$ small, a^2 is even smaller $a^2 << 0$.

Hence Grothendieck associates infinitesimals

with nilpotent non-zero elements $x \in A$ such that $x^2 = 0$.

- **E. Cartan's theory of differential n-forms** in differential geometry.

Rather than being evanescent quantities on the verge of nothingness (infinitesimals),

differentials are a net form of quantities that can be used to detect *holes*,

that is, localized forms of nothingness

(De Rham cohomology).

Carnot on Differentials

Differentials are useful « auxiliary quantities »

that ultimately disappear in the final result

though a process of « compensations of errors ».

L. Carnot, *Réflexions sur la métaphysique du calcul infinitésimal,* Paris, 1797.

Post-Kantian Takes on Infinitesimals

- **S. Maimon** (1753-1800), *Essay on Transcendental Philosophy* (1790). Leibnizian extension of Kant's constitution of objective experience: the very sensible/empirical matter of appearances is algo generated by a spontaneous synthesis ruled by differential calculus.

- **J.H. Wronski** (1776-1893), *Philosophie de l'infini* (1814), *Philosophie de la technie algorithmique* (1815). Eidetic nature of differentials: whereas *understanding* is the faculty of the *finite/discontinuous*, *reason* is the faculty of the *infinite/continuous* (differentials are « ideal differences »).

- **H. Cohen** (1842-1918), *The Principle of the Infinitesimal Method and its History* (1883); Second edition of *Kant's Theory of Experience* (1885); *Logic of Pure Knowledge* (1902). Infinitesimals dx are understood as intensive magnitudes (« fluxiones ») that generate the continuous extensive ones (« fluents »).

Derivatives as limits

The key idea underlying the notion of a *limit*

is the distinction between the *value* $f(x_0)$ *of a function* f *at a given point* x_0

and the *value that the function* f *approaches as* x *approaches* x_0

which is denoted

$$lim_{x \to x_0} f(x).$$

The function is said to be ***continuous*** at x_0 when

$$lim_{x \to x_0} f(x) = f(x_0).$$

Now, the function f might be discontinuous at x_0 or not even defined at x_0.

(ϵ, δ)-definition of limits (Bolzano, Cauchy)

The limit of a function $f(x)$ when tends x to x_0 is the number L

— noted $lim_{x \to x_0} f(x) = L$ —

if for any « error » $\epsilon > 0$ it is possible to find a number $\delta > 0$

such that $0 < |L - f(x)| < \epsilon$ for $0 < |x - x_0| < \delta$,

which means that the difference between $f(x)$ and L

can be made smaller than any « error » ϵ whatsoever

by fine-tunning the control variable δ,

i.e. by making x sufficiently close to x_0.

Given two functions f and g,

the notion of limit satisfies the following properties:

- $\lim_{x \to x_0} \big(f(x) + g(x)\big) = \lim_{x \to x_0} f(x) + \lim_{x \to x_0} g(x)$

- $\lim_{x \to x_0} \big(f(x) \cdot g(x)\big) = \lim_{x \to x_0} f(x) \cdot \lim_{x \to x_0} g(x)$

- $\lim_{x \to x_0} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)}$ **provided that** $\lim_{x \to x_0} g(x) \neq 0$

The key point in the definition of the derivative

is that limits of quotients cannot be simplified

into quotients of limits when the denominator tends to zero.

In other words,

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} \neq \frac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)}$$

in the indeterminate case $\frac{0}{0}$,

which is precisely the relevant case in the definition of a derivative.

Examples

$$\lim_{x \to 0} \frac{x^4}{x^2} = 0$$

— where the numerator x^4 goes to 0 faster than the denominator x^2 so that the whole quotient tends to 0 —

$$\lim_{x \to 0} \frac{x^2}{x^4} = +\infty$$

— where now the denominator x^4 goes to 0 faster than the numerator x^2 so that the whole quotient tends to 0 —

Even if the evaluation of the two functions at 0

yields the same indeterminate form $\frac{0}{0}$,

the two expressions have different limits.

Example

$$f(x) = \frac{x^2 - 1}{x - 1}$$

The direct evaluation of $f(x)$ at $x = 1$ yields

$$f(1) = \frac{1^2 - 1}{1 - 1} = \frac{0}{0}$$

But the limit of $f(x)$ when $x \to 1$ is

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} \frac{(x + 1)(x - 1)}{x - 1} = \lim_{x \to 1} x + 1 = 2$$

Definition of Derivatives via Limits

$$\frac{df}{dx}(x) \doteq lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon} .$$

According to this definition,

a derivative should not be understood as a quotient

(as Leibniz's notation suggests),

but rather as the *limit of a quotient*

(as Newton had already understood).

Notation

$\frac{df}{dx}(x)$ (Leibniz's notation) $\equiv \dot{f}(x)$ (Newton's notation) $\equiv f'(x)$ (usual notation)

The instantaneous velocity as the slope of the tangent to the graph

Whereas the « mean rate of change »

$$\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

is the *slope* of the straight line (called ***secant***)

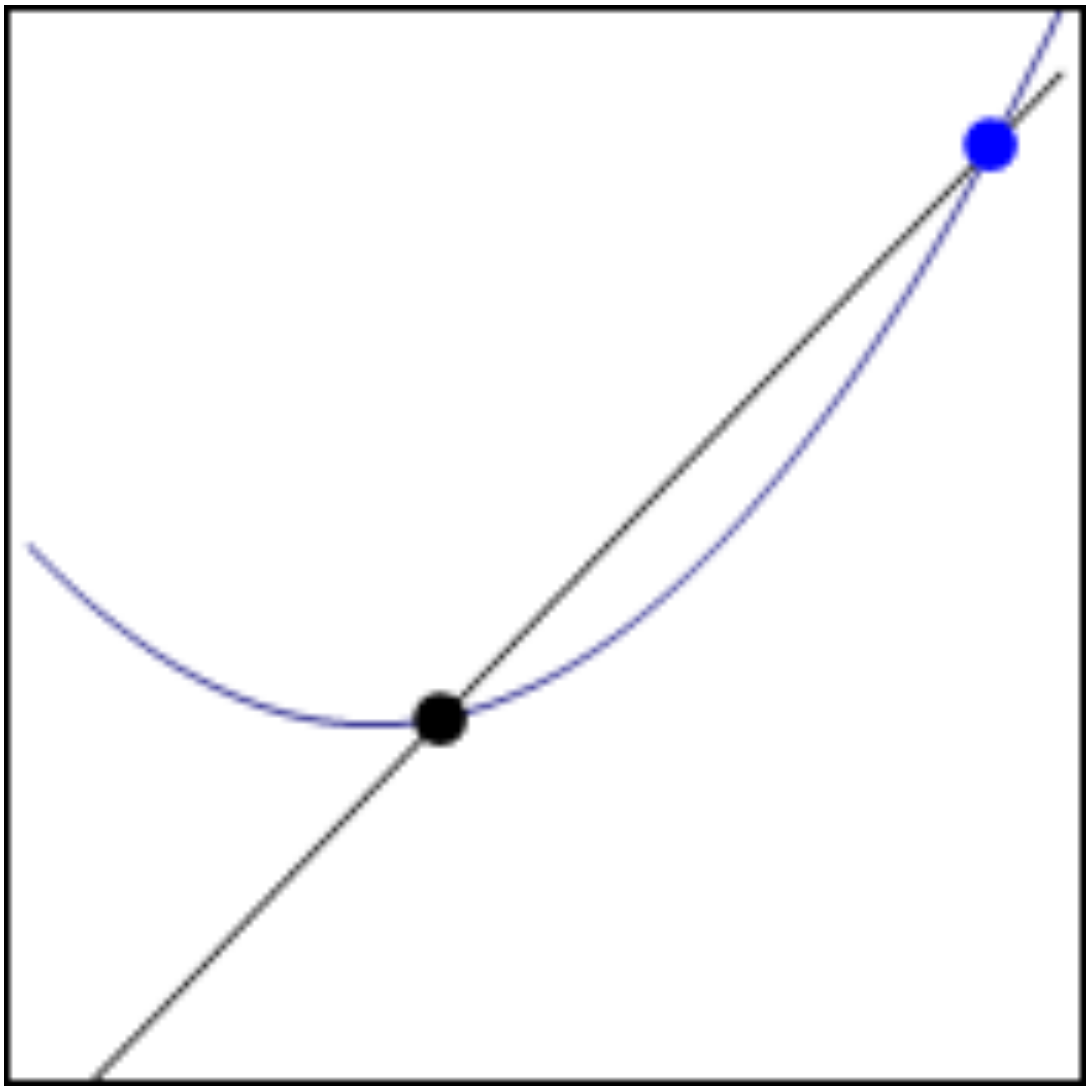
that passes trough the events $(x, f(x))$ and $(x + \Delta x, f(x + \Delta x))$ in the graph $f(x)$ vs. x,

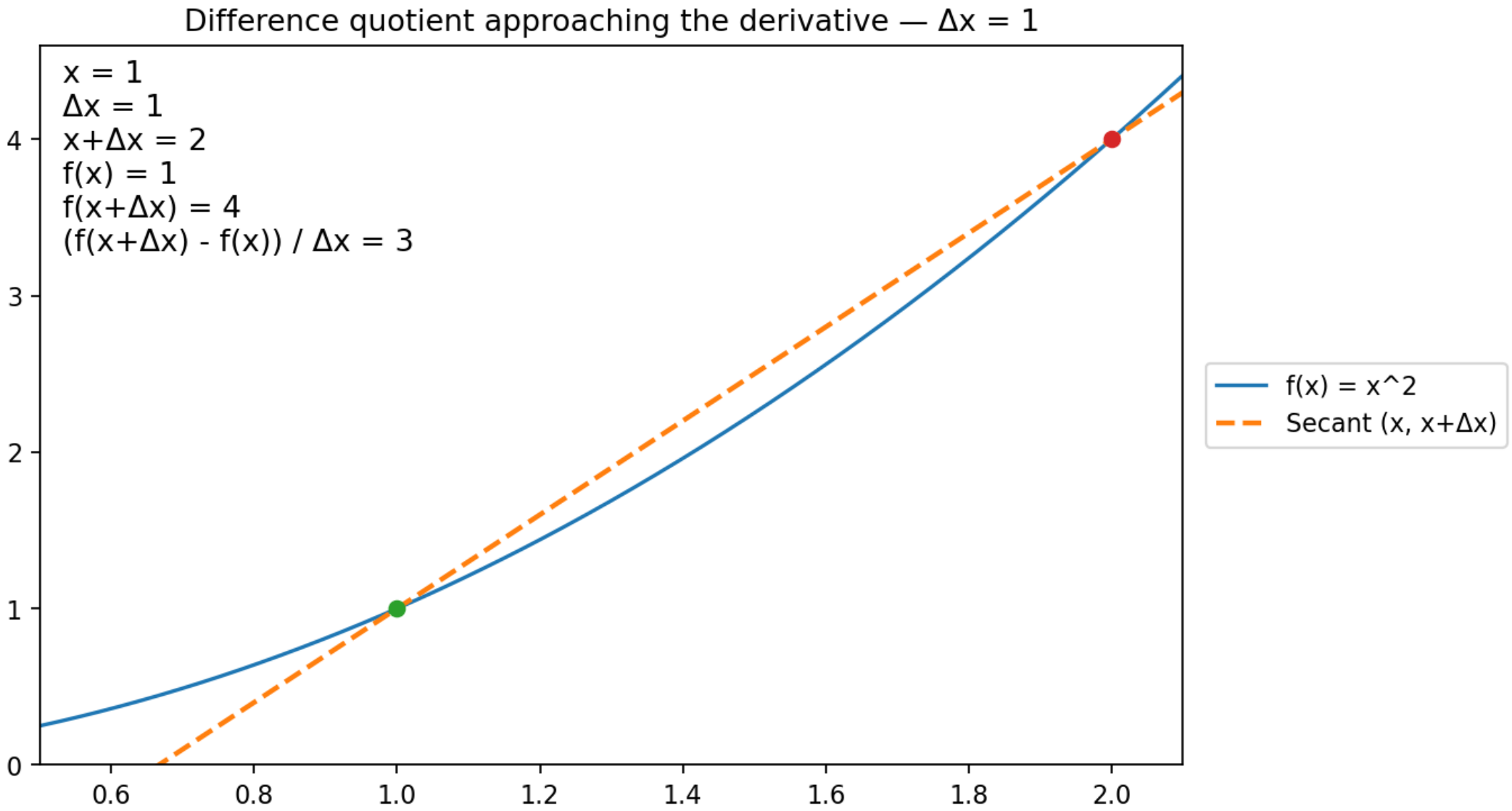
the limit

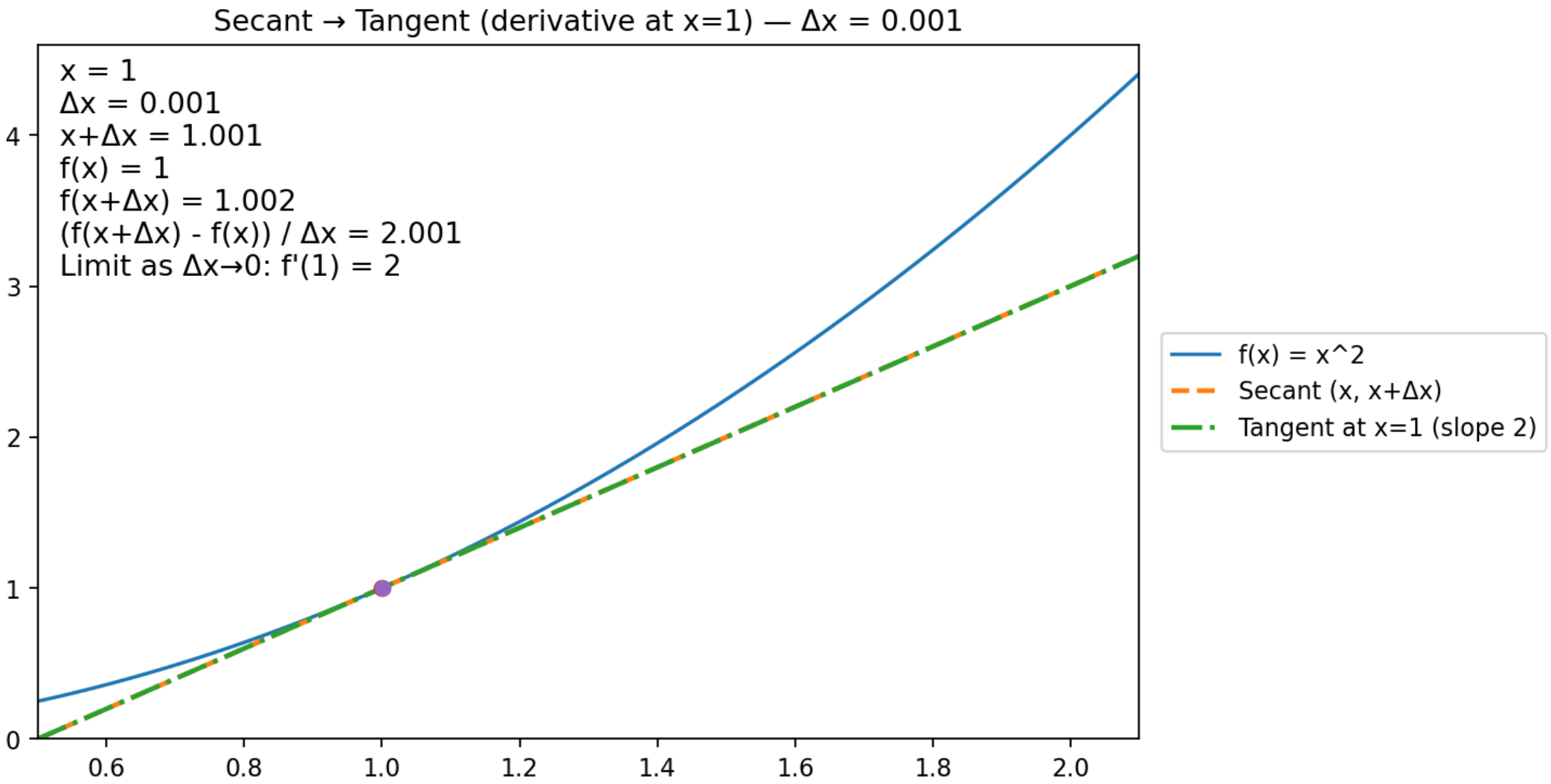
$$\frac{df}{dx}(x) \doteq lim_{\epsilon \to 0} \frac{f(x + \epsilon) - f(x)}{\epsilon}$$

is the slope of the ***tangent*** to the curve at x.

The slope of the tangent line is the tangent of the angle it forms with the positive x-axis.




Difference quotient approaching the derivative — Δx = 1
x = 1
Δx = 1
x+Δx = 2
f(x) = 1
f(x+Δx) = 4
(f(x+Δx) - f(x)) / Δx = 3
f(x) = x^2
Secant (x, x+Δx)
0
1
2
3
4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0


Secant → Tangent (derivative at x=1) — Δx = 0.001
x = 1
Δx = 0.001
x+Δx = 1.001
f(x) = 1
f(x+Δx) = 1.002
(f(x+Δx) - f(x)) / Δx = 2.001
Limit as Δx→0: f'(1) = 2
f(x) = x^2
Secant (x, x+Δx)
Tangent at x=1 (slope 2)
0
1
2
3
4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0

The expression

$$\frac{df}{dx}(x) \doteq lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}$$

is a function that sends each x

to the slope of the graph of $f(x)$ at each x.

Newton's proton-definition of limits.

« Perhaps it may be objected, that there is no ultimate proportion, of evanescent quantities ; because the proportion, before the quantities have vanished, is not the ultimate, and when they are vanished, is none. [...] by the ultimate ratio of evanescent quantities is to be understood the ratio of the quantities not before they vanish, nor afterwards, but with which they vanish. [...] For those ultimate ratios with which quantities vanish are [...] limits towards which the ratios of quantities decreasing without limit do always converge ; and to which they approach nearer than by any given difference [...] till the quantities are diminished *in infinitum*. [...] Therefore if in what follows [...], I should happen to mention quantities as least, or evanescent, or ultimate, you are not to suppose that quantities of any determinate magnitude are meant, but such as are conceived to be always diminished without end. »

I. Newton, *Mathematical Principles of Natural Philosophy and his System of The World,* Book 1. The Motion of Bodies, Section I, 1687.

D'Alembert against infinitesimals

« It is not a question, as is still commonly said, of infinitely small quantities in differential *calculus*: it is solely a question of limits of finite quantities. Thus the metaphysics […] of infinitely small quantities, some larger or smaller than others, is completely unnecessary for *differential* calculus. The term *infinitely small* is used only to abbreviate expressions. »

[« Il ne s'agit point, comme on le dit encore ordinairement, de quantités infiniment petites dans le calcul *différentiel*: il s'agit uniquement de limites de quantités finies. Ainsi la métaphysique […] des quantités infiniment petites plus grandes ou plus petites les unes que les autres est totalement inutile au calcul *différentiel*. On ne se sert du terme d'*infiniment petit*, que pour abréger les expressions. »]

D'Alembert, *Calcul Différentiel*, *Encyclopédie, ou, Dictionnaire raisonné des sciences, des arts et des métiers*, 1754.

Russell against Infinitesimals

« There are in the Calculus no such magnitudes as dx and dy. There are finite differences Δx and Δy, but no view, however elementary, will make x equal to $x + \Delta x$. There are ratios of finite differences, $\Delta y/\Delta x$, and in cases where the derivative of y exists, there is one real number to which $\Delta y/\Delta x$ can be made to approach as near as we like by diminishing Δx and Δy. This single real number we choose to denote by dy/dx; but it is not a fraction, and dx and dy are nothing but typographical parts of one symbol. [...] infinitesimals as explaining continuity must be regarded as unnecessary, erroneous and self-contradictory. »

B. Russell, *Principles of Mathematics*, 1903, §350, p.347.

Properties of Derivation

$$\frac{d(c)}{dx} = 0$$

$$\frac{d(f+g)}{dx} = \frac{df}{dx} + \frac{dg}{dx}$$

$$\frac{d(fg)}{dx} = f\frac{dg}{dx} + g\frac{df}{dx} \text{ (Leibniz rule)}$$

$$\frac{d(cf)}{dx} = c\frac{df}{dx}$$

Derivatives of Elementary Functions

$$\frac{d(x^n)}{dx} = nx^{n-1}$$

$$\frac{d(e^{kx})}{dx} = ke^x$$

(in particular $\frac{d(e^x)}{dx} = e^x$!!!)

$$\frac{d(sin(x))}{dx} = cos(x)$$

$$\frac{d(cos(x))}{dx} = -sin(x)$$

Derivatives of Compositions (Chain Rule)

Given two functions $f(x)$ and $g(x)$,

we can apply g to x and then apply f to $g(x)$.

In other terms, we can define the composition

$$(f \circ g)(x) : x \mapsto f(g(x)).$$

By manipulating nonchalantly and irresponsibly Leibniz' notation,

we can compute the derivative of $(f \circ g)(x)$:

$$\frac{df(g(x))}{dx} = \frac{df(g)}{1}\frac{dg}{dg}\frac{1}{dx} = \frac{df}{dg}\frac{dg}{dx}$$

Derivative of a quotient

(using Leibniz rule and the chain rule)

$$\left(\frac{f(x)}{g(x)}\right)' = (fg^{-1})' = f'g^{-1} + f(g^{-1})' = f'g^{-1} + f(-1g^{-2}g') = \frac{f'g - fg'}{g^2}$$

Higher Derivatives

Let's consider an unidimensional motion

described by the expression

$$s(t) = t^2$$

And let's compute its position and velocity at $t = 0$:

$$s(0) = 0,$$

$$s'(t) = 2t \Rightarrow s'(0) = 0$$

Now, if the position and the velocity at $t = 0$ are zero,

why the car is somewhere else an instant later?

Higher Derivatives

Given a function $f(x)$,

its derivative $\frac{df}{dx}$ is itself a function

such that the value $\frac{df}{dx}(x)$ at a given x

is the slope of the tangent line to $f(x)$ at x.

Since $\frac{df}{dx}(x)$ is itself a function, we can compute its own derivative:

$$\frac{d(\frac{df}{dx})}{dx} \equiv \frac{d^2f}{dx^2}$$

which is called the *2nd-order derivative of* $f(x)$.

For instance, given a motion $s(t)$,

the ***instantaneous velocities***

— i.e. the instantaneous rates of change of the position $s(t)$ —

are given by the derivative function:

$$v(t) = \frac{ds(t)}{dt}$$

In turn, the ***instantaneous accelerations***

— the instantaneous rates of change of the velocities $v(t)$,

i.e. the instantaneous rates of change

of the the instantaneous rates of change of position $s(t)$—

are given by the second derivative function of the positions $s(t)$:

$$a(t) = \frac{dv(t)}{dt} = \frac{d^2s(t)}{dt^2}.$$

Analysis of functions (I)

If $f'(x_0) > 0$

the function f is ***increasing*** at x_0,

i.e. $f(x_0) < f(x_1)$ if $x_0 < x_1$.

If $f'(x_0) < 0$

the function f is ***decreasing*** at x_0,

i.e. $f(x_0) > f(x_1)$ if $x_0 < x_1$.

Analysis of functions (II): Maxima and Minima

- The points x_0 such that $f'(x_0) = 0$ are called ***critical values*** of f.

- In order to determine the nature of the critical x_0 (a *maximum* or a *minimum)*, we have to look at $f''(x_0)$:

  .If $f''(x_0) > 0$, the derivative $f'(x)$ is increasing, i.e. f passes from negative to positive slopes at x_0, which means that x_0 is a ***minimum*** (concave upwards).

  Eg.: $f(x) = x^2$ at $x_0 = 0$.

  .If $f''(x_0) < 0$, the derivative $f'(x)$ is decreasing, i.e. f passes from positive to negative slopes at x_0, which means that x_0 is a ***maximum*** (concave downwards).

  Eg.: $f(x) = -x^2$ at $x_0 = 0$.

Analysis of functions (III): Inflection Points

If $f''(x_0) = 0$ and changes sign at $x_0 = 0$,

x_0 is an ***inflection point***

i.e. a point at which the concavity of the curve changes sign.

Examples:

$f(x) = x^3$ at $x_0 = 0$ (concave downwards to the left of $x_0 = 0$ and concave upwards to the right);

$f(x) = x^3 - x$.

Analysis of functions (IV): Summary

- The sign of $f'(x)$ is associated to the increasing/decreasing behavior of $f(x)$:

  $f'(x_0) > 0$: increasing at x_0.
  $f'(x_0) < 0$: decreasing at x_0.
  $f'(x_0) = 0$: critical point, neither increasing not decreasing.

- The sign of $f''(x)$ is associated to the concavity of $f(x)$ at x:

  $f''(x_0) > 0$: concavity upwards.
  $f''(x_0) < 0$: concavity downwards.
  $f''(x_0) = 0$ and changes sign at x_0: inflection point.

Analysis of functions (V): Exercises

Determine the nature of $x_0 = 0$ for the function

$$f(x) = x^4.$$

Analyze the critical points of the function

$$f(x) = \frac{1}{3}x^3 - x^2 - 3x + 1.$$

Continuity and Differentiability Classes

- A function $f(x)$ is ***continuous*** (there are no gaps or jumps of its values) if

$$lim_{x \to x_0} f(x) = f(x_0)$$

(a continuous function might not be differentiable, like $f(x) = |x|$, which is continuous but not differentiable at $x = 0$).

- A function $f(x)$ is ***differentiable*** if $f'(x)$ exists at every point x in its domain.

- A function $f(x)$ is ***continuously differentiable*** if $f'(x)$ exists at every x and is continuous.

- A function $f(x)$ is of ***differentiable class*** C^k if $f'(x), f''(x), \ldots, f^k(k)$ exist at every x and are continuous (but $f^k(k)$ is not necessarily differentiable).

Towards the Climax of Smoothness

- A C^0-function is a continuous function.

- A C^1-function is a continuous function such that $f'(x)$ exists and is continuous (e.g. $f(x) = x^{4/3}$, where $f'(x)$ exists and is continuous but not differentiable).

- A C^∞-function embodies the climax of smoothness.

Partial Derivatives

Generalizing the Notion of Derivative to Higher Dimensions

A function on two variables (x, y)

$$f(x, y) : \mathbb{R}^2 \to \mathbb{R}$$

defines a surface in $\mathbb{R}^3$.

The rate of change of $f(x, y)$ at a point (x_0, y_0)

depends on the direction in $\mathbb{R}^2$ that we want to consider

(e.g. a saddle $f(x, y) = x^2 - y^2$).

Hence, the notion of « the derivative of $f(x, y)$ at a point (x_0, y_0) »

remains indeterminate, i.e. further information (a direction) has to be provided.

But we can introduce the notion of « the derivative of $f(x, y)$ at a point (x_0, y_0)

along a certain direction in $\mathbb{R}^2$ ».

Partial Derivatives

For instance, the derivative of $f(x, y)$ at a point (x_0, y_0)

along the direction defined by the x-axis

— noted $\dfrac{\partial f(x, y)}{\partial x}(x_0, y_0)$ —

can be computed by considering $f(x, y)$ as a family of functions $f_y(x)$

on one variable x depending on a parameter y.

We can then consider the standard derivative with respect to x

of the function $f_{y_0}(x)$ in this family:

$$\frac{\partial f(x, y)}{\partial x}(x_0, y_0) = \frac{df_{y_0}(x)}{dx}(x_0)$$

Example of Partial Derivatives

For

$$f(x, y) = x^2 + xy + y^3,$$

we have

$$\frac{\partial f(x, y)}{\partial x} = \frac{df_y(x)}{dx} = 2x + y$$

and

$$\frac{\partial f(x, y)}{\partial y} = \frac{df_x(y)}{dy} = x + 3y^2$$

Exercise: evaluate the nature of the critical point $(x, y) = (0,0)$ of the function $f(x, y) = x^2 - y^2$.

Gradient Vector Field

The partial derivatives

define a vector field on $\mathbb{R}^2$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

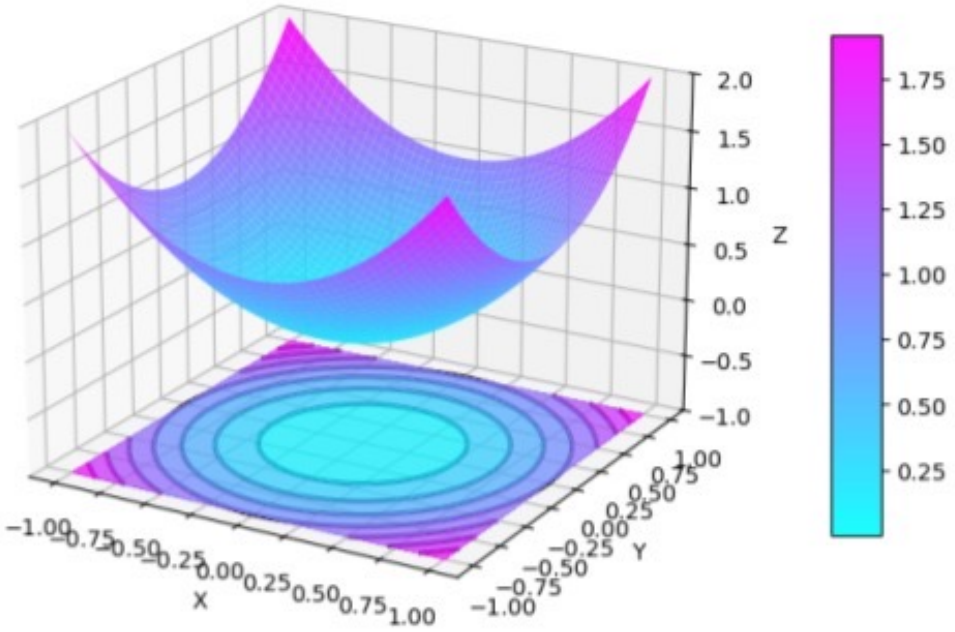
that is always perpendicular to the ***level lines*** of $f(x, y)$

— lines in the plane (x, y) with a constant value of f —

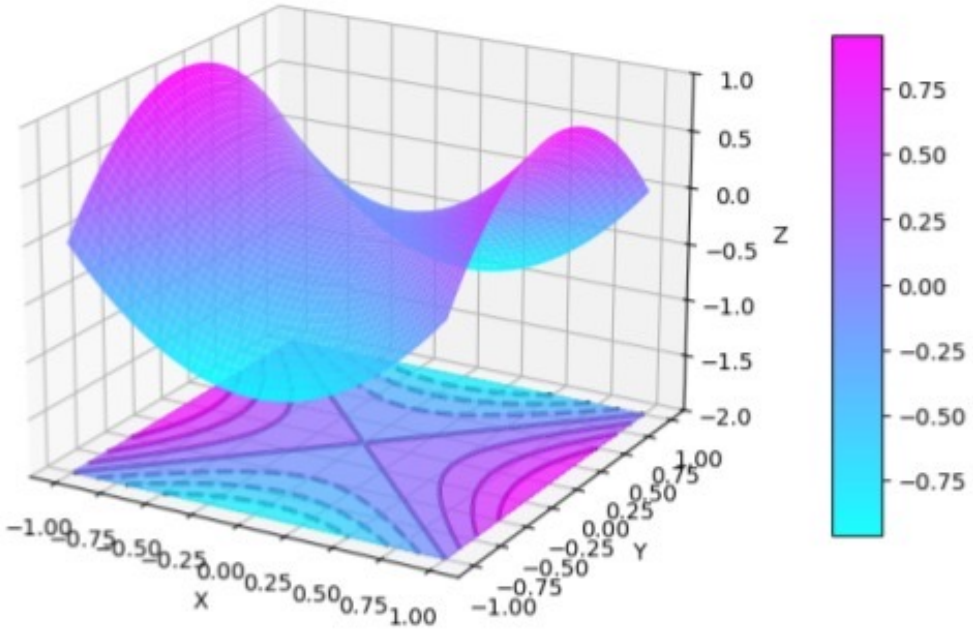
and points towards the direction in which $f(x, y)$

is *increasing* the fastest.

$f(x, y) = x^2 + y^2$



$f(x, y) = x^2 - y^2$



Directional Derivatives

Given a unit vector $\vec{u}$

— i.e. a vector of length 1 —,

the projection of the gradient vector field

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

onto $\vec{u}$

— i.e. the inner product $\langle \vec{u}, \nabla f \rangle$ —

defines the ***directional derivative*** of $f(x, y)$

$$\langle \vec{u}, \nabla f \rangle \doteq \partial_{\vec{u}} f$$

in the direction of $\vec{u}$.

Series

Decimal expansions as linear combinations

Real numbers

admit a *decimal positional representation* of the form

$$376.491 = 300 + 70 + 6 + 0{,}4 + 0{,}09 + 0{,}001 + \dots$$

$$376.491 = 3 \times 10^2 + 7 \times 10^1 + 6 \times 10^0 + 4 \times 10^{-1} + 9 \times 10^{-2} + 1 \times 10^3$$

We could rephrase this by saying

that any real number is a linear combination

of the elements in the « basis »

$$(\dots, 10^2, 10^1, 10^0.10^{-1}, 10^{-2}, 10^3 + \dots)$$

with certain coefficients between 0 and 9; in our case:

$$(\dots 0,0,3,7,6.4,9,1,0,0,...).$$

Real numbers can be expressed by means of positional representations based on other bases, like for instance the binary basis

$$(\ldots, 2^2, 2^1, 2^0, 2^{-1}, 2^{-2}, 2^3 + \ldots).$$

Newton (around 1690) had the idea

that the same could be done with functions,

i.e. that any function $f(x)$ can be expressed

as a « linear combination »

of a given set of canonical functions

forming a « basis ».

By choosing different bases of functions,

we can define different kinds of series expansions of function:

Taylor series: the basis is given by the monomials

$$x^0, x^1, x^2, x^3, \ldots$$

Fourier series: the basis is given by trigonometric functions with different amplitudes, frequencies and phases.

Series

We then have two alternative definitions of a function $f : A \to B$:

- By specifying the values $f(a)$ for all $a \in A$ (by means, for instance, of a functional expression).

- By describing the function as a unique point in an infinite-dimensional vector space, where the base is given by an infinite number of « canonical functions » (e.g. monomials, trigonometric functions, etc.).

Taylor Series

« O instante é semente viva. […] Mais que um instante, quero o seu fluxo. »

C. Lispector, *Água Viva*

The derivative

$$\frac{df}{dx}$$

of a function $f(x)$

conveys information defined at each single x

about the local behavior of the function *around* x.

Conceptually, the idea behind Taylor expansions

is that we can (partially or totally) reconstruct a function $f(x)$ for any x

out of ***local information***

— given by the derivatives of $f(x)$ —

about the behavior of $f(x)$

around a given point x_0

(in what follows, we shall choose $x_0 = 0$ to simplify the presentation).

From a physical standpoint,

we can rephrase this as a problem of

rational ***predictability.***

If we consider a function $f(t)$,

what kind of local information about the present $t = 0$

we need to retrodict the past and predict the future (the other values of f)?

Can we define a « germ » of local information

that encodes the global (« past » and « future ») unfolding of the function?

If we do not have such a full « germ », but only a truncated one,

what is the most rational prediction that we can make

on the basis of this partial information?

Formally, this idea will allows us to express a function $f(x)$

as an infinite series of the form

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_nx^n + \ldots$$

where

$$(x^0 = 1, x^1 = x, x^2, x^3, \ldots, x^n, \ldots)$$

is a basis of the ∞-dimensional functional vector space

and $(a_0, a_1, a_2, a_3, \ldots, a_n, \ldots)$

are the coordinates that determine the « vector » $f(x)$.

The problem is then to find the coefficients $(a_0, \ldots, a_n, \ldots)$

that define a given function $f(x)$.

0**th-order (constant) approximation** $f_0(x)$ **of a function** $f(x)$

The most basic kind of information about a function $f(x)$ at 0

that we can have is the value $f(0)$

— that we shall call 0-***germ*** *of* $f(x)$ at 0 —

We cannot globally unfold $f(x)$ from its 0-germ at 0,

but we can make the most reasonable prediction

based on this information, namely

$$f(x) \approx f_0(x) = f(0),$$

which amounts to presuppose that $f(x)$ is constant.

The fact that this prediction is reasonable is attested by the fact that even

if the real function $f(x)$ is not constant,

the value $f(0)$ provides an approximation

$$f(x) \approx f_0(x) = f(0)$$

that becomes more accurate as x becomes smaller

(under the assumption that $f(x)$ is continuous).

$f_0(x) = f(0)$ is the ***best constant approximation*** to $f(x)$ at 0.

1**st-order (linear) approximation** $f_0(x)$ **of a function** $f(x)$

Let's suppose now that we know the 1-***germ*** of $f(x)$ at 0

that is, $f(0)$ and $f'(0)$.

Now, the most reasonable prediction

that we can make for the value of $f(x)$ at any x is that

$$f(x) \approx f_1(x) = f(0) + f'(0)x$$

which amounts to presuppose that $f'(x)$ is constant,

i.e. that $f(x)$ is a linear function with slope $f'(x)$.

Even if $f'(x)$ is not constant, the approximation

$$f(x) \approx f_1(x) = f(0) + f'(0)x$$

becomes more accurate as x becomes smaller.

$f_1(x) = f(0) + f'(0)x$ is the ***best linear approximation*** to $f(x)$ at 0.

$2nd$-order (parabolic) approximations $f_2(x)$ of a function $f(x)$

• 0th-order (or constant) approximation $f_0(x)$ based on the knowledge of the 0 -germ $f(0)$:

$$f(x) \approx f_0(x) = f(0)$$

• 1st-order (linear) approximation $f_1(x)$ based on the knowledge of the 1-germ $f(0)$ and $f'(0)$:

$$f(x) \approx f_1(x) = f(0) + f'(0)x$$

• 2nd-order (parabolic) approximation $f_2(x)$ based on the knowledge of the 2 -germ $f(0), f'(0)$ and $f''(0)$

$$f(x) \approx f_2(x) = f(0) + f'(0)x + \frac{1}{2} f''(0)x^2,$$

where $f_2(x)$ fulfills the conditions $f_2(0) = f(0), f_2'(0) = f'(0)$ and $f_2''(0) = f''(0)$.

We can then define a full-fledged « germ » of local information

that faithfully encodes the global unfolding of the function.

Such a full-fledged ***germ*** of $f(x)$ is the data given

by the value of the function and all its derivatives at 0

$$f(0), f'(0), \ldots, f^n(0), \ldots$$

The germ of the function fully encodes its « future » unfolding.

Full Taylor Expansion of a Function $f(x)$

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3 + \ldots + \frac{1}{n!}f^n(0)x^n + \ldots$$

or

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(0)x^n$$

Such an infinite series perfectly matches $f(x)$

under certain conditions (e.g. $f(x)$ being C^∞)

and within a certain ***radius of convergence***.

Technicalities (beyond the reach of this introduction)

The Taylor expansion

$$\sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(0) x^n$$

of a smooth function $f(x)$ at 0

might diverge beyond a certain *radius of convergence* R,

that is, diverge for $x \geqslant R$.

And even if the Taylor series converges,

it might not converge to the value of the function at x.

Natura non facit saltus

By construction, Taylor expansions exist

if the function $f(x)$ is C^{∞}, i.e. maximally continuously differentiable.

Only if $f(x)$ and all its derivatives $f^{k}(x)$

are maximally smooth,

the future predictably unfolds

from the « germ » of the present.

Prediction relies on maximal smoothness.

Physical interpretation of the 1st-order (linear) approximation

In physical terms, a 1st-order (linear) approximation

$$s(t) \approx s(0) + s'(0)t$$

amounts to compute the position $s(t)$ for any t

from the knowledge of the position $s(0)$ and the velocity $s'(0)$ at 0 (called *initial conditions*).

This remains just an approximation since it presupposes

that the velocity remains constant.

The smaller t, the better the approximation.

Newtonian Mechanics

Now, in Newtonian mechanics,

a trajectory can be fully reconstructed

from a « truncated germ » given by such *initial conditions,*

namely the position $f(0)$ and the velocity $f'(0)$ at 0.

This « truncated germ » does not fully determine its future,

an indeterminacy that holds space for external contingencies so to speak.

How can we determine, unfold or predict a future trajectory

from the partial information contained in the 1-germ $(f(0), f'(0))$ at $t = 0$?

Newton's answer to this problem amounts to a new entity $F(s, t)$

— called *force* —

that externally fixes the second derivative $s''(t)$ of $s(t)$

everywhere and at every instant of time

through the so-called ***second law of motion***:

$$F = ma$$

or

$$F(s, t) = ms''(t).$$

The main problem of mechanics is that of « integrating »

this so-called « equation of motion »

— a reverse operation to derivation —

in order to obtain the trajectory $s(t)$.

Examples of Taylor Series

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \ldots$$

Derivatives of Trigonometric and Exponential Functions

$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f''''(x)$
$sin(x)$	$cos(x)$	$-sin(x)$	$-cos(x)$	$sin(x)$
$cos(x)$	$-sin(x)$	$-cos(x)$	$sin(x)$	$cos(x)$
e^x	e^x	e^x	e^x	e^x
e^{ax}	ae^{ax}	a^2e^{ax}	a^3e^{ax}	a^4e^{ax}
e^{ix}	ie^{ix}	$-e^{ix}$	$-ie^{ix}$	e^{ix}

Taylor Series of Trigonometric Functions

$$sin(x) = \cancel{sin(0)} + cos(0)x - \cancel{\frac{sin(0)}{2!}x^2} - \frac{cos(0)}{3!}x^3 + \cancel{\frac{sin(0)}{4!}x^4} + \frac{cos(0)}{5!}x^5 - \cancel{\frac{sin(0)}{6!}x^6} - \frac{cos(0)}{7!}x^7 + \dots$$

$$cos(x) = cos(0) - \cancel{sin(0)x} - \frac{cos(0)}{2!}x^2 + \cancel{\frac{sin(0)}{3!}x^3} + \frac{cos(0)}{4!}x^4 - \cancel{\frac{sin(0)}{5!}x^5} - \frac{cos(0)}{6!}x^6 + \cancel{\frac{sin(0)}{7!}x^7} + \dots$$

$$\sin(0) = 0$$

$$\cos(0) = 1$$

$$sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

$$cos(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$$

Taylor Series of Exponential Function

$$e^x = e^0 + e^0 x + \frac{e^0}{2!}x^2 + \frac{e^0}{3!}x^3 + \frac{e^0}{4!}x^4 + \frac{e^0}{5!}x^5 + \frac{e^0}{6!}x^6 + \frac{e^0}{7!}x^7 + \dots$$

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 + \frac{1}{6!}x^6 + \frac{1}{7!}x^7 + \dots$$

$$e^{ix} = 1 + ix - \frac{1}{2!}x^2 - i\frac{1}{3!}x^3 + \frac{1}{4!}x^4 + i\frac{1}{5!}x^5 - \frac{1}{6!}x^6 - i\frac{1}{7!}x^7 + \dots$$

$$e^{ix} = \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) + i\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots\right)$$

$$e^{ix} = \cos(x) + i\sin(x)$$

Euler's Formula

By combining the Taylor expansions

for e^x, $sin(x)$ and $cos(x)$ with $x = i\theta$,

we obtain

$$e^{i\theta} = cos(\theta) + isin(\theta).$$

$$e^{i\pi} + 1 = 0$$

When exponential growth

becomes purely imaginary,

be sure you've entered a loop.

Differentials

Differentials vs. Infinitesimals.

There is another philosophical lineage

— exemplified by Hegel and Deleuze —

that seeks to dissociate the notion of the ***differential***

from that of the ***infinitesimal***,

in order to preserve the former

from the criticisms directed at the latter.

Hegel on Differentials

« […] x and y as such are still supposed to signify quanta; now this whole significance is entirely lost in the so-called *infinitesimal differences*. dx, dy, are no longer quanta […] but have meaning only in connection, *the meaning of mere moments*. They no longer are a *something*, the something taken as quantum, nor are they finite differences; but *neither* are they *nothing*, not a null void of determination. Outside their relation they are pure nullities; but they are to be taken only as moments of the relation, as *determinations* of the differential coefficient $\frac{dy}{dx}$. […] that which exists *only* in relation is not a quantum; quantum is a determination which should possess a perfectly indifferent existence outside the relation it is in and be indifferent to its difference from another quantum; the qualitative, on the contrary, is what it is only in its distinction from an other. The infinite magnitudes of calculus […] exist only as terms of comparison, in relation. »

G.W.F. Hegel, *The Science of Logic*, pp.215-6.

Deleuze on Differentials

« *Dx* est tout à fait indéterminé par rapport à *x*, *dy* par rapport à *y*, mais ils sont parfaitement déterminables l'un par rapport à l'autre. […] Le rapport $\frac{dy}{dx}$ n'est pas comme une fraction qui s'établit entre quanta particuliers dans l'intuition, mais n'est pas davantage un rapport général entre grandeurs variables ou quantités algébriques. Chaque terme n'existe absolument que dans son rapport avec l'autre ; il n'est plus besoin, ni même possible d'indiquer une variable indépendante. »

G. Deleuze, *Différence et répétition*, p. 223.

A.L. Cauchy (1789–1857)

proposed a rigorous mathematical definition of differentials,

disentangling this notion

from the problematic idea of the infinitesimal.

Unlike Hegel and Deleuze,

Cauchy's definition implies that differentials are

well-defined (generalized) quantities.

Cauchy's Rigorous Definition of Differentials (I)

Cauchy proposed a rigorous definition of the notion of *differential dx*

- that is independent of the (now unnecessary) notion of infinitesimal;

- that relies on the definition of a derivative as a limit (rather than as a quotient of infinitesimals);

- and such that Leibniz notation

$$\frac{df}{dx}$$

can be rigorously interpreted as a quotient of differentials.

Cauchy's Rigorous Definition of Differentials (II)

Differentials can be defined as « functions »

— noted df —

of two variables,

— noted x and Δx —

given by the expression:

$$(df)(x, ; \Delta x) \doteq \frac{df}{dx}(x)\Delta x,$$

where the derivative $\frac{df}{dx}$ is defined through the usual limit definition.

Linear or 1st-Order Approximation of a Function

The differential df yields

the increment of the first-order (linear) approximation to the function $f(x)$

at x for a finite displacement Δx:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x = f(x) + df(x; \Delta x)$$

The smaller Δx, the smaller the difference

between the real increment in the value of the function

$$\Delta f = f(x + \Delta x) - f(x)$$

and

$$df(x; \Delta x) = f'(x)\Delta x$$

Comments on Cauchy's definition

- The value $(df)(x; \Delta x)$ defined by a finite displacement Δx is not necessarily (infinitely) small, i.e. it is not an *infinitesimal*.

- The differential df is not a new class of number, but rather a function on two arguments.

- Cauchy's idea will be fully developed by Élie Cartan in the ***theory of differential n-forms***.

Making Leibniz Notation Rigorous

For the particular case

$$f(x) = x,$$

we obtain

$$dx(x; \Delta x) = \frac{dx}{dx}\Delta x = \Delta x$$

and then we can write

$$(df)(x; \Delta x) \equiv f'(x)\Delta x = f'(x)dx$$

or

$$\frac{df}{dx} = f'(x)$$

Contrary to Deleuze's and Hegel's intuitions,

Cauchy's notion of differential

transforms the problematic symbol *dx*

into a well-defined quantity of a higher order, namely

— in Cartan's later terminology —

a ***differential*** 1***-form***

(whereas a function *f* is a *differential* 0-*form*).

« In elementary textbooks, one is presented with the differential df as representing 'an infinitesimal change in the function $f(p)$' associated with some infinitesimal displacement of the point p; but one will recall that the displacement of p is left arbitrary […]. Thus df represents a change in f in some unspecified direction. […] The only failing of the textbook presentation, then, was its suggestion that df was a scalar or a number; the explicit recognition of the need for specifying a direction $\vec{v}$ to reduce df to a number $df(\vec{v})$ shows that in fact df is a 1-form, the gradient of f. »

C.W. Misner, K.S. Thorne, J.A. Wheeler, *Gravitation*, 1973, p.63.

« In order to retain the *operational* facility which the differential point of view affords, the concept of the differential has been logically defined, not in terms of mathematical atomism, but as a notion derived from that of the derivative. [...] The differentials as thus defined are only new variables, and not fixed infinitesimals, or indivisibles, or ‘ultimate differences,’ or ‘quantities smaller than any given quantity,’ or ‘qualitative zeros,’ or ‘ghosts of departed quantities,’ as they have been variously considered in the development of the calculus. »

C.B. Boyer, *The History of the Calculus and its Conceptual Development*, 1949, p.12.

« Classical differential geometers (and classical analysts) did not hesitate to talk about 'infinitely small' changes dx^i of the coordinates x^i, just as Leibnitz had. No one wanted to admit that this was nonsense, because true results were obtained when these infinitely small quantities were divided into each other (provided one did it in the right way).

Eventually it was realized that the closest one can come to describing an infinitely small change is to describe a direction in which this change is supposed to occur, i.e., a tangent vector. Since df is supposed to be the infinitesimal change of f under an infinitesimal change of the point, df must be a function of this change, which means that df should be a function on tangent vectors. The dx^i themselves then metamorphosed into functions, and it became clear that they must be distinguished from the tangent vectors $\partial/\partial x^i$.

Once this realization came, it was only a matter of making new definitions, which preserved the *old* notation, and waiting for everybody to catch up. In short, all classical notions involving infinitely small quantities became functions on tangent vectors, like df, except for quotients of infinitely small quantities, which became tangent vectors, like dc/dt. »

M. Spivak, *A Comprehensive Introduction to Differential Geometry*, volume 1, p. 111.

Differentials in Dimension 2

Whereas in dimension 1,

Cauchy' notion of differential

$$(df)(x;\Delta x) \doteq \frac{df}{dx}(x)\Delta x$$

yields the increment of the linear approximation of $f(x)$

under a displacement Δx,

in dimension 2, the differential is such that

$$(df)(x,y;\vec{v}) \doteq \frac{\partial f}{\partial x}(x,y)v_x + \frac{\partial f}{\partial y}(x,y)v_y,$$

which yields the increment of the linear approximation of $f(x,y)$

along a vectorial displacement $\vec{v} = (v_x, v_y)$.

In dimension 2, the differential df at a point (x,y) « eats » vectors $\vec{v}$

and yields the directional derivative along $\vec{v}$.

Directional Derivatives

Whereas in dimension 1,

Cauchy' notion of differential

yields the derivative of $f(x)$ when $\Delta x = 1$

$$(df)(x; \Delta x = 1) \doteq \frac{df}{dx}(x),$$

the general higher dimensional notion of differential

yields the ***directional derivative of*** $f(x, y)$ ***along*** $\overrightarrow{v} = (v_x, v_y)$

$$(df)(x, y; \overrightarrow{v} = (v_x, v_y)) \doteq \frac{\partial f}{\partial x}(x, y)v_x + \frac{\partial f}{\partial y}(x, y)v_y = \partial_{\overrightarrow{v}} f(x, y).$$

Later on, E. Cartan will generalize

the notion of differential of a function df

to a more general notion

— called ***differential k-forms*** —

that « eats » k vectors to yield numbers.

Differential k-forms provide

the most general notion of quantity

that we shall consider here.

Whereas numbers need no evaluation whatsoever

to yield ***numbers***

(because they are already numbers)

and ***functions*** $f(x)$ yield numbers

when evaluated on points,

we shall see that ***differential*** k***-forms*** are new kind of quantities

that yield numbers

when « evaluated » on k-dimensional objects

— like curves or surfaces —

This generalized form of evaluation is called ***integration***.

Roughly speaking,

a gradient differential df encodes information

about the rate of change of $f(x)$ at every x.

Summing up the corresponding « infinitesimal » or local changes

yields the overall or global change,

which is the numerical outcome of « evaluating/integrating »

the differential on the curve.

Integrals

While

derivatives

formalize the ***local notion*** of

instantaneous rate of change,

integrals

formalize the ***global notion*** of

accumulated change

along a finite interval.

Integrals

« integrate »

or « sum up »

the infinite number of

« infinitesimal » increments

defined by the

« instantaneous » rates of change

at each moment.

Let's suppose that we are in a moving car

such that the only information which is accesible to us

is the reading of the speedometer $v(t)$.

Is it possible to compute the distance traveled by the car

between between $t_i = a$ and $t_f = b$?

The Golden Rule of Calculus

Split a complex problem

into an infinite number of infinitely small, simpler problems,

and then « integrate » their solutions

into the solution of the original problem.

Given an object moving with a ***constant velocity*** v

between $t_i = a$ and $t_f = b$

the product

$$v\Delta t = v(b - a)$$

is both

- the total distance traversed by the object between $t_i = a$ and $t_f = b$.

- the signed area of the rectangle « under » the graph $v(t)$ vs. t between t_i and t_f.

♥ Nota bene: if $v < 0$, the signed area $v(b - a)$ is negative which reflects the fact that the object is moving leftwards.

Conjecture

Even if the velocity is a non-trivial function of time $v(t)$,

the overall distance traveled between $t_i = a$ and $t_f = b$

coincides with the signed area

between the graph of $v(t)$ and the t-axis

between a and b.

Riemann's demonstration of the conjecture (I)

- Let's divide the time interval $[a, b]$ in n subintervals of the same length Δt such that

$$n \times \Delta t = b - a.$$

- The interval $[a, b]$ is then divided as

$$t_0 = a, t_1, \ldots, t_n = b$$

where

$$t_{i+1} - t_i = \Delta t.$$

Riemann's demonstration of the conjecture (II)

- We shall approximate the distance traveled in a temporal sub-interval $[t_i, t_{i+1}]$ by assuming that the velocity during that interval is constant with value $v(t_i)$.

- Under this approximation, the overall distance traveled in each sub-interval $[t_i, t_{i+1}]$ is

$$v(t_i)\Delta t.$$

- The approximated total distance traveled between a and b is

$$\sum_{i=0}^{n-1} v(t_i)\,\Delta t\,.$$

Riemann's demonstration of the conjecture (II)

The expression

$$\sum_{i=0}^{n-1} v(t_i)\,\Delta t$$

amounts to approximate the area under the graph of the function $v(t)$ vs t

as the sum of the areas of a finite collection of rectangles,

each with height $v(t_i)$ and width Δt.

Under happy circumstances, this approximation tends

to the signed area defined by the graph $v(t)$ and the t-axis between $t = a$ and $t = b$

when the partition of this interval is refined

into an infinite number of rectangles of « infinitesimal » width $\Delta t \to 0$.

Riemann Integral

The area defined by

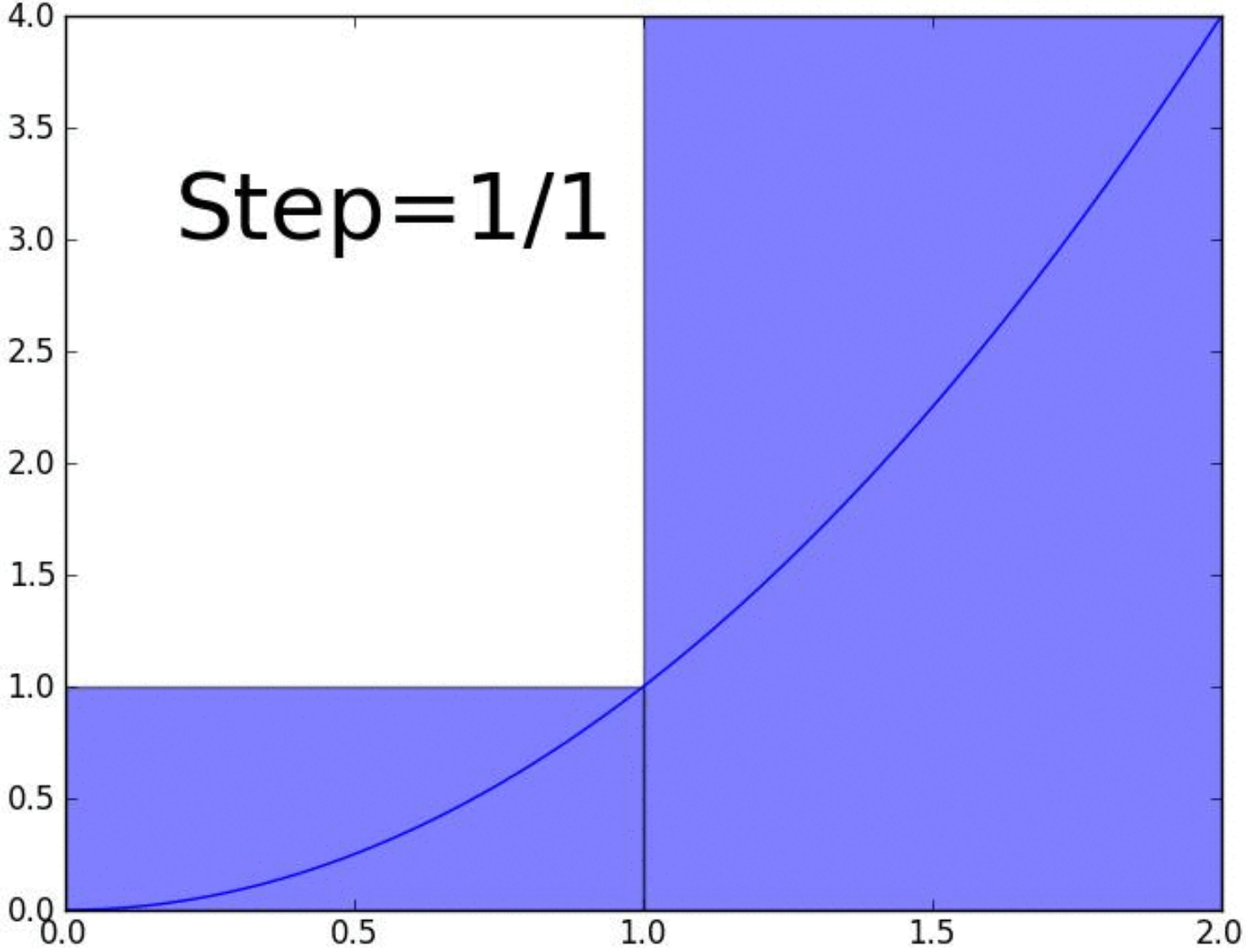
the graph of the function $v(t)$,

the t-axis

and the vertical lines defined by $t = a$ and $t = b$

is given by the expression:

$$\int_a^b v(t)dt \doteq lim_{\Delta t \to 0, n \to \infty} \sum_{i=0}^{n-1} v(t_i)\, \Delta t\,.$$


Step=1/1
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0

Leibniz's notation

$$\int_a^b v(t)dt$$

suggest that the overall traveled distance Δs

between $t_i = a$ and $t_f = b$

is a sum $\int$ of an infinite number

of « infinitesimal » displacements $ds = v(t)dt$

taking place during « infinitesimal » lapses of time dt.

According to Zeno's first (*The Dichotomy*)

and second paradox (*Achilles and the Turtle*),

continuous motion

— i.e. motion through an extension that can be infinitely divided —

is impossible

since it seems to require completing

an infinite number of intermediate steps

in finite time.

Zeno's First Paradox: The Dichotomy

« The first [paradox] asserts the non-existence of motion on the ground that that which is in locomotion must arrive at the half-way stage before it arrives at the goal. »

Aristotle, *Physics*, 239b11-13.

« [...] if there is motion, it is necessary that a moving thing will entirely traverse an infinite number of things in a finite time; but this is impossible; consequently, there is no motion. »

Simplicius, *Commentary on Aristotle's* Physics, 1013,3-14.

Zeno's Second Paradox: Achilles and the Turtle

« The second is the so-called Achilles, and it amounts to this, that in a race the quickest runner can never overtake the slowest, since the pursuer must first reach the point whence the pursued started, so that the slower must always hold a lead. »

Aristotle, *Physics*, 239b14-29.

Aristotle's solution: time is also (as space) infinitely divisible

« […] Zeno's argument is wrong in assuming that it is not possible to traverse an infinite number of points […] in a finite time. For both magnitude and time, and in general every continuum, may be said to be infinite […] in respect of division. Now a thing can […] in a finite time be in contact with things infinite […] in respect of divisibility; for the time itself is infinite in this sense.

Thus it is in an infinite, not a finite, time that it is possible to traverse what is infinite, and the contact is by means of parts of time that are infinite, not finite, in number. Therefore a thing cannot traverse either what is infinite in a finite time, or what is finite in an infinite time, but if either is infinite, so is the other. »

Aristotle, *Physics*, Book VI, 233a.

Riemann's anti-Zeno Strategy

Rather than pulverizing a finite motion

into an infinite number of infinitely small steps

in order to show that motion is rationally impossible,

Riemann does perform such a decomposition

in order to compute the overall distance

effectively traversed.

The advantage of Riemann's strategy

is that the infinite infinitesimal problems produced by such a decomposition

— an infinite number of infinitesimal rectangular areas —

are trivial to compute.

The Newton-Leibniz's Formula

or the Second Fundamental Theorem of Calculus (I)

$$\int_a^b v(t)dt \doteq lim_{\Delta t \to 0, n \to \infty} \sum_{i=0}^{n-1} v(t_i)\,\Delta t$$

$$= lim_{\Delta t \to 0} \sum_{i=0}^{n-1} s(t_{i+1}) - s(t_i)$$

$$= lim_{\Delta t \to 0} s(t_1) - s(t_0) + s(t_2) - s(t_1) + \ldots + s(t_{n-1}) - s(t_{n-2}) + s(t_n) - s(t_{n-1})$$

$$= lim_{\Delta t \to 0} s(t_n) - s(t_0)$$

$$= s(t_n) - s(t_0) = s(b) - s(a).$$

The Newton-Leibniz's Formula

or the Second Fundamental Theorem of Calculus (II)

In order to compute the area under the curve of the function $v(t)$

between $t_i = a$ and $t_f = b$,

we consider the function $s(t)$ of which $v(t)$ is the derivative (i.e. $v(t) = \frac{ds}{dt}$)

and compute the difference $s(b) - s(a)$

between the values of $s(t)$ at the extremes of the interval.

Second Fundamental Theorem of Calculus (FTC)

$$\int_a^b f(t)dt = F(b) - F(a),$$

where the function $F(t)$

is a ***primitive***, an ***antiderivative*** or a ***potential*** of $f(t)$,

that is:

$$\frac{dF(t)}{dt} = f(t)$$

(in our example $\frac{ds(t)}{dt} = v(t)$)

The Second Fundamental Theorem of Calculus from a Conceptual Standpoint

The sum $\int$ of « infinitesimal » changes dF of a quantity $F(x)$

between $t = a$ and $t = b$

— i.e. the sum of the « infinitesimal » contributions $dF = \frac{dF(t)}{dt}dt = f(t)dt$ —

adds up to the net change $\Delta F = F(b) - F(a)$ in the quantity F

between $t = a$ and $t = b$.

Primitives, antiderivatives or potentials

are defined up to a constant.

If $F(t)$ is a a potential for $f(t)$

— i.e. $\dfrac{dF(t)}{dt} = f(t)$ —

then, $\tilde{F}(t) = F(t) + C$ is also a potential:

$$\frac{d\tilde{F}(t)}{dt} = \frac{d(F(t)+C)}{dt} = \frac{dF(t)}{dt} + \frac{dC}{dt} = \frac{dF(t)}{dt} = f(t).$$

Potentials are not uniquely defined (« gauge freedom »).

Rule for computing areas

$$\int_a^b f(t)dt$$

1) Find an antiderivative or a potential $F(t)$ of $f(t)$, noted also

$$F(t) = \int f(t).$$

2) Compute $F(b) - F(a)$.

Antiderivatives of Elementary Polynomial, Exponential and Trigonometric Functions

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

Properties of Integrals

The 2nd FTC $\int_a^b f(t)dt = F(b) - F(a)$ implies that

♣ $\int_a^a f dx = 0,$

which reflects the evident fact that the « area » beneath the graph of f from a to a is zero.

♥ $\int_a^b f dx = -\int_b^a f dx,$

which means that integrals do depend on the orientation of the interval $[a, b]$ and which trivially implies $\int_a^b f dx + \int_b^a f dx = 0$, which means that integrals around a closed loop are zero.

♠ $\int_a^c f dx = \int_a^b f dx + \int_b^c f dx$ for any $a, b, c \in \mathbb{R},$

which means that integrals do not depend on the path followed between the limits of integration.

How to compute the average of a continuous function? (I)

Given a discrete sample of values $(x_1, \ldots, x_n)$ of a quantity A,

we can compute its average

by means of the expression

$$\langle A \rangle = \frac{\sum_{i=1,\ldots,n} x_i}{n}.$$

How to compute the average of a continuous function? (II)

In order to calculate the average of a continuous function $f(x)$

in a certain interval $[a, b]$ of length $L = b - a$

$$\langle f \rangle_{[a,b]}$$

we would need to add the infinite values of the function f at each point x

and divide the sum by ∞,

which is mathematically ill-defined.

But we can apply the golden rule of calculus:

We renormalize the continuous problem by reducing it to a manageable discrete case

and we recover the continuous nature of the original problem by taking a limit.

How to compute the average of a continuous function? (III)

To make the problem discrete,

we divide the interval L in finite intervals of length $\Delta x = x_{i+1} - x_i$ where

$$x_0 = a, x_1, \ldots, x_n = b$$

and we consider the discrete sample of $n = \dfrac{L}{\Delta x}$ values $f(x_0), f(x_1), \ldots f(x_{n-1})$.

We can then compute the average of the discrete sample by using the standard formula:

$$\frac{\sum_{i=0}^{n-1} f(x_i)}{\frac{L}{\Delta x}} = \frac{\Delta x \sum_{i=0}^{n-1} f(x_i)}{L} = \frac{\sum_{i=0}^{n-1} f(x_i)\Delta x}{L}$$

We can now recover the continuous case by taking the limit $\Delta x \to 0, n \to \infty$

$$\langle f \rangle_L = = \frac{\int_a^b f(x)dx}{L} = \frac{\text{Area beneath the graph}}{\text{Width of the interval}}$$

Graphically, we substitute $f(x)$ by a constant function $f'(x) = c$ such that the area of the resulting rectangle $c \times L$ coincides with the area $\int_a^b f(x)dx$ under the curve of $f(x)$.

The average of $f(x)$ is then given by $\dfrac{c \times L}{L} = c$, i.e. by the value of the constant function.

The mean velocity computed as an integral

By definition of the average of a continuous function,

$$\langle v \rangle_{\Delta t} = \frac{1}{\Delta t} \int_a^b v(t) dt$$

Comments about the limits of calculus over $\mathbb{R}$

In the big picture provided by

Cartan's theory of differential forms,

it will not necessarily hold

♦ **that integrals around closed loops are zero**

and

♠ **that integrals are path-independent.**

The Fundamental Theorems of Calculus

or

Why Derivatives (Slopes)

and

Definite Integrals (Areas)

are Inverse Operations?

Area Function

Given a function $f(t)$ we shall define the ***area function***

$$A(x) = \int_0^x f(t)dt$$

which yields the signed area defined by the graph of $f(t)$

between 0 and the variable limit of integration x.

We have thus far defined two « operators »

acting on functions $f(t)$

and producing other functions,

namely the ***differentiation operator***

$$\frac{d}{dt} : f(t) \mapsto \frac{df}{dt}$$

and the ***integration operator***

$$\int_0^x dt : f(t) \mapsto \int_0^x f(t)dt$$

We shall now see in which sense these two operators

are inverses of each other.

If we choose an antiderivative $\tilde{F}(t)$ such that $\tilde{F}(0) = 0$,

then, the expression

$$\int_0^x f(t)dt = \int_0^x \frac{d\tilde{F}(t)}{dt}dt = \tilde{F}(x) - \tilde{F}(0)$$

can be rewritten as

$$\left(\int_0^x dt \circ \frac{d}{dt} \right) \tilde{F}(t) = \tilde{F}(x),$$

i.e. as if the differentiation operator $\frac{d}{dt}$

and the integral operator $\int_0^x dt$

were inverses of (or cancel) each other.

Applying $\dfrac{d}{dt}$ to $\tilde{F}(t) = F(t) + C$

erases the information about the value of C.

Hence, we cannot in general recover $\tilde{F}(t) = F(t) + C$

when we apply successively $\displaystyle\int_0^x dt$ and $\dfrac{d}{dt}$ to $\tilde{F}(t)$,

but only the difference $\tilde{F}(x) - \tilde{F}(0) = F(x) - F(0)$.

We have thus far seen that

$$\left(\int_0^x dt \circ \frac{d}{dt} \right) \tilde{F}(t) = \tilde{F}(x)$$

We shall now show that the composition

of the differentiation operator and the area operator

in the other order

$$\left(\frac{d}{dx} \circ \int_0^x dt \right) f(t) = f(x)$$

also gives the original function.

First Fundamental Theorem of Calculus

The function

$$A(x) = \int_0^x f(t)dt$$

which gives the area defined by the graph of f between 0 and x

is a particular *antiderivative*, a *primitive* or a *potential* of f

that is, a function whose derivative is f:

$$\frac{dA(x)}{dx} = \frac{d}{dx}\int_0^x f(t)dt = f(x)$$

Demonstration of the First Fundamental Theorem of Calculus

Since we want to compute the rate of change of the area $A(x)$

under an « infinitesimal » variation of x,

let's start by computing the change of $A(x)$ under a finite variation $x \to x + \Delta x$:

$$\Delta A = A(x + \Delta x) - A(x) \approx f(x)\Delta x$$

or

$$\frac{\Delta A}{\Delta x} = \frac{A(x + \Delta x) - A(x)}{\Delta x} \approx f(x)$$

In the limit $\Delta x \to 0$ in which the variation becomes « infinitesimal »,

$$\frac{dA(x)}{dx} = lim_{\Delta x \to 0} \frac{A(x + \Delta x) - A(x)}{\Delta x} = f(x).$$

Note: in the abusive Leibniz's notation, we allow Δx and $A(x + \Delta x) - A(x)$

to become the « infinitesimals » dx and dA. Then,

$$\Delta A = A(x + \Delta x) - A(x) \approx f(x)\Delta x$$

becomes

$$dA(x) = f(x)dx \Rightarrow \frac{dA}{dx} = f(x)$$

The area/integral operator $\int_0^x dt$ that computes areas

and the differentiation operator $\frac{d}{dx}$ that computes slopes

are inverse operations:

- First fundamental theorem of calculus:

$$\left(\frac{d}{dx} \circ \int_0^x dt\right) f(t) = f(x)$$

- Second fundamental theorem of calculus:

$$\left(\int_0^x dt \circ \frac{d}{dt}\right) F(t) = F(x) - F(0)$$

$$A(x) \xrightarrow{d} f(x) \xrightarrow{d} \frac{df}{dx}$$

$$\text{Area under the curve} \longrightarrow \text{Curve} \longrightarrow \text{Slope of the curve}$$

$$\text{Anti-derivative} \longrightarrow \text{Function} \longrightarrow \text{Derivative}$$

$$A(x) = sin(x) \longrightarrow f(x) = cos(x) \longrightarrow \frac{df}{dx} = -sin(x)$$

$$A(x) = e^x - 1 \longrightarrow f(x) = e^x \longrightarrow \frac{df}{dx} = e^x$$

A vicious circle maybe?

The function

$$A(x) = \int_0^x f(t)dt$$

is an antiderivative of $f(t)$.

Now, in order to have an explicit functional expression for $A(x)$,

we must compute the integral $\int_0^x f(t)dt$.

Yet, our recipe for computing such integrals

requieres us to already know an explicit antiderivative of the integrand $f(t)$.

Thus, to find an explicit antiderivative of $f(t)$ through the expression $\int_0^x f(t)dt$,

we must begin by knowing an explicit antiderivative of $f(t)$.

Way out of the vicious circle

- First, the antiderivative of $f(t)$

$$A(x) = \int_0^x f(t)dt$$

is well-defined as a Riemann integral independently of the 2nd FTC.

- Second, the 2nd FTC

$$\int_a^b f(t)dt = F(b) - F(a)$$

can be used to compute the area on the left

only when we can find an explicit elementary functional expression

for the antiderivative F.

The Gaussian Integral

Even if the function $f(x) = e^{-x^2}$

does not have an antiderivative

given by an « elementary function » (which can be proved)

that could be used to compute

the integral of $f(x) = e^{-x^2}$ over the entire real line

$$\int_{-\infty}^{+\infty} e^{-x^2} dx$$

through the 2nd FTC,

the value of the integral can be computed (by other means) and yields $\sqrt{\pi}$.

The function $f(x) = e^{-x^2}$ is integrable

but does not have an elementary primitive.

Double Integral

Given a function $f(x, y) : \mathbb{R}^2 \to \mathbb{R}$,

we can compute the volume determined

by the surface defined by $f(x, y)$

and the xy-plane between the limits

$$a \leqslant x \leqslant b \text{ and } c \leqslant y \leqslant d$$

through the double integral

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy,$$

where the variable y in $\int_a^b f(x, y) dx$ is treated as a parameter

and the result of this integration is a function of y.

Change of Variables Theorem or Integration by Substitution

$$\int_a^b f(x(t))\frac{dx}{dt}\,dt = \int_{x(a)}^{x(b)} f(x)\,dx$$

where

$$x = x(t) \quad \& \quad dx = \frac{dx}{dt}\,dt$$

Example:

$$\int_a^b 2tcos(t^2)dt = \int_{a^2}^{b^2} cos(x)dx = sin(b^2) - sin(a^2),$$

where $x = t^2 \quad \& \quad dx = 2tdt$

The formula

$$\int_a^b f(x(t))\,\frac{dx}{dt}\,dt = \int_{x(a)}^{x(b)} f(x)\,dx$$

also encodes the fact that any change of variables

— e.g. a change of variables from x to a new variable t —

can be performed under the condition of introducing

a change-of-coordinates factor $\frac{dx}{dt}$

that converts dx into dt.

Coordinates like x or t do not have any intrinsic meaning;

they can be exchanged at will

provided that the corresponding change-of-coordinates factors are taken into account.

Integrals of « Gradient » Differential 1-forms

We shall now define the integral $\int$

of the differential

$$df$$

of a function

$$f(x, y) : \mathbb{R}^2 \to \mathbb{R}$$

along a curve γ in $\mathbb{R}^2$,

that we shall note

$$\int_\gamma df$$

We shall compute this integral by « pulling it back » to a situation we know:

the integral of a function along an interval in $\mathbb{R}$

To do so, we shall introduce a ***parameterization*** of the curve γ:

$$t \in [0,1] \mapsto (\gamma_x(t), \gamma_y(t))$$

Conceptually, this amounts to describe the « static » curve γ

as a « dynamical » unfolding $\gamma(t)$

« parameterized » by a temporal variable t.

Such a parameterization is not intrinsic to the static curve

and the value of the integral should not depend

on the chosen (orientation-preserving) parameterization.

Definition of the integral of a « gradient » 1-form along a curve γ

$$\int_{\gamma} df \doteq \int_{0}^{1} df(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})dt$$

• We are « feeding » the gradient 1-form df at each point $(\gamma_x(t), \gamma_y(t))$ of γ with the speed vector $\vec{\dot{\gamma}}$ tangent to $\gamma(t)$, i.e. df « eats » its way along γ.

• By definition of the differential df, the result $df(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})$ of such a manducation is the linear increment of the function $f(x, y)$ with respect to the displacement vector $\vec{\dot{\gamma}}$.

• The integral $\int_{\gamma} df$ adds all these linear increments of the function $f(x, y)$ all along the curve γ.

Fundamental Theorem of Calculus
for the Integrals of Differentials Along Curves

In analogy with the second fundamental theorem of calculus,

we could expect that

the addition of all these tiny increments of $f(x, y)$

along the curve γ

adds up to the total difference

between the final and initial values of the « potential » $f(x, y)$

$$f(\gamma_x(1), \gamma_y 1)) - f(\gamma_x(0), \gamma_y 0)).$$

This « fundamental theorem of calculus »

for integrals of « gradients » along curves implies that

$$\int_\gamma df = 0$$

when

♣ when γ coincides with a level line of $f(x, y)$

♥ γ is a loop, i.e. when $\gamma(1) = \gamma(0)$.

Path-Independence

The fact that

$$\int_{\gamma} df = 0$$

when γ is a loop

implies path-independence.

If γ_1 and γ_2 are two paths connecting the same endpoints,

we can form a loop $\gamma_1 - \gamma_2$. Then:

$$\int_{\gamma_1 - \gamma_2} d\eta = \int_{\gamma_1} d\eta - \int_{\gamma_2} d\eta = 0.$$

Hence,

$$\int_{\gamma_1} d\eta = \int_{\gamma_2} d\eta.$$

Physics Incarnation

For ***conservative force fields***

— i.e. force fields described by differential 1-forms

that derive from a potential f —

the total amount of ***work*** done by the force field on a particle moving

along a path γ in the force field

is independent of the path

and equal to the difference in potential energy

Differential Geometry

Main Historial References

♠ Hermann Grassmann (1844), *Die Lineale Ausdehnungslehre, ein neuer Zweig der Mathematik* [*The Theory of Linear Extension, a New Branch of Mathematics*]. See analysis of Grassman's text in G. Châtelet, *Les enjeux du mobile*, chap.4.

♥ Élie Cartan (1899), Sur certaines expressions différentielles et le problème de Pfaff, *Annales Scientifiques de l'École Normale Supérieure*, 16, 239-332.

♣ Élie Cartan,(1945), *Les Systèmes Différentiels Extérieurs et leurs Applications Géométriques*, Paris: Hermann.

We have thus far introduced two kinds of quantities:

numbers and ***functions***.

We shall now introduce a more general notion of quantity

namely, the so-called ***differential n-forms***.

Whereas ***numbers*** are particular cases of functions

— namely ***constant functions*** —,

functions f are particular cases of differential forms

— namely, ***differential 0-forms*** —

and ***differentials*** df are also particular cases of differential forms

— namely, ***differential 1-forms*** —

New insights arising from Cartan's theory of differential forms (I)

• Cartan's theory extends differential and integral calculus on $\mathbb{R}$ to general manifolds.

• We shall consider for instance calculus on a circle S^1.

• We shall see that in this case, it is not true

.that every function has a potential (1st FTC);

.that integrals can always be computed as the difference of the values of the potential at the extremes of the integration interval, since there might be no potential to do the job (2nd FTC).

• All in all, it is not true that the so-called « fundamental » theorems of calculus necessarily hold beyond calculus on $\mathbb{R}$.

• Since $\mathbb{R}$ is a very particular manifold, we cannot search for the « essence » of calculus by looking at a particular example.

New insights arising from Cartan's theory of differential forms (II)

• We shall introduce a general notion of ***integration*** that can be understood as a generalization of both the evaluation of functions on points and the integrals of calculus on $\mathbb{R}$.

• The 2nd fundamental theorem of calculus is generalized to the so-called ***Stokes theorem***.

• Stokes' theorem allows us to glimpse a deeper essence of calculus beyond the classical statement that differentiation and integration are inverse operations:

The ultimate « duality » is between differentiation

and the geometric operation of extracting the boundary of a manifold.

• The restriction of the validity of the 1st FTC elicits a ***generalization of the notion of quantity*** from differentials $d\eta$ to more general « differential » forms $\omega \neq d\eta$ that do not « derive » from a potential.

• This more general notion of quantity, extending beyond the realm of the FTCs, becomes suitable for detecting ***holes*** (***De Rham cohomology***, which establishes a connection between the differential forms defined on a manifold and its global topological properties).

Reformulation of the

Second Fundamental Theorem of Calculus

$$\int_a^b f(t)dt = F(b) - F(a),$$

$$\text{where } \frac{dF(t)}{dt} = f(t)$$

Reormulation of the

Second Fundamental Theorem of Calculus (I)

$[a, b]$ is an oriented 1-dimensional segment

(going from a to b)

with a 0-dimensional boundary

that inherits an orientation

from the orientation of $[a, b]$:

$$\partial[a, b] = \{-a, +b\}.$$

where the orientation of an extreme point b of an oriented curve

is defined as positive $+$ if the oriented curve points outwards at b

(and negative otherwise).

Reormulation of the

Second Fundamental Theorem of Calculus (II)

Using Cauchy's notion of differential

the integrand of $\int_a^b f(x)dx$ can be written as

$$f(x)dx = \frac{dF(x)}{dx}dx = dF$$

Strictly speaking, we do not integrate *functions* $f(x)$

but *differentials* $dF(x)$

$$\int_a^b dF(x)$$

Reformulation of the

Second Fundamental Theorem of Calculus (III)

We shall define the « integration » of a function $f(x)$

on an oriented 0-dimensional manifold

given by a set of signed (oriented) points $\Omega = \{\pm p_i\}$ as:

$$\int_{\Omega} f \doteq \Sigma_i \text{sign}(p_i) f(p_i)$$

The evaluation of a function $f(x)$ on a point p

becomes a particular case of the broader notion of integration.

Reformulation of the

Second Fundamental Theorem of Calculus (IV)

By putting all these elements together,

$$\int_a^b f(x)dx = F(b) - F(a)$$

can be recasted as

$$\int_{[a,b]} dF = \int_{\partial[a,b]} F\,.$$

This shows that the operations $\int$ and d are not

strictly speaking inverses of each other (as is often claimed),

but rather that d and ∂ form an ***adjoint pair***.

Reformulation of the

Second Fundamental Theorem of Calculus (V)

In the expression

$$\int_{[a,b]} dF = \int_{\partial[a,b]} F\,.$$

we have the following ingredients:

- Oriented extended objects (intervals and points). This will be generalized to ***oriented n-dimensional manifolds***.

- Quantities F defined on extended objects (functions). This will be generalized to ***differential n-forms.***

- Gradient differential forms will be generalized to differential forms that do not « derivate » from other forms.

- An operation d acting on quantities. This will be generalized to the ***exterior derivative*** d acting on differential n-forms.

- A ***boundary operator*** ∂ acting on manifolds and yielding their boundary.

- An ***integration operation*** $\int$ (generalizing the evaluation of function on points) that couples differential n-forms and oriented n-dimensional manifolds giving numbers.

Stokes Theorem

The expression

$$\int_{[a,b]} dF = \int_{\partial[a,b]} F$$

presents the 2nd fundamental theorem of calculus

as a particular case of a much more general expression (called ***Stokes Theorem***)

$$\int_{\Sigma} d\omega = \int_{\partial\Sigma} \omega$$

where

- ω is a differential $(n-1)$-form.
- d is the exterior derivative that increases the degree of a differential form in 1.
- Σ is a n-dimensional manifold.
- $\partial\Sigma$ is the $(n-1)$-dimensional boundary of Σ.

Oriented n-Dimensional Manifolds

An n-dimensional manifold is a topological space M

such that each point $p \in M$ has a neighborhood

that is identical *qua* topological space

(i.e. there exists a continuous invertible mapping called ***chart***)

to an open subset of $\mathbb{R}^n$.

A collection of charts that cover the manifold define an ***atlas*** (a « patchwork » of charts).

Informally, an n-dimensional manifold is a space that can be *modeled*

locally as a piece of an n-dimensional Euclidean space.

Topology against Terraplanism: locally the surface of a sphere looks like a portion of flat space $\mathbb{R}^2$,

but globally the sphere does not.

Counterexample: the figure 8 is not a manifold.

Descent

n-dimensional manifolds can also be described

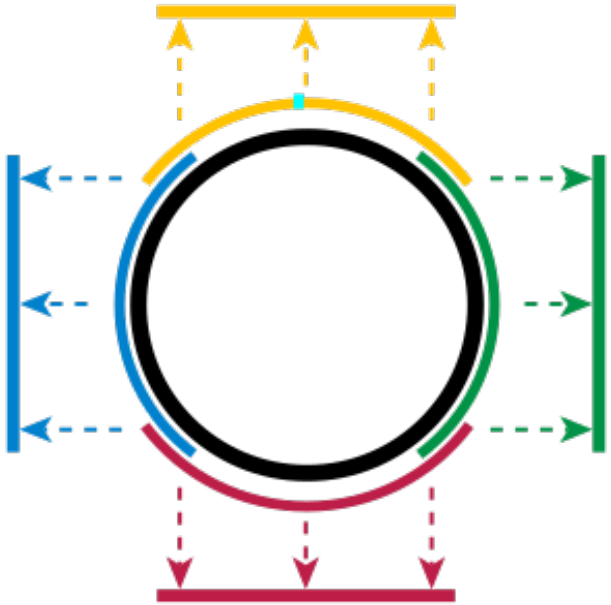
as a collection of balls U_i in $\mathbb{R}^n$

that are partially glued together by means of *gluing data*

defined along their intersections $U_i \cap U_j$,

thereby giving rise to a « patchwork », namely the manifold.

Atlas of a circle *qua* 1-dimensional manifold



Note: A circle is a 1-dimensional object

that cannot be embedded in $\mathbb{R}^1$.

Orientation of a Manifold

- Orientation of a 0-dimensional manifold (collection of points): choosing a sign $+$ or $-$ for each point.

- Orientation of a 1-dimensional manifold (curve): choosing a direction along the curve as positive orientation.

- Orientation of a 2-dimensional manifold (surface): choosing a clockwise (negative) or anti-clockwise (positive) « orientation » at every point in a consistent manner.

- Orientation of a 3-dimensional manifold (surface): choosing a right-handed or left-handed basis at every point in a consistent manner.

Nota Bene: a manifold is *non-orientable* if we can change the orientation of an oriented (« chiral ») object (i.e. go « through the looking glass ») by moving it in the space (e.g. convert a left hand into a right hand).

The orientation of a manifold M with boundary

induces an orientation on its boundary ∂M

- The boundary of a curve oriented positively from a to b is $\{-a, +b\}$.

- The 1-dimensional boundary of a surface oriented anti-clockwise is oriented positively in the anti-clockwise direction.

Differential k-forms

Why should we extend the notion of *quantity*?

• Quantities *qua* functions privilege points to the detriment of sub-manifolds of higher dimensions (curves, surfaces, etc.).

But why should we cling

to this reductionist point-centric conception of manifolds?

• We want a notion of quantity that can be integrated non-trivially over manifolds without boundary, i.e. quantities such that the « fundamental theorem of calculus » does not apply.

• Such quantities will be given by « differentials » that do not derive from a potential f, i.e. that cannot be written in the form df.

• In physics, not all force fields derive from a potential f, i.e. not all force fields can be written in the form df.

The generalized notion of *quantity*

on an oriented n-dimensional manifold M

— generally called *differential k-forms* —

will be *graded*, i.e. organized in a hierarchy of degrees:

- (-1)-forms aka *numbers*.
- 0-forms aka *functions*.
- 1-forms (e.g. gradient differentials of the form df).
- k-forms for $k \leq n$ (e.g. multivariable differentials).

In the 1970s,

differential k-forms

made their way into ***quantum field theory*** (BRST formalism),

where they appeared under the name ***ghosts*****,**

a term that resonates with the *irrational*, the *imaginary*, and the *fictitious*

kinds of quantities we have encountered so far.

A differential k-form ω is an object

that can be *evaluated* or *integrated*

on a k-dimensional oriented manifold M

This coupling between

a differential k-form ω and a k-dimensional oriented manifold M

will be denoted

$$\int_M \omega$$

or

$$\langle \omega, M \rangle$$

• (-1)-forms (*numbers*) no need to be evaluated in order to obtain numbers (since they are already numbers).

• 0-forms (*functions*) can be evaluated on oriented 0-dimensional manifolds (sets of signed *points*):

$$\int_M f \doteq \Sigma_i \text{sign}(p_i) f(p_i) \in \mathbb{R}$$

The evaluation of a function on a point

is understood as a particular case of a generalized notion of integration.

• 1-forms ω can be evaluated on oriented 1-dimensional curves γ (generalization of the integration of a « gradient » differential df along a curve):

$$\int_\gamma \omega \in \mathbb{R}$$

• 2-forms η can be evaluated on oriented 2-dimensional surfaces.

• k-forms can be integrated along oriented k-dimensional manifolds.

Introducing differential 1-forms

Whereas a function $f : M \to \mathbb{R}$ yields a number $f(p) \in \mathbb{R}$

when evaluated at a point $p \in M$,

a « gradient » differential df

— that encodes the rates of variations of f in every possible direction —

yields a number when evaluated at both

a point $p \in M$ (the gradient changes in general from point to point)

and a vector $\vec{v} \in T_pM$ tangent to M at p.

The number $df(p; \vec{v})$

is the linear increment in the value of f at p along the displacement v.

In general, a differential 1-form ω

(note: not necessarily of the form df)

is an $\mathbb{R}$-valued map defined on the so-called

tangent bundle TM on M:

$$\omega : TM \to \mathbb{R}$$

that sends each vector $\vec{v} \in T_pM$

tangent to M at each point p of M

to a real number $\omega(p; \vec{v}) \in \mathbb{R}$.

More generally, a differential k-form ω on M

is an object

that « eats » a point $p \in M$

and k tangent vectors to M at p

$$\vec{v}_1, \ldots, \vec{v}_k \in T_p M$$

and gives a real number $\omega(p; \vec{v}_1, \ldots, \vec{v}_k) \in \mathbb{R}$ as output.

The vocation of a differential k-form

is to transform k tangent vectors to any $p \in M$ into a number.

Differential Forms are Antisymmetric

An essential property of differential forms is that they are ***anti-symmetric***.

For instance, if ω is a 2-form, its action on a couple of vectors obeys

$$\omega(\vec{v}, \vec{u}) = -\omega(\vec{u}, \vec{v}),$$

which implies

$$\omega(\vec{v}, \vec{v}) = 0.$$

In other terms, the 2-form acts non-trivially only when the couple of vectors $(\vec{v}, \vec{u})$

are not collinear, i.e. when they span a 2-dimensional surface (a parallelepiped).

The antisymmetry property encodes the fact that the form

is sensitive to the ***orientation*** of the resulting parallelepiped,

where « the orientation reveals that the juxtaposition [of the vectors that generate the parallelepiped]

has been made in a certain order. » (G. Châtelet, *Figuring Space*, p.105)

All in all, a **differential k-form** acts non-trivially

in an orientation-sensitive manner

on a set of k vectors $(\vec{v}_1, \ldots, \vec{v}_k)$

when the latter span a k-dimensional parallelepiped.

Properties of Differential Forms: Linearity

- Differential forms can be added:

$$(\omega + \eta)(\vec{v}) = \omega(\vec{v}) + \eta(\vec{v})$$

and multiplied by constants:

$$(k\omega)(\vec{v}) = k\omega(\vec{v}) \text{ with } k \in \mathbb{R}$$

- Differential forms behave as ***linear maps*** when they act on vectors

$$\omega(a\vec{v} + b\vec{u}) = a\omega(\vec{v}) + b\omega(\vec{u})$$

Properties of Differential Forms: Wedge Product (I)

• Differential forms are equipped with a product called ***wedge or exterior product*** denoted $\wedge$, which results from an ***antisymmetrization of the tensor product*** $\otimes$.

• If ω and η are two 1-forms (each of which « eats » by definition a single vector), their tensor and wedge products yield 2-forms whose digestive action on a couple of vectors $(\overrightarrow{v}, \overrightarrow{u})$ is given by the expressions:

$$(\omega \otimes \eta)(\overrightarrow{v}, \overrightarrow{u}) = \omega(\overrightarrow{v})\eta(\overrightarrow{u})$$

$$(\omega \wedge \eta)(\overrightarrow{v}, \overrightarrow{u}) = \omega(\overrightarrow{v})\eta(\overrightarrow{u}) - \omega(\overrightarrow{u})\eta(\overrightarrow{v})$$

• The wedge product of a k-form ω and a l-form η is a $(k+l)$-form $\omega \wedge \eta$.

• The wedge product is graded-commutative:

$$\omega \wedge \eta = (-1)^{deg(\omega)deg(\eta)} \eta \wedge \omega.$$

It follows that $\omega \wedge \omega = 0$ if $deg(\omega)$ is odd.

Properties of Differential Forms: Wedge Product (II)

It follows from the definition of the wedge product $\wedge$ that for two 1-forms ω and η

$$(\omega \wedge \eta)(\vec{v}, \vec{u}) = -(\omega \wedge \eta)(\vec{u}, \vec{v})$$

$$(\omega \wedge \eta)(\vec{v}, \vec{v}) = \omega(\vec{v})\eta(\vec{v}) - \omega(\vec{v})\eta(\vec{v}) = 0$$

and that

$$(\omega \wedge \omega)(\vec{v}, \vec{u}) = \omega(\vec{v})\omega(\vec{u}) - \omega(\vec{u})\omega(\vec{v}) = 0$$

The Exterior Product in Grassmann's Terms

« […] we call the product to which we are led here the *exterior* [or *outer*] product, since by this nomenclature we mean that this type of product only represents a new extension and has a nonzero value insofar as the factors diverge from one another, and becomes zero if the factors remain together. »

Hermann Grassmann (1844), *Die Lineale Ausdehnungslehre, ein neuer Zweig der Mathematik* [*The Theory of Linear Extension, a New Branch of Mathematics*], § 36, p.84.

Cartesian basis 1-forms and 2-forms in $\mathbb{R}^2$

The 1-forms dx, dy and dz

acting on a vector $\overrightarrow{v} = (v_x, v_y, v_z)$ yield the length of its projections

to the Cartesian axes

$$dx(\overrightarrow{v}) = v_x,\ dy(\overrightarrow{v}) = v_y,\ dz(\overrightarrow{v}) = v_z.$$

Analogously, the 2-form

$$dx \wedge dy$$

acting on a couple of vectors $(\overrightarrow{v}, \overrightarrow{u})$

yields the oriented area of the projection of the oriented parallelepiped

spanned by $\overrightarrow{v}$ and $\overrightarrow{u}$ to the xy-plane.

If $\overrightarrow{v}$ and $\overrightarrow{u}$ are collinear, the area is zero.

Properties of Differential Forms: Description in Coordinates of Forms in $\mathbb{R}^3$

General differential k-forms can be written

as linear combinations of k-fold wedge products of the basis1-forms dx, dy and dz:

.0-forms (aka functions):

$$f = f(x, y, z)$$

.1-forms:

$$\omega = \omega_x(x, y, z)dx + \omega_y(x, y, z)dy + \omega_z(x, y, z)dz,$$

E.g. $\omega = adx$ can be represented as a stack of parallel planes transversal to the x-axis

such that $1/a$ is the distance between successive planes in the stack.

Hence, the higher the « strength » a of ω the more densely packed the planes of the stack.

.2-forms:

$$\eta = \eta_{xy}(x, y, z)dx \wedge dy + \eta_{yz}(x, y, z)dy \wedge dz + \eta_{zx}(x, y, x)dz \wedge dx$$

.3-forms (forms of maximal degree in $\mathbb{R}^3$, also called *volume forms* when they are nowhere vanishing):

$$\xi = \xi_{xyz}(x, y, z)dx \wedge dy \wedge dz$$

There are no 4-forms in $\mathbb{R}^3$

since any 4-fold wedge product of dx, dy and dz

would have to repeat one of the differentials

and the wedge product of a form with itself is always zero.

In general, there are differential k-forms on an n-dimensional manifold M

only for $0 \leqslant k \leqslant n$.

Differentials 1-Forms as Stacks

1-forms define at each point of M

an oriented stack of parallel straight lines (in $\mathbb{R}$)

or an oriented stack of parallel planes (in $\mathbb{R}^2$).

In the case of a gradient 1-form, the stack

is the linear approximation of the level curves at each point.

A 1-form ω acting on a vector $\vec{v}$ at a point $p \in M$

yields the number $\omega(\vec{v})$ of times

that $\vec{v}$ crosses the lines/planes of the stack

(where the sign of $\omega(\vec{v})$ indicates whether $\vec{v}$ crosses the stack

in the positive or negative orientation).

If $\vec{v}$ is parallel to the stack $\omega(\vec{v}) = 0$,

i.e. $\omega(\vec{v}) \neq 0$ when $\vec{v}$ crosses the stack.

Differentials 2-Forms as Honeycombs or Faraday Tubes

A 2-form define an oriented *tubular structure*

obtained by superimposing the 2 stacks associated to the basis 1-forms.

A 2-form ω acting on a pair of vectors $\vec{v}$ and $\vec{u}$ count

how many tubes cross the parallelepiped $P(\vec{v}, \vec{u})$ spanned by $\vec{v}$ and $\vec{u}$.

$\omega(\vec{v}, \vec{u}) \neq 0$ only if the parallelepiped $P(\vec{v}, \vec{u})$

crosses or cuts transversally the tubular structure

and the sign specifies whether the orientation of the tubular structure

is aligned or opposed to that of $P(\vec{v}, \vec{u})$.

In other terms, $\omega(\vec{v}, \vec{u})$ measures the *flux* of the tubular structure across $P(\vec{v}, \vec{u})$.

Ref.: C.W. Misner, K.S. Thorne, J.A. Wheeler, *Gravitation*, 1973, p.100.

Exterior Derivative

Properties of Differential Forms: Exterior Derivative

We shall introduce a new operator

called ***exterior derivative*** and denoted d

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

that converts k-forms into $(k+1)$-forms

and generalizes Cauchy's notion of differential df in calculus on $\mathbb{R}$

and the gradient differential df in calculus on $\mathbb{R}^n$

as objects that « eats » displacements (i.e. vectors)

and gives back the linear variation of f along the displacements.

The Four Definitions of the Exterior Derivative d

♥ **Axiomatic definition.**

♦ **Local or Coordinate definition.**

♥ **Global or Intrinsic definition.**

♣ **Geometric definition.**

Axiomatic Definition of the Exterior Derivative

The exterior derivative is the unique map

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

such that

1) $d : \Omega^0(M) \to \Omega^1(M)$ is the gradient 1-form df;

2) $d(c\omega + \eta) = cd\omega + d\eta$ (linearity);

3) $d(\omega \wedge \eta) = d\omega + (-1)^p d\eta$ where $\omega \in \Omega^p(M)$ (graded Leibniz Law);

4) $d(d\omega) = 0$ (this encodes the fact that partial derivatives commutes or, more geometrically, it is the dual of $\partial^2 = 0$).

Local or Coordinate Definition of the Exterior Derivative (I)

Let's compute the exterior derivative of a generic 1-form in $\mathbb{R}^2$

$$\omega = \omega_x(x, y)dx + \omega_y(x, y)dy$$

By applying the previous axioms, we obtain:

$$d\omega = d\omega_x(x, y) \wedge dx + d\omega_y(x, y) \wedge dy$$

$$= (\frac{\partial \omega_x}{\partial x}dx + \frac{\partial \omega_x}{\partial y}dy) \wedge dx + (\frac{\partial \omega_y}{\partial x}dx + \frac{\partial \omega_y}{\partial y}dy) \wedge dy$$

$$= \frac{\partial \omega_x}{\partial y}dy \wedge dx + \frac{\partial \omega_y}{\partial x}dx \wedge dy$$

$$= (\frac{\partial \omega_y}{\partial x} - \frac{\partial \omega_x}{\partial y})dx \wedge dy$$

Local or Coordinate Definition of the Exterior Derivative (II)

The exterior derivative of a general k-form in $\mathbb{R}^n$

$$\omega = \sum_I \omega_I dx^I$$

— where I stands for a k-tuple $(i_1, \ldots, i_k)$ of distant integers from 1 to n

and $dx^I \doteq dx^{i_1} \wedge \ldots \wedge dx^{i_k}$ —

is

$$d\omega = \sum_I d\omega_I \wedge dx^I$$

or

$$d\omega = \sum_I \frac{\partial \omega_I}{\partial x^i} dx^i \wedge dx^I$$

This expression shows that the exterior derivative of a differential form

encodes the variations of the functional coefficients ω_I that define the form in every possible direction.

Global or Intrinsic Definition of the Exterior Derivative

$$(d\omega)(\overrightarrow{u}, \overrightarrow{v}) = (d(\omega(\overrightarrow{v})))(\overrightarrow{u}) - (d(\omega(\overrightarrow{u})))(\overrightarrow{v}) - \omega([\overrightarrow{u}, \overrightarrow{v}])$$

- The first term computes the variation of the function $\omega(\overrightarrow{v})$ in the direction $\overrightarrow{u}$.
- The second term guarantees the antisymmetry.
- The third term « removes » the variations of the vector fields $\overrightarrow{u}$ and $\overrightarrow{v}$.

Geometric Definition of the Exterior Derivative (I)

Let's reconsider the definition of the differential of a function

$$df(x,h) = f'(x)h \approx f(x+h) - f(x) = \int_{\partial[x,x+h]} f.$$

Hence, the standard definition of the differential of f

is an object df

that « eats » both points and 1-dimensional displacement

such that the result $df(x,h)$ of such a « manducation »

is ultimately the integral of f over the 0-dimensional boundary of the displacement:

$$df(x,h) = \int_{\partial[x,x+h]} f$$

Geometric Definition of the Exterior Derivative (II)

The integral recasting of Cauchy's differential

$$df(x,h) = \int_{\partial[x,x+h]} f$$

can be generalized to the exterior derivative of a k-form φ

through the expression

$$d\varphi(x; v_1, \ldots, v_{1+k}) = \int_{\partial P(v_1,\ldots,v_{1+k})} \varphi$$

where $P(v_1, \ldots, v_{1+k})$ is the $(k+1)$-dimensional parallelepiped

spanned by the vectors $(\overrightarrow{v}_1, \ldots, \overrightarrow{v}_{k+1})$.

Algebraic Structure of Differential Forms

The set of $\Omega^*(M)$ of differential k-forms for $0 \leqslant k \leqslant n$

on an oriented n-dimensional manifold M

is a ***graded differential associative algebra over*** $\mathbb{R}$

(called ***De Rham complex***), where

- an ***associative algebra*** is a ring (addition $\omega + \eta$ and wedge product of forms $\omega \wedge \eta$) endowed with a scalar multiplication $k\omega$, where $k \in \mathbb{R}$.

- the ***grading*** is given by the form degree k.

- and the ***differential structure*** is given by the exterior derivative

$$d : \Omega^k(M) \to \Omega^{k+1}(M).$$

The De Rham Complex of Generalized Quantities

Given an oriented n-dimensional (connected) manifold M,

the (augmented) De Rham complex takes the form

$$0 \longrightarrow \mathbb{R} \hookrightarrow \Omega^0(M) \xrightarrow{d_0} \Omega^1(M) \xrightarrow{d_1} \Omega^2(M) \xrightarrow{d_2} \ldots \xrightarrow{d_{n-2}} \Omega^{n-1}(M) \xrightarrow{d_{n-1}} \Omega^n(M)$$

where $\Omega^k(M)$ is the space of differential k-forms.

Integrals of Differential Forms

In standard calculus,

it is important to make the distinction between

Riemann's definition of an integral as a limit of a sum

$$\int_a^b f(x)dx \doteq lim_{\Delta x \to 0, n \to \infty} \sum_{i=1}^{n} f(x_{i-1})\,\Delta x$$

and the second fundamental theorem of calculus

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is an *antiderivative* or a *potential* of f:

$$\frac{dF}{dx} = f.$$

There is a general formula

for integrating a k-form ω over an k-dimensional manifold N:

$$\int_N \omega$$

through a Riemann integral.

Only in certain cases

— namely, when ω is exact (i.e. when it derives from a potential $\omega = d\eta$) —

such an integral can be computed

by means of a generalization

of the second fundamental theorem of calculus

called ***Stokes theorem***.

Geometric intuition of the integral of a 1-form along a curve

At each point of the curve $\gamma(t)$,

we compute how many level lines

of the local stack defined by the 1-form ω at that point

are crossed by the tangent vector to $\gamma(t)$

and then we add all these numbers for all the points of $\gamma(t)$.

Formal definition of the integral of a 1-form along a curve

We have seen before that the integral of a gradient 1-form over a curve

is given by the expression

$$\int_\gamma df \doteq \int_0^1 df(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})dt = f(\gamma_x(1), \gamma_y 1)) - f(\gamma_x(0), \gamma_y 0))$$

where $t \in [0,1] \mapsto (\gamma_x(t), \gamma_y(t))$ is a parameterization of γ.

The integral of a 1-form ω in $\mathbb{R}^2$along a curve γ

is defined similarly as

$$\int_\gamma \omega = \int_0^1 \omega(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})dt$$

i.e. we evaluate the 1-form ω along the parameterized path $\gamma(t)$

and feed it with the tangent $\vec{\dot{\gamma}}$ vectors to $\gamma(t)$.

Then we perform a standard integration of the resulting function $\omega(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})$ of t

over the integration variable t.

The first expression

$$\int_\gamma df \doteq \int_0^1 df(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})dt = f(\gamma_x(1), \gamma_y 1)) - f(\gamma_x(0), \gamma_y 0))$$

is a particular case of the second one

$$\int_\gamma \omega = \int_0^1 \omega(\gamma_x(t), \gamma_y(t))(\vec{\dot{\gamma}})dt$$

when the 1-form that we are integrating

« derives » from a potential f,

in which cases a « fundamental theorem of calculus » applies

and the integral can be computed by evaluating

the potential on the endpoints of γ.

Geometric intuition of the integral of a 2-form over a surface

At each point $(\Phi_x(t, l), \Phi_y(t, l), \Phi_z(t, l))$ of the parameterized surface S,

we compute the flux of the local tubular structure

defined by the 2-form η at that point

through the parallelepiped tangent to S

defined by the tangent vectors $\dot{\Phi}_t$ and $\dot{\Phi}_l$.

By adding all these local fluxes,

we obtain the total flux of the tubular structure

defined by the 2-form η through S.

Formal definition of the integral of a 2-form over a surface

Analogously, the integral of a 2-form η over a surface $S \subset \mathbb{R}^3$

equipped with a parameterization $\Phi(t, l) : \mathbb{R}^2 \to \mathbb{R}^3$

is defined as

$$\int_S \eta \doteq \iint_{t,l} \eta(\Phi_x(t,l), \Phi_y(t,l), \Phi_z(t,l))(\dot{\Phi}_t, \dot{\Phi}_l) dt dl$$

where

$$\dot{\Phi}_t = (\partial_t \Phi_x(t,l), \partial_t \Phi_y(t,l), \partial_t \Phi_z(t,l))$$

and

$$\dot{\Phi}_l = (\partial_l \Phi_x(t,l), \partial_l \Phi_y(t,l), \partial_l \Phi_z(t,l))$$

are the two vectors tangent to the surface S at each point.

Stokes' Theorem

or

the Generalized Fundamental Theorem of Calculus

(E. Cartan, 1945)

Given a k-form ω and a $(k+1)$-dimensional manifold N, it holds

$$\int_N d\omega = \int_{\partial N} \omega$$

where ∂N is the a k-dimensional boundary of N.

Adjunction between d and ∂

In synthetic notation, the coupling

$$\langle \omega, N \rangle \doteq \int_N \omega \in \mathbb{R}$$

between k-forms and k-dimensional manifolds obeys

$$\langle d\omega, M \rangle = \langle \omega, \partial M \rangle$$

which means that the exterior derivative d

is the ***adjoint*** of the boundary operator ∂.

Properties of Exact Forms: Applicability of the Fundamental Theorem of Exterior Calculus

Thanks to Stokes theorem, the integral of an exact form

$$\omega = d\eta$$

over a manifold M

can be converted into an integral over its boundary ∂M:

$$\int_M \omega = \int_M d\eta = \int_{\partial M} \eta$$

Hence, exactness (the fact that ω « derives » from a potential η) guarantees

that the fundamental theorem of calculus

is valid in the generalized setting.

Conversely, when ω is not exact

$$\int_M \omega$$

cannot be converted into an integral over ∂M.

Faraday's Law of Electromagnetic Induction (I)

Let's start with one of the four Maxwell's equations (1873):

$$dE = -\frac{\partial B}{\partial t}$$

where the electric field E and the magnetic field B

are described by 1- and 2-forms respectively.

This equation encodes the fact

that a varying magnetic field B

generates an electric field E.

Faraday's Law of Electromagnetic Induction (II)

By integrating $dE = -\frac{\partial B}{\partial t}$ over a surface S on both sides, we get

$$\int_S dE = -\int_S \frac{\partial B}{\partial t}$$

Applying Stokes to the LHS

$$\int_S dE = \int_{\partial S} E$$

we obtain

$$\int_{\partial S} E = -\frac{\partial}{\partial t}\int_S B$$

Physically, this means that a varying magnetic flux

through a surface S bounded by a wire

induces an electric field on the wire.

This generalized « fundamental » theorem of calculus

can be applied only when we are integrating a

form $d\omega$ that « derives » from a potential ω.

Since we know now how to integrate general differential forms

— even when they do not « derive » from a potential —

the fundamental theorem of calculus is not that fundamental.

Homology

Boundary Operator

The boundary operator ∂ « eats » manifolds with boundary M

such that the output is the boundary ∂M of M.

- The boundary of a 1-dimensional interval $[a, b]$ is the 0-dimensional set $\{a, b\}$.

- The boundary of a circle is the empty set $\varnothing$. Manifolds without boundary are called ***cycles***.

- The boundary of a 2-dimensional disk D is a 1-dimensional circle C.

In general the boundary of a n-dimensional manifold (with boundary)

is a $(n-1)$-dimensional manifold,

that is, the boundary operator ∂ reduces the dimension in 1.

The boundary of a boundary is zero

or

A boundary has no boundary

or

A boundary is necessarily a cycle.

$$\partial^2 = 0$$

For example,

$$\partial^2[\text{Disk}] = \partial(\partial[\text{Disk}]) = \partial(\text{Circle}) = \varnothing$$

Homology in a Nutshell

The fact that a manifold M that is a boundary

$$M = \partial N$$

is itself boundless (a cycle):

$$\partial M = \partial\partial N = \partial^2 N = 0$$

does not imply that every boundless manifold

— M such that $\partial M = 0$ —

is necessarily the boundary of another manifold N.

Every boundary is a cycle,

but not every cycle is a boundary.

A boundless manifold (a cycle)

— M such that $\partial M = 0$ —

is not the boundary of another manifold N

when it encircles a hole.

A boundless manifold (a cycle)

that is not a boundary

is a symptom of a hole.

De Rham Cohomology

From Homology to Cohomology

Thanks to the duality (or adjointness)

between the boundary operator ∂ and the exterior derivative d,

the homological characterization of holes

by means of cycles (M such that $\partial M = 0$)

that are not boundaries ($M \neq \partial N$)

can be « dualized »

into a cohomological characterization of holes

by certain kinds of differential forms.

$$\partial^2 = 0 \text{ implies } d^2 = 0$$

$$\int_M d^2\omega = \int_M d(d\omega) = \int_{\partial M} d\omega = \int_{\partial\partial M=\varnothing} \omega = 0,$$

where we have applied Stokes theorem twice.

Hence,

$$d^2 = 0.$$

« […] if I could only understand

the beautiful consequence following from the concise proposition

$$d^2 = 0. \text{ »}$$

From Henri Cartan Laudatio on receiving the degree of Doctor Honoris Causa, Oxford University, 1980.

« The boundary of a boundary is zero. This central principle of algebraic topology, identity, triviality, tautology, though it is, is also the unifying theme of Maxwell electrodynamics, Einstein geometrodynamics and almost every version of modern field theory. That one can get so much from so little, almost everything from almost nothing, inspires hope that we will someday complete the mathematization of physics and derive everything from nothing, all law from no law. »

J.A. Wheeler, *Information, Physics, Quantum: The Search for Links*.

Closed Differential Forms

A n-form ω is ***closed*** if

$$d\omega = 0.$$

Exact Forms

A n-form ω is ***exact*** if

$$\omega = d\eta$$

where η is a $(n-1)$-form.

The form η is called a *potential* for ω.

Gauge Freedom

If η is a potential for ω

— i.e. if $\omega = d\eta$ —

then $\eta + d\theta$ is also a potential.

Indeed, $d(\eta + d\theta) = d\eta + d^2\theta = d\eta = \omega$.

Potentials are defined up to an exact form.

This is the generalization of the fact

that anti-derivatives are defined up to a constant.

Exact Forms are Close

Exact forms

$$\omega = d\eta$$

are necessarily closed:

$$d\omega = d(d\eta) = 0$$

since

$$d^2 = 0.$$

Now, are closed forms necessarily exact?

Poincaré Lemma

Every form ω that is closed

$$d\omega = 0$$

is *locally* exact

$$\omega = d\eta.$$

Proof of the Poincaré Lemma for 1-forms (I)

Locally, every cycle $\mathscr{C}$ is a boundary $\mathscr{C} = \partial S$

and (by using Stokes) the integral of a closed closed 1-form φ ($d\varphi = 0$)

over such a cycle is zero:

$$\int_{\mathscr{C}=\partial S} \varphi = \int_S d\varphi = 0.$$

If γ_1 and γ_2 are two paths connecting an « origin » o and a point p,

we can form a cycle $\mathscr{C} = \gamma_1 - \gamma_2$. We then have

$$\int_{\mathscr{C}} \varphi = \int_{\gamma_1-\gamma_2} \varphi = \int_{\gamma_1} \varphi - \int_{\gamma_2} \varphi = 0$$

or

$$\int_{\gamma_1} \varphi = \int_{\gamma_2} \varphi,$$

i.e. the line integral of φ from o to p does not depend on the chosen path.

Proof of the Poincaré Lemma for 1-forms (II)

We shall now prove that the function

$$f(p) = \int_{\gamma: o \to p} \varphi$$

is a potential for φ, i.e. that $df = \phi$.

Nota Bene: $f(p)$ is well-defined since the integrals of φ are path-independent.

If p_1 and p_2 are two points separated by a tiny curve $\delta : p_1 \to p_2$

whose tangent vector at p_1 is $\overrightarrow{v}$

and K_1 and K_2 are two paths from o to p_1 and p_2 respectively, then

$$df(\overrightarrow{v}) \approx \int_{K_2} \varphi - \int_{K_1} \varphi = \int_{K_2 - K_1} \varphi = \int_{\delta} \varphi \approx \varphi(\overrightarrow{v})$$

i.e. $df = \varphi$.

The topological obstructions to the global extension

of the Poincaré lemma

— classified by the so-called ***De Rham cohomology*** —

are given the presence of ***holes***.

Homologically,

holes are associated to cycles that do not bound.

We shall now see what kind of quantities

can be attributed to such non-bounding cycles

in order to detect the corresponding holes.

Summary

- Holes are homologically associated to non-bounding cycles.

- We want to attribute properties to such cycles (quantities « evaluated » on them) that indicate that there is a hole within.

- If the quantities ω are exact, Stokes' Theorem applies and the integral of ω over the cycle is always zero (independently of the presence of holes within the cycle). Hence, we need to focus on non-exact quantities.

- If the quantities are closed, they integrate to zero over boundaries.

- Hence, closed non-exact forms integrate to non-zero values over a cycle when the cycle does not bound anything, i.e. when it bounds nothing.

Exact Forms Have Zero Integrals over Cycles

By Stokes theorem, the integral of an exact form

$$\omega = d\eta$$

over a boundless manifold M (a *cycle*)

(where $\partial M = \varnothing$)

is always zero:

$$\int_M \omega = \int_M d\eta = \int_{\partial M = \varnothing} \eta = 0$$

independently of whether M is a boundary or not.

$$\langle \text{Exact}, \text{Cycle} \rangle = 0.$$

Exact forms

do not make the distinction

between cycles that bound

and cycles that do not.

$\int$	**Boundaries** $M = \partial N$	**Cycles** $\partial M = 0$
Exact forms $\omega = d\eta$	$\langle \text{Exact}, \text{Boundary} \rangle = 0$ (since every boundary is a cycle)	$\langle \text{Exact}, \text{Cycle} \rangle = 0$
Closed forms $d\omega = 0$		

Consequence: Integrals of Exact Forms are Path-Independent

If γ_1 and γ_2 are two paths connecting the same endpoints a and b,

we can form a cycle (or a loop) $L = \gamma_1 - \gamma_2$.

Then,

$$\int_L d\eta = \int_{\gamma_1 - \gamma_2} d\eta = \int_{\gamma_1} d\eta - \int_{\gamma_2} d\eta = 0$$

since $\langle \text{Exact}, \text{Cycle} \rangle = 0$

Hence,

$$\int_{\gamma_1} d\eta = \int_{\gamma_2} d\eta,$$

i.e. the integral of $d\eta$ between a and b

does not depend on the path.

Gradient Theorem

By applying Stokes Theorem,

$$\int_{\gamma} \omega = \int_{\gamma} d\eta = \int_{\partial\gamma} \eta = \eta(b) - \eta(a),$$

where a and b are the initial and final endpoints

of the oriented curve γ.

In physics jargon, the integral of a field ω

that « derives » from a potential η along a curve γ

is given by the difference between the values of the potential η

at the endpoints a and b of γ.

If our goal is to detect holes,

exact forms $\omega = d\eta$ are not good

since their integrals along cycles are always zero:

$$\int_M \omega = 0 \text{ (where } \partial M = \varnothing)$$

regardless of whether the cycle M is a boundary or not.

Hence, we should look at forms that are not exact,

i.e. forms that block the application

of the « fundamental theorem of calculus »

(Stokes theorem) so that their integral over cycles is not necessarily zero.

Closed Forms have Vanishing Integrals over Boundaries

By Stokes theorem, the integral of closed form ω

$$(d\omega = 0)$$

over a boundary $M = \partial N$

is always zero:

$$\int_{M=\partial N} \omega = \int_{M} d\omega = 0$$

Synthetically,

$$\langle \text{Closed}, \text{Boundary} \rangle = 0$$

$\int$	**Boundaries** $M = \partial N$	**Cycles** $\partial M = 0$
Exact forms $\omega = d\eta$	$\langle \text{Exact}, \text{Boundary} \rangle = 0$ (since every boundary is a cycle)	$\langle \text{Exact}, \text{Cycle} \rangle = 0$
Closed forms $d\omega = 0$	$\langle \text{Closed}, \text{Boundary} \rangle = 0$	

Hence, only a ***non-exact closed form*** ω

might integrate non-trivially

over a cycle M,

and this will be the case

only when the cycle M is not a boundary,

i.e. when it encircles a hole.

$\int$	**Boundaries** $M = \partial N$	**Cycles** $\partial M = 0$
Exact forms $\omega = d\eta$	**0** **An exact form is necessarily closed and closed forms integrate to zero over boundaries**	**0** $\langle \text{Exact}, \text{Cycle} \rangle = 0$
Closed forms $d\omega = 0$	**0** $\langle \text{Closed}, \text{Boundary} \rangle = 0$	Non-zero only when the closed form is not exact and the cycle is not a boundary.

Holes from a Cohomological Standpoint

Since a cycle M

that is not a boundary

$$M \neq \partial N$$

is a symptom of a hole (homology),

closed non-exact forms detect holes.

Periods

The integrals

$$\int_M \omega$$

of closed forms over cycles

are called ***periods***.

The fact that a period is not zero

indicates that the cycle is not a boundary,

i.e. that it encircles a hole.

Forms ω and $\omega + d\theta$

differing in an exact form $d\theta$

define equivalent periods:

$$\int_M \omega + d\theta = \int_M \omega + \int_M d\theta = \int_M \omega + \int_{\partial M = \varnothing} \theta = \int_M \omega$$

Hence, the relevant objects

are closed non-exact forms up to exact forms,

i.e. equivalence classes under the equivalence relation

$$\omega \sim \eta$$

if

$$\eta = \omega + d\theta$$

Periods only depend on the homology class of the cycle

Two cycles γ and γ' are in the same homology class

— and said to be *homologous* —

if $\gamma' - \gamma = \partial\lambda$, i.e.

if together they form the boundary of some higher-dimensional object

(which means that cycles that are boundaries are representatives of the trivial homology class).

The period of a form over a cycle only depends on the homology class of the cycle:

$$\int_{\gamma'} \omega = \int_{\gamma+\partial\lambda} \omega = \int_{\gamma} \omega + \int_{\partial\lambda} \omega = \int_{\gamma} \omega + \int_{\lambda} d\omega = \int_{\gamma} \omega,$$

where we have used Stokes theorem and $d\omega = 0$.

Conceptually, this means that periods

only « see » the hole-related topological properties of the cycle,

namely whether it loops around a hole or not.

De Rham Cohomology (I)

— after Georges De Rham (1903–1990) —

If very closed k-form ω ($d\omega = 0$)

is exact

— i.e. $\omega = d\eta$ —,

then the De Rham complex

$$0 \longrightarrow \mathbb{R} \hookrightarrow \Omega^0(M) \xrightarrow{d_0} \Omega^1(M) \xrightarrow{d_1} \Omega^2(M) \xrightarrow{d_2} \ldots \xrightarrow{d_{n-2}} \Omega^{n-1}(M) \xrightarrow{d_{n-1}} \Omega^n(M)$$

is said to be an « exact » sequence,

which encodes the fact that the manifold M has no holes.

De Rham Cohomology (II)

Closed n-forms ω ($d\omega = 0$)

that are not exact

— that is, $\omega \neq d\eta$ —

obstruct the « exactness » of the complex

$$0 \longrightarrow \mathbb{R} \hookrightarrow \Omega^0(M) \xrightarrow{d_0} \Omega^1(M) \xrightarrow{d_1} \Omega^2(M) \xrightarrow{d_2} \ldots \xrightarrow{d_{n-2}} \Omega^{n-1}(M) \xrightarrow{d_{n-1}} \Omega^n(M)$$

The k-cohomology group of M is defined as

$$H^k(M) = \frac{Ker(d^{k+1})}{Im(d^k)}$$

where an element in $H^k(M)$

is an equivalence class $[\omega]$

of closed k-forms under the equivalence relation $\omega \sim \eta$ if $\eta = \omega + d\theta$,

where θ is a $(k-1)$-form.

This means that the exact forms are representatives of the trivial class [0]

Example: Calculus on a Circle S^1 (I)

• Since we are in dimension 1, the highest form degree is 1 (i.e. we can only have 0-forms and 1 -forms). Then $d\omega = 0$ for any 1-form ω, that is, all 1-forms are closed.

• In order to compute the cohomology $H^1(S^1)$, we need to evaluate whether a 1-form ω (necessarily closed) is exact or not.

• Given a 1-form $\omega = f(\theta)d\theta$ with $0 \leqslant \theta \leqslant 2\pi$, we want to define an antiderivative or potential as we do it in standard calculus on $\mathbb{R}$.

• As in standard calculus, we propose the potential

$$F(\theta) = \int_0^\theta f(\phi)d\phi$$

such that

$$\frac{dF}{d\theta} = f(\theta) \text{ or } dF(\theta) = f(\theta)d\theta = \omega.$$

Hence, $\omega = dF$ is exact.

• Since on the circle $0 = 2\pi$, the potential $F(\theta)$ must satisfy the periodicity condition

$$F(2\pi) = F(0) = \int_0^0 f(\phi)d\phi = 0.$$

Example: Calculus on a Circle S^1 (II)

• Let's consider now the family of 1-forms $\omega = \frac{c}{2\pi} d\theta$ with $c \in \mathbb{R} - \{0\}$.

• The potential that would make the 1-forms $\omega = \frac{c}{2\pi} d\theta$ exact would be $F(\theta) = \frac{c}{2\pi} \int_0^\theta d\phi = \frac{c}{2\pi} \theta$ so that $\frac{dF}{d\theta} = \frac{c}{2\pi}$.

• Now, the potential $F(\theta) = \frac{c}{2\pi} \theta$ does not fulfill the periodicity condition $F(2\pi) = F(0)$ since $F(2\pi) = c$.

• Therefore, the closed 1-forms $\omega = \frac{c}{2\pi} d\theta$ are not exact and have non-zero periods

$$\int_{S^1} \omega = \frac{c}{2\pi} \int_{S^1} d\theta = c \in \mathbb{R}.$$

• As expected, the closed 1-forms that are exact have zero periods since $\langle \text{Exact}, \text{Cycle} \rangle = 0$. Indeed, if $\omega = dF$:

$$\int_{S^1} \omega = \int_{S^1} dF = \int_{\partial S^1 = \varnothing} dF = 0$$

Differential Geometry in a Nutshell (I)

• Quantities are graded and called ***differential k-forms***.

• A differentiation operator d called ***exterior derivative*** encodes the variations of differential forms in any direction:

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

• n-quantities ω can be « coupled » to oriented n-dimensional *manifolds* N through an operation called ***integration***

$$\langle \omega, N \rangle \doteq \int_N \omega \in \mathbb{R}$$

• ***Stokes theorem***: d is « adjoint » to the boundary operator ∂:

$$\langle d\omega, P \rangle = \langle \omega, \partial P \rangle$$

or

$$\int_P d\omega = \int_{\partial P} \omega$$

Differential Geometry in a Nutshell (II)

- $\partial^2 = 0$ or « A boundary has no boundary » or « Boundaries are cycles ».

- By using Stokes theorem, $\partial^2 = 0$ implies $d^2 = 0$.

- $d^2 = 0$ means that every *exact form*

$$\omega = d\eta$$

is a *closed form*

$$d\omega = d(d\eta) = 0.$$

- **Poincaré lemma**: a closed form $d\omega = 0$ is *locally* exact $\omega = d\eta$.

- **De Rham cohomology** encodes the topological obstructions (given by holes) to the global extension of the Poincaré lemma.

De Rham Cohomology in a Nutshell

.The integral of forms ω that are not (globally) exact $\omega \neq d\eta$ over cycles are not necessarily zero since Stokes theorem only applies to exact forms.

.The integral of closed forms ω over cycles is zero when the cycle is a boundary (by Stokes theorem).

.Hence, the integral of closed non-exact forms ω over cycles might be non-zero only when the cycle is not a boundary, i.e. when the cycle encircles a hole.

Closed non-exact forms are the adequate quantities to detect holes.

We began with hypothetical objects

— the infinitesimals dx —

conceived as an extension of the notion of number

that are as close as possible

to nothingness without falling into it,

and we ended with a generalization of the notion of quantity

— the differential forms —

through which we can detect localized manifestations

of nothingness,

that is, holes.

Thanks

for your kind

and patient

attention!

Appendices

Integration along a curve (I)

We want to generalize the integration of a function on a single variable x

$$f : \mathbb{R} \to \mathbb{R}$$

along an interval

$$[a, b] \subset \mathbb{R}$$

to an integration of a function on n variables $(x, y, z, \dots)$

$$f : \mathbb{R}^n \to \mathbb{R}$$

along a more general curve

$$\gamma \subset \mathbb{R}^n.$$

Integration along a curve (II)

Given a function

$$f(x, y) : \mathbb{R}^2 \to \mathbb{R}$$

and a curve γ in $\mathbb{R}^2$,

we want to compute the integral

$$\int_\gamma f(x, y) ds,$$

where ds denotes an « infinitesimal » interval along the curve γ,

i.e. a differential length of γ.

This integral yields the area of the « curtain »

defined by the graph of $f(x, y)$ (a surface)

and the curve γ.

Integration along a curve (III)

We shall compute this integral by « pulling it back » to a situation we know:

the integral of a function along an interval in $\mathbb{R}$

To do so, we shall introduce a ***parameterization*** of the curve γ:

$$t \in [0,1] \mapsto (\gamma_x(t), \gamma_y(t))$$

Conceptually, this amounts to describe the « static » curve γ

as a « dynamical » unfolding of γ

« parameterized » by a temporal variable t.

Such a parameterization is not intrinsic to the static curve

and the value of the integral should not depend

on the chosen (orientation-preserving) parameterization.

Integration along a curve (III)

The integral of a function

$$f(x, y) : \mathbb{R}^2 \to \mathbb{R}$$

along a curve γ in $\mathbb{R}^2$ parameterizaed by the map

$$t \in [0,1] \mapsto (\gamma_x(t), \gamma_y(t)),$$

is defined as

$$\int_\gamma f(x, y)ds \doteq \int_0^1 f(\gamma_x(t), \gamma_y(t))\sqrt{\left(\frac{d\gamma_x}{dt}\right)^2 + \left(\frac{d\gamma_y}{dt}\right)^2}\, dt$$

where $\sqrt{\left(\frac{d\gamma_x}{dt}\right)^2 + \left(\frac{d\gamma_y}{dy}\right)^2}\, dt = \sqrt{\left(\frac{d\gamma_x}{dt}\right)^2 dt^2 + \left(\frac{d\gamma_y}{dt}\right)^2 dt^2} = \sqrt{\left(d\gamma_x\right)^2 + \left(d\gamma_y\right)^2} = ds$

is the « infinitesimal » interval along the curve.

Nota Bene: the RHS integral in the definition is a standard integral

of a function on a single variable t

over an interval $[0,1] \subset \mathbb{R}$.

Reparameterization Invariance

Given a map

$$h(t') = [a', b'] \to I = [a, b],$$

the composition

$$\gamma \circ h : [a', b'] \to \mathbb{R}^2$$

is a new parameterization of the same static curve γ.

The integral is invariant under this change of parameterization

$$\int_\gamma f(x,y)ds = \int f(x(t),y(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

$$= \int f(x(t),y(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,\frac{dt}{dt'}dt'$$

$$= \int f(x(h(t')),y(h(t')))\sqrt{\left(\frac{dx}{dh}\frac{dh}{dt'}\right)^2 + \left(\frac{dy}{dh}\frac{dh}{dt'}\right)^2}\,dt'$$

$$= \int f(x(h(t')),y(h(t')))\sqrt{\left(\frac{dx}{dt'}\right)^2 + \left(\frac{dy}{dt'}\right)^2}\,dt'$$